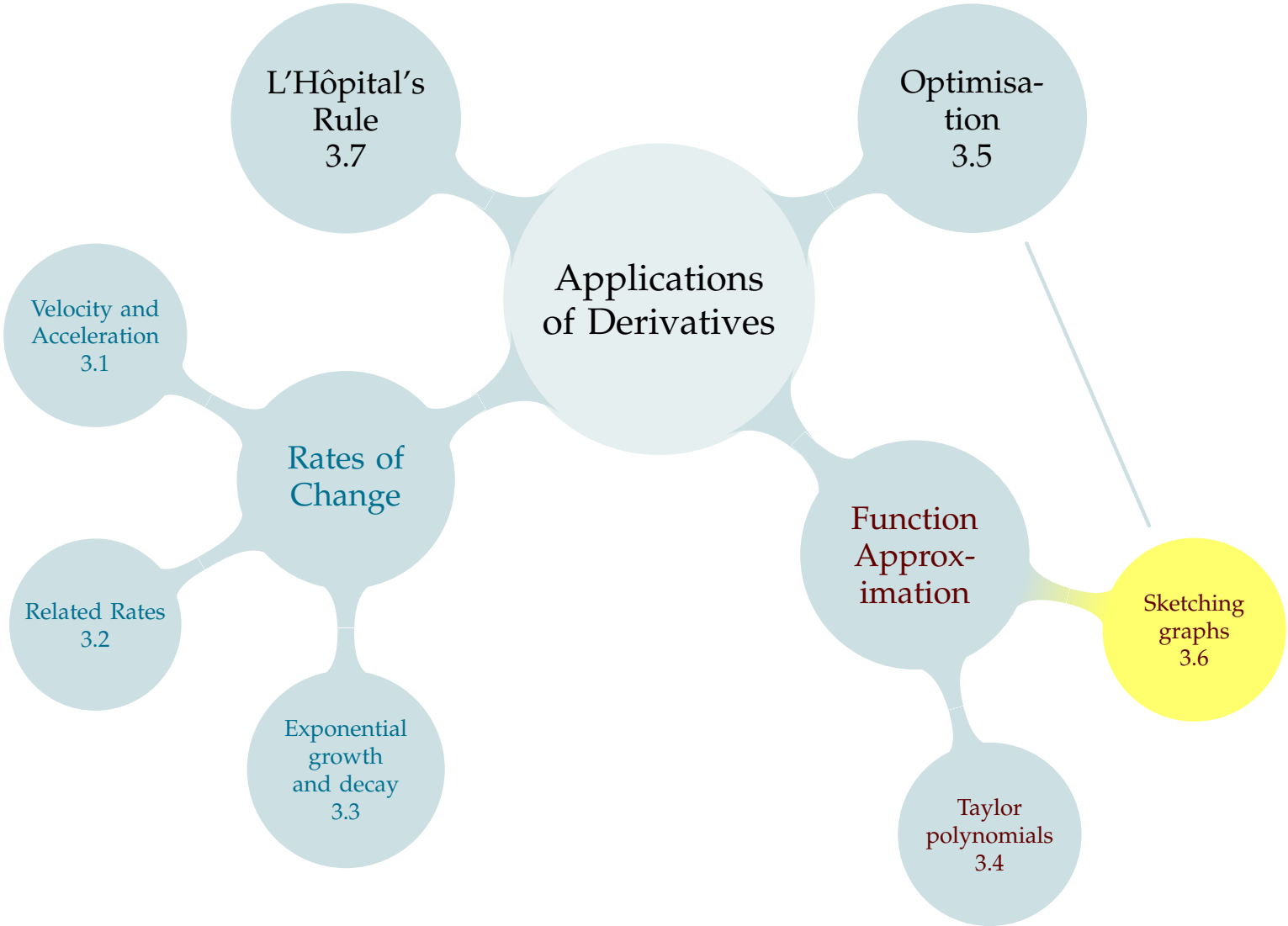
Table of Contents



CURVE SKETCHING

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Curve Sketching

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Where might you expect $f(x)$ to have a vertical asymptote? What does the function look like nearby?
(Recall: a vertical asymptote occurs at $x = a$ if the function has an infinite discontinuity at a. That is, $\lim_{x \to a^{\pm}} f(x) = \pm\infty$.)

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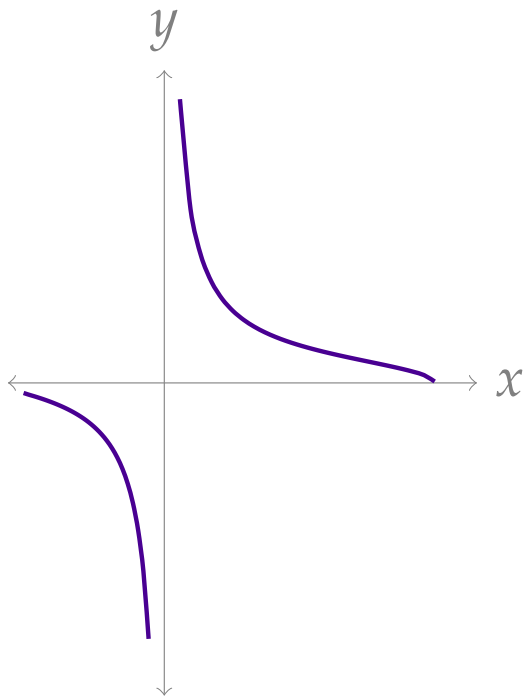
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Where is $f(x) = 0$?

What happens to $f(x)$ near its other endpoint, $x = -1$?


y
x

Curve Sketching

Good things to check:

- Domain
- Vertical asymptotes: $\lim_{x \to a} f(x) = \pm\infty$
- Intercepts: $x = 0, f(x) = 0$
- Horizontal asymptotes and end behaviour: $\lim_{x \to \pm\infty} f(x)$

CURVE SKETCHING

Identify: domain, vertical asymptotes, intercepts, and horizontal asymptotes

$$f(x) = \frac{x-2}{(x+3)^2}$$

Curve Sketching

Identify: domain, vertical asymptotes, intercepts, and horizontal asymptotes

$$f(x) = \frac{x-2}{(x+3)^2}$$

- Domain: $x \neq -3$
- Vertical asymptote: $x = -3$
- Intercepts: $(2, 0)$, $(0, -\frac{2}{9})$
- Horizontal asymptote: $y = 0$ in both directions

https://www.desmos.com/calculator/hyzl5cyq7i

CURVE SKETCHING

Identify: domain, vertical asymptotes, intercepts, and horizontal asymptotes

$$f(x) = \frac{(x+2)(x-3)^2}{x(x-5)}$$

Q

Curve Sketching

Identify: domain, vertical asymptotes, intercepts, and horizontal asymptotes

$$f(x) = \frac{(x+2)(x-3)^2}{x(x-5)}$$

- Domain: $x \neq 0, 5$
- Vertical asymptotes: $x = 0$, $x = 5$
- Intercepts: $(-2, 0)$, $(3, 0)$
- Horizontal asymptote: none

https://www.desmos.com/calculator/ploa0q7bxn

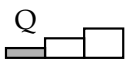
First Derivative

Add complexity: Increasing/decreasing, critical and singular points.

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$$f(x) = \frac{1}{2}x^4 - \frac{4}{3}x^3 - 15x^2$$



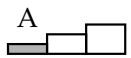
First Derivative

Add complexity: Increasing/decreasing, critical and singular points.

$$f(x) = \frac{1}{2}x^4 - \frac{4}{3}x^3 - 15x^2$$

Domain	all real numbers
Intercepts	Factor $f(x) = x^2\left(\frac{1}{2}x^2 - \frac{4}{3}x - 15\right)$. Intercept at the origin; use the quadratic formula to find x-intercepts at $x = \frac{4 \pm \sqrt{286}}{3}$, so $x \approx 7$ and $x \approx -4.3$.
End behaviour	$\lim_{x\to\infty} f(x) = \infty$ and $\lim_{x\to-\infty} f(x) = \infty$

This is the information we've already talked about gathering. Now let's add the first derivative.



First Derivative

Add complexity: Increasing/decreasing, critical and singular points.

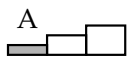
$$f(x) = \frac{1}{2}x^4 - \frac{4}{3}x^3 - 15x^2$$

$$\begin{aligned} f'(x) &= 2x^3 - 4x^2 - 30x \\ &= 2x(x^2 - 2x - 15) \\ &= 2x(x-5)(x+3) \end{aligned}$$

So, critical points are $x = 0$, $x = -3$, and $x = 5$. No singular points. At the critical points $f(0) = 0, f(-3) = -58.5, f(5) = -229.1\overline{6}$.

$x \approx -4.3$	$x < -3$	x=-3	$-3 < x < 0$	x=0	$0 < x < 5$	x=5	$x > 5$	$x \approx 7$
$f(x) = 0$	$f' < 0$	CP	$f' > 0$	CP	$f' < 0$	CP	$f' > 0$	f(x)=0
intercept	decr	loc min	incr	loc max	decr	loc min	incr	intercept

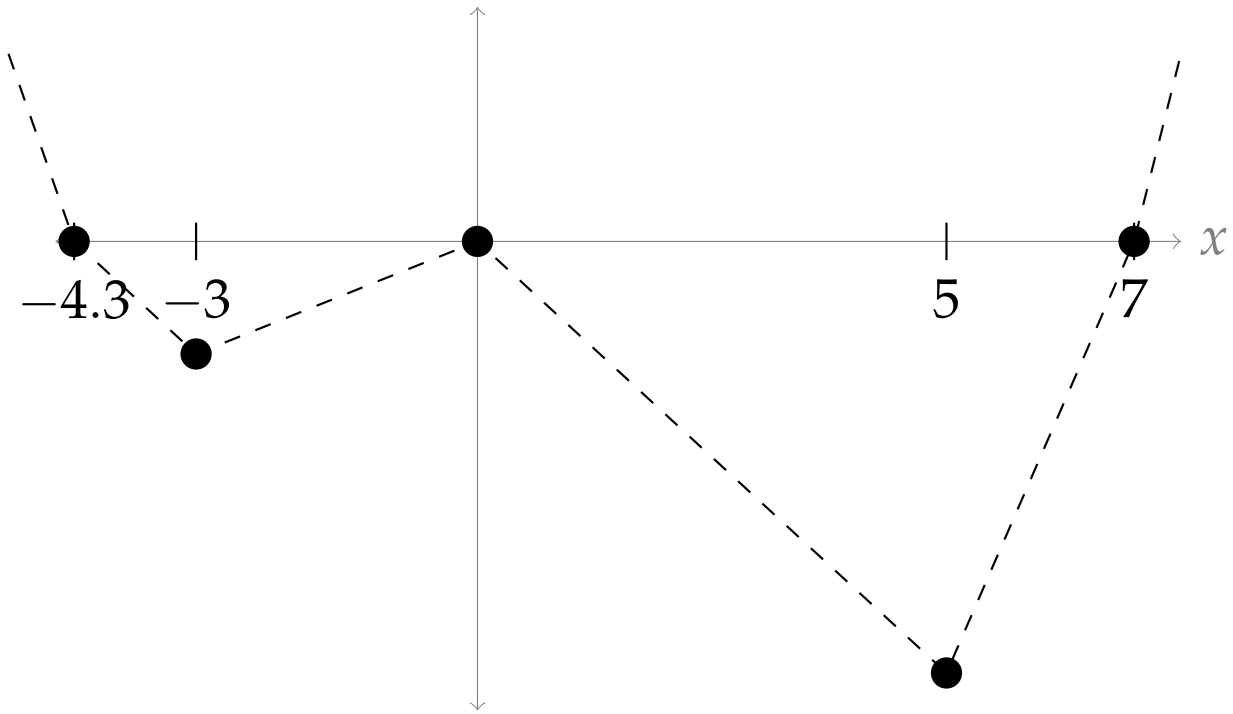
This gives us enough information to draw a skeleton.



First Derivative

Add complexity: Increasing/decreasing, critical and singular points.

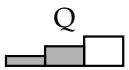
$$f(x) = \frac{1}{2}x^4 - \frac{4}{3}x^3 - 15x^2$$



https://www.desmos.com/calculator/lxdlgmhnsl

What does the graph of the following function look like?

$$f(x) = \frac{1}{3}x^3 + 2x^2 + 4x + 24$$



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Domain all real numbers.

End behaviour $\lim_{x\to-\infty} f(x) = -\infty$ and $\lim_{x\to\infty} f(x) = \infty$

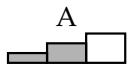
Intercepts

$$f(0) = 24$$

$$f(x) = \frac{1}{3}x^2(x+6) + 4(x+6)$$

$$= \left(\frac{1}{3}x^2 + 4\right)(x+6)$$

Only two intercepts: $(0, 24)$ and $(-6, 0)$.



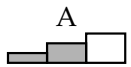
What does the graph of the following function look like?

$$f(x) = \frac{1}{3}x^3 + 2x^2 + 4x + 24$$

Critical points $f'(x) = x^2 + 4x + 4 = (x+2)^2$ so $x = -2$ is the location of the only critical point.
$f(-2) = 24 - \frac{8}{3} = 21 + \frac{1}{3}$.

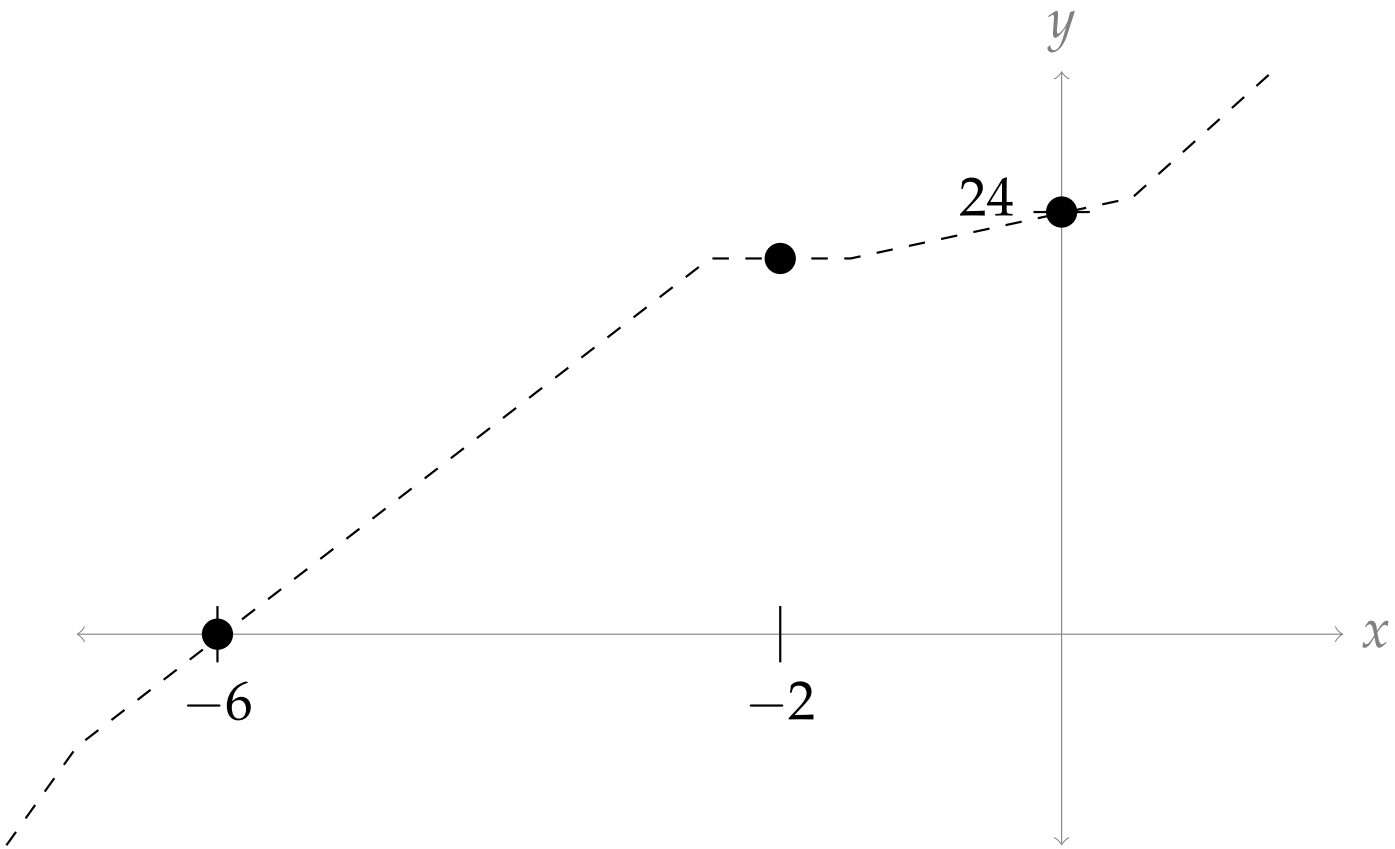
Increasing, decreasing $f'(x) > 0$ except at $x = -2$, so apart from the critical point, $f(x)$ is increasing

This is enough for us to draw a skeleton.



What does the graph of the following function look like?

$$f(x) = \frac{1}{3}x^3 + 2x^2 + 4x + 24$$



https://www.desmos.com/calculator/xum0mstmiv

What does the graph of the following function look like?

$$f(x) = e^{\frac{x+1}{x-1}}$$



What does the graph of the following function look like?

$$f(x) = e^{\frac{x+1}{x-1}}$$

Domain $x \neq 1$

We need to consider what happens as x approaches 1 from the left and the right.

Vertical asymptotes

$$\lim_{x\to 1^-} \frac{x+1}{x-1} = -\infty \implies \lim_{x\to 1^-} f(x) = \lim_{A\to -\infty} e^A = 0$$

$$\lim_{x\to 1^+} \frac{x+1}{x-1} = \infty \implies \lim_{x\to 1^+} f(x) = \lim_{A\to +\infty} e^A = \infty$$

End behaviour $\lim_{x\to\pm\infty} \frac{x+1}{x-1} = 1$, so $\lim_{x\to\pm\infty} f(x) = e$

Intercepts $f(0) = \frac{1}{e}$; there are no roots



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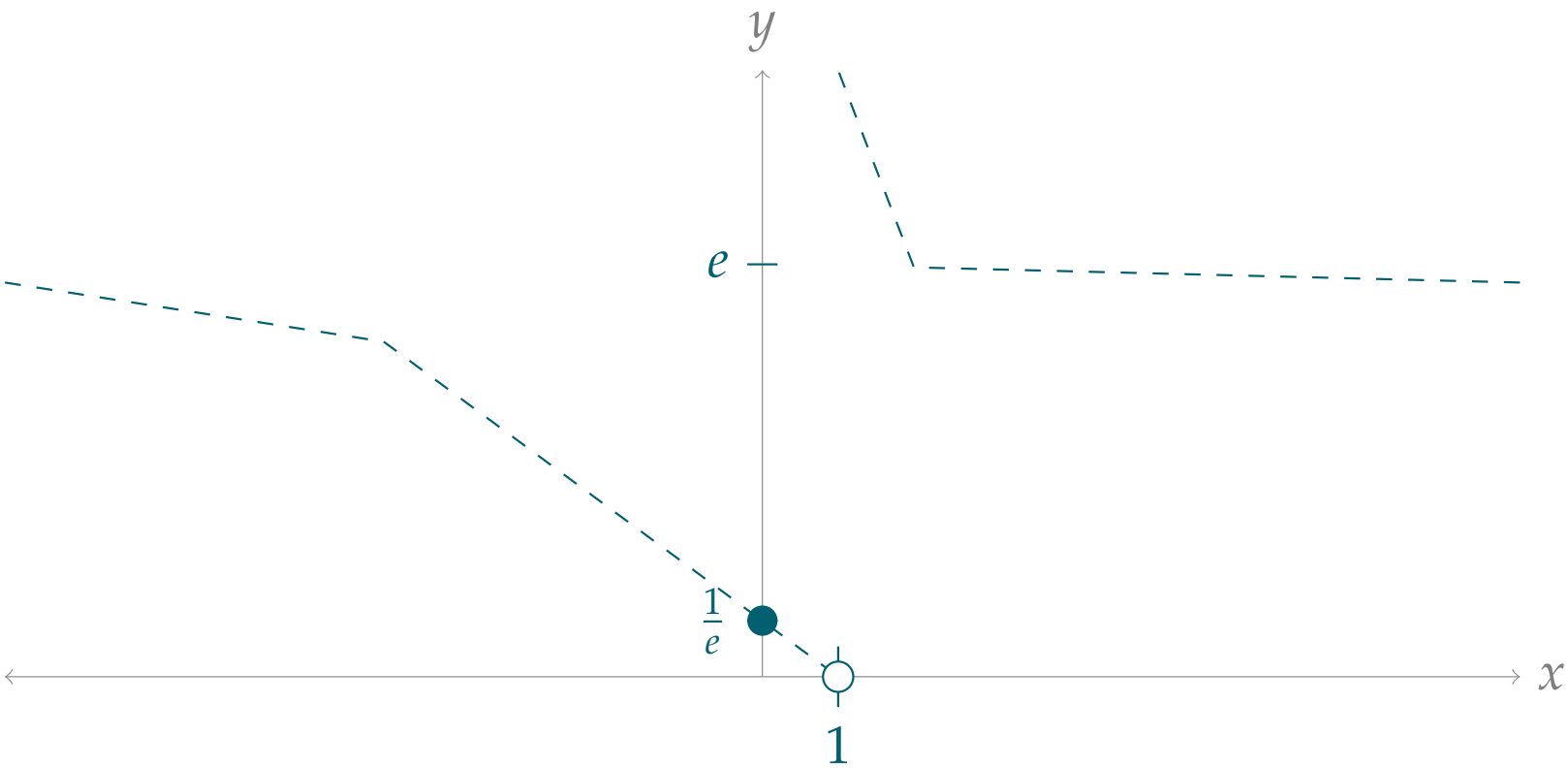
Critical points	$f'(x) = e^{\frac{x+1}{x-1}} \left(\frac{-2}{(x-1)^2} \right)$	no critical points
Increasing, decreasing	$f(x)$ is decreasing everywhere it is defined	

This information is enough to draw a skeleton.

A

What does the graph of the following function look like?

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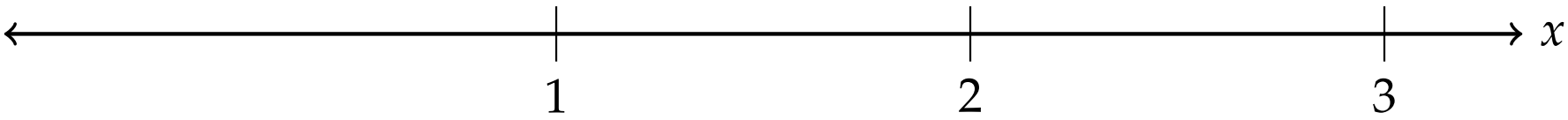


https://www.desmos.com/calculator/x0cccy1ggj

Signs of Factored Functions

SKIP SIGN CHANGES

$$f(x) = \quad (x-1) \quad (x-2) \quad (x-3)$$

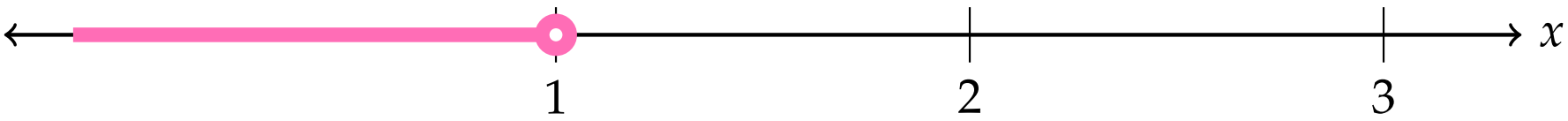


Signs of Factored Functions

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$$\quad\; - \qquad\quad\;\; \qquad\quad\;\;$$



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$$f(x) = \quad (x-1) \quad (x-2) \quad (x-3)$$
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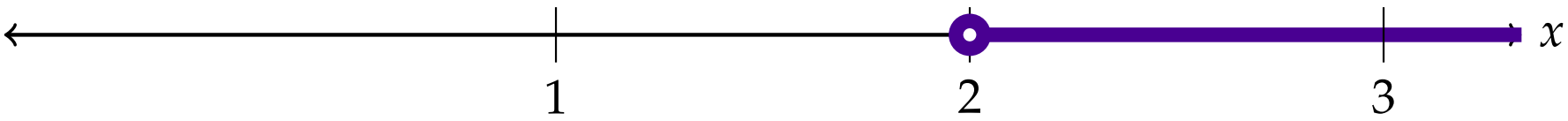
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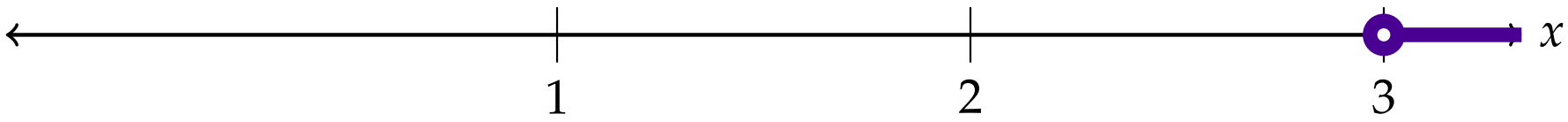
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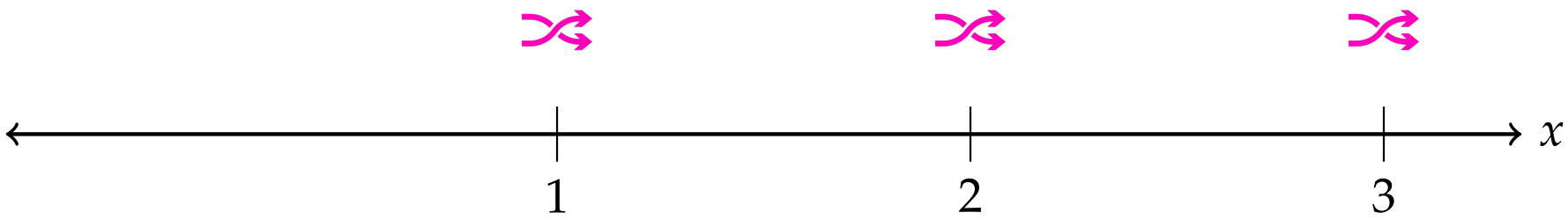
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Signs of Factored Functions

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Sign of entire function:

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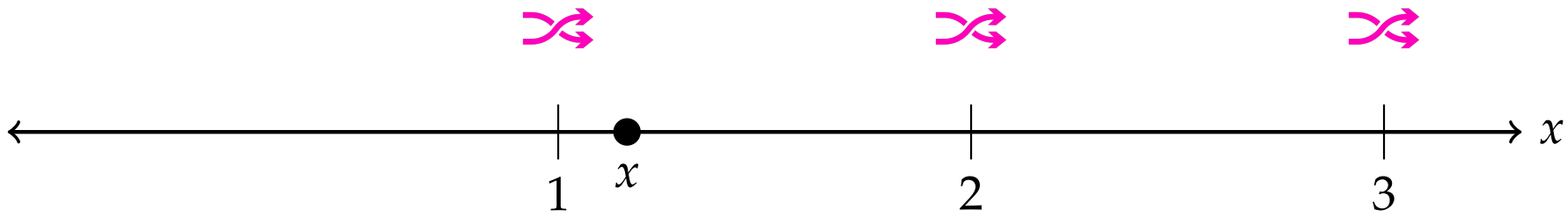
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$$f(x) = \underset{+}{(x-1)} \quad \underset{-}{(x-2)} \quad \underset{-}{(x-3)}$$

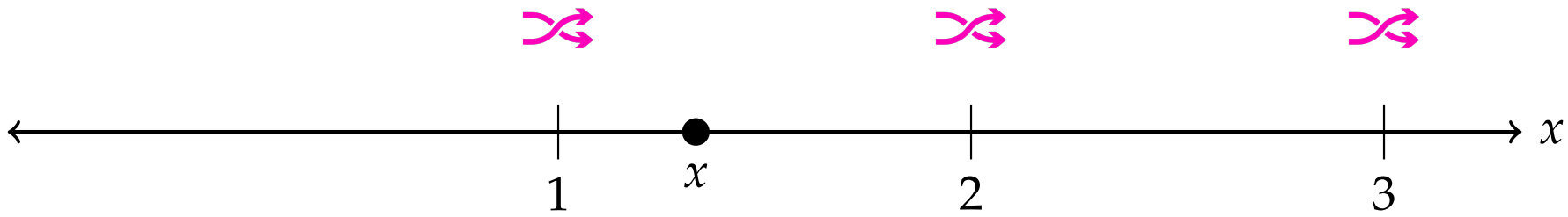


Sign of entire function: $+$

Signs of Factored Functions

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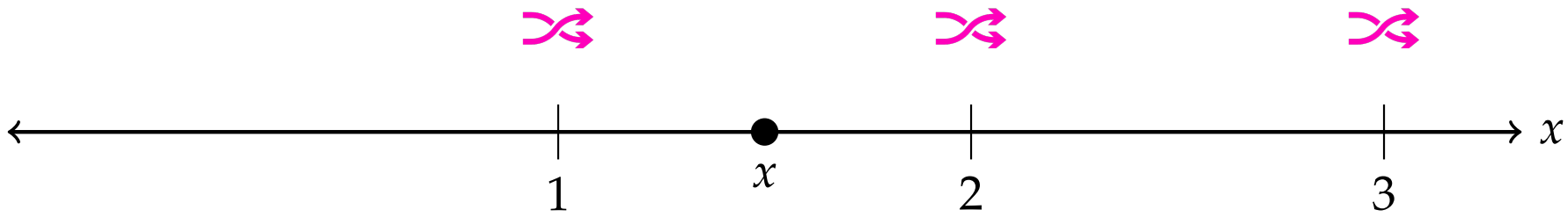


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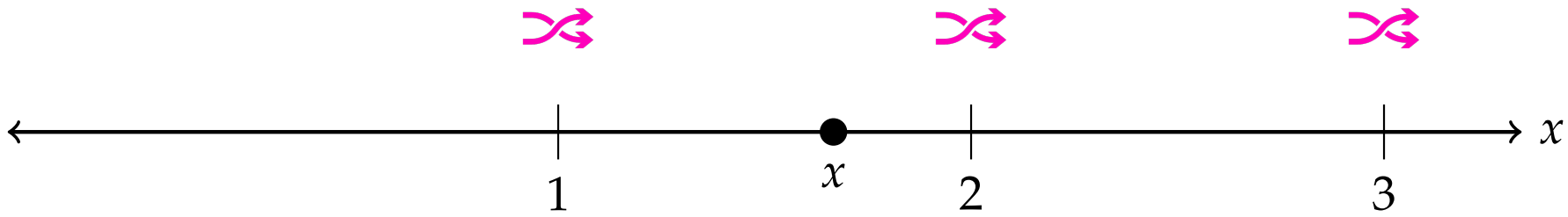


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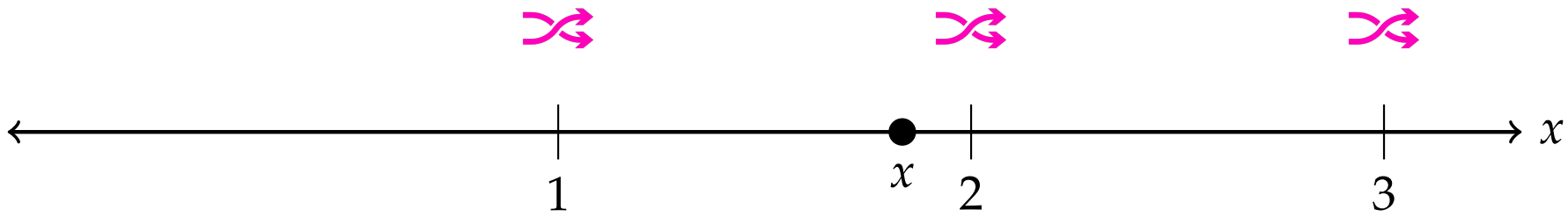


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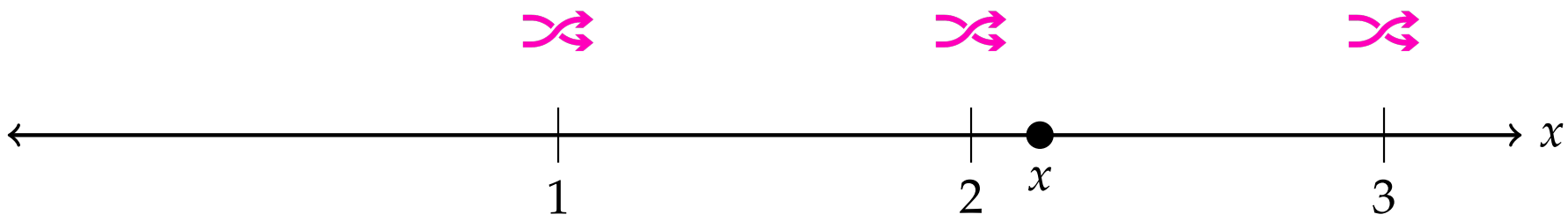


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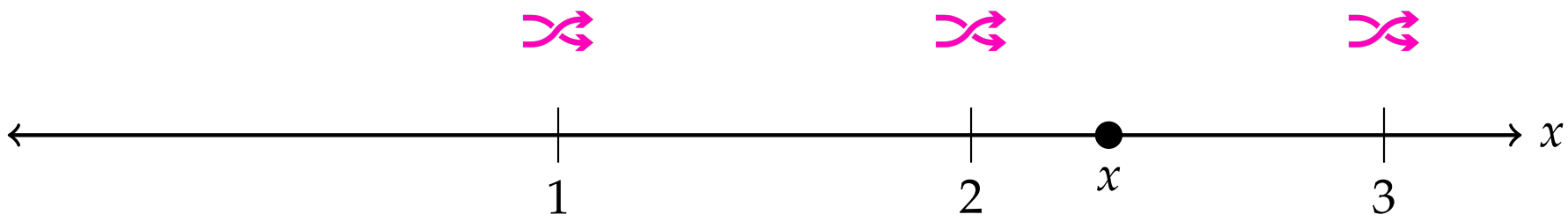


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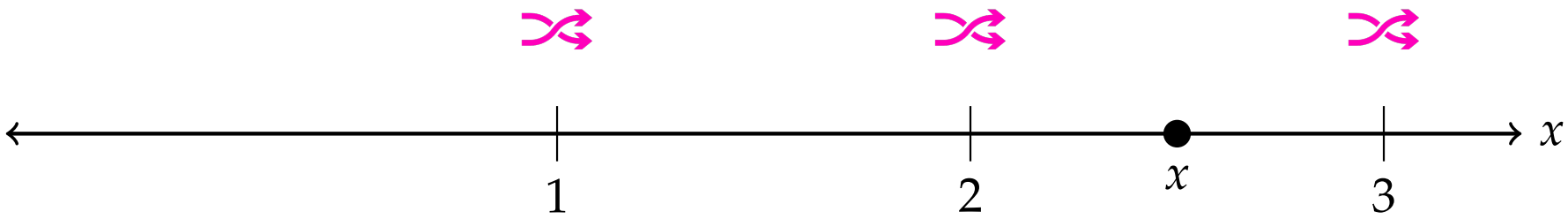


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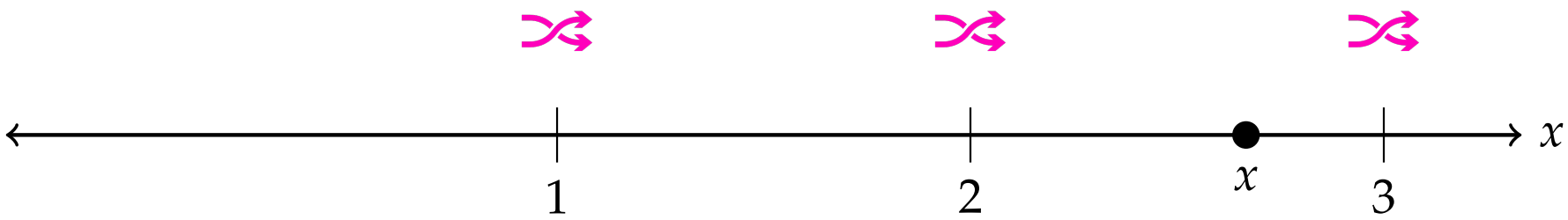


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Signs of Factored Functions

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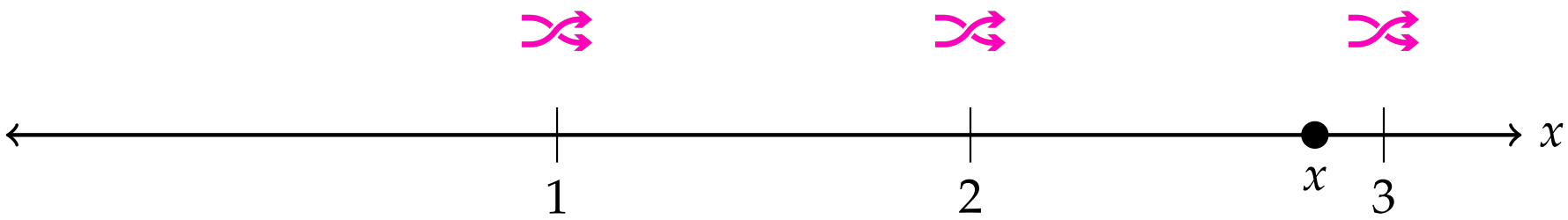


Sign of entire function: —

Signs of Factored Functions

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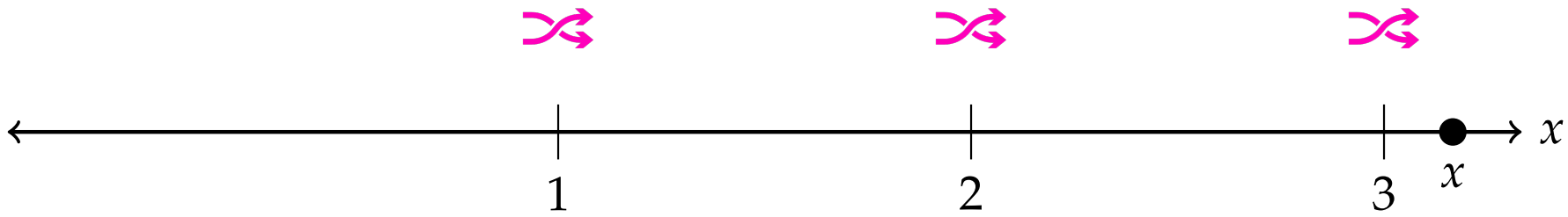


Sign of entire function: —

Signs of Factored Functions

SKIP SIGN CHANGES

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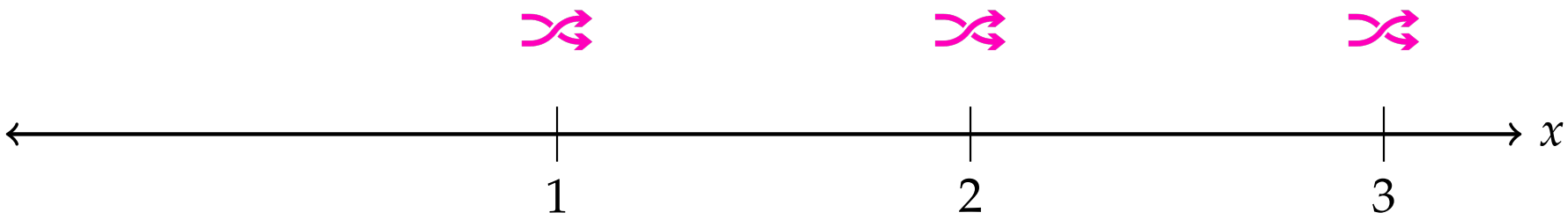


Sign of entire function: +

Signs of Factored Functions

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$$f(x) = \quad (x-1) \quad (x-2) \quad (x-3)$$

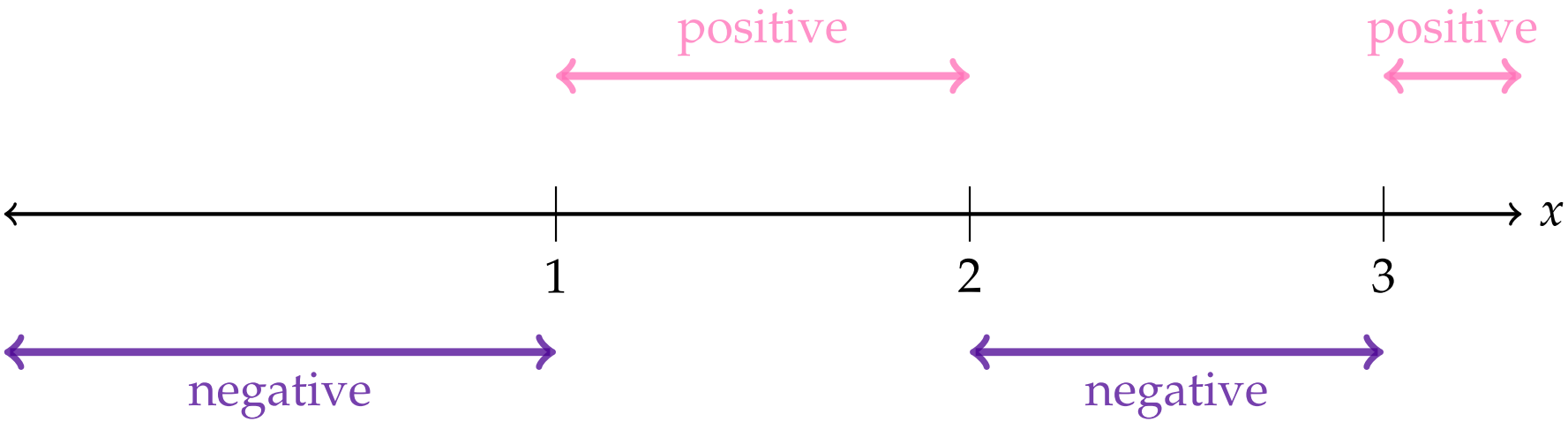


Sign of entire function:

Signs of Factored Functions

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$$f(x) = \quad (x-1) \quad (x-2) \quad (x-3)$$

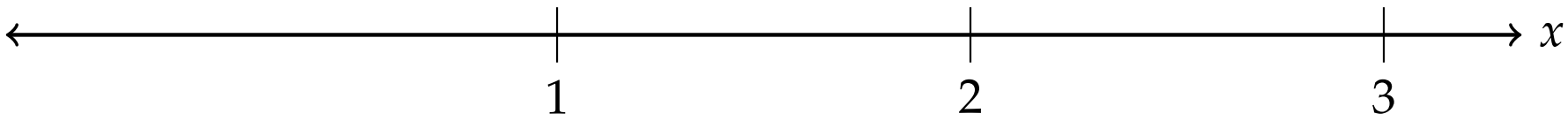


Sign of entire function:

Signs of Factored Functions

SKIP SIGN CHANGES

$$f(x) = \quad (x-1) \quad (x-2)^2 \quad (x-3)$$

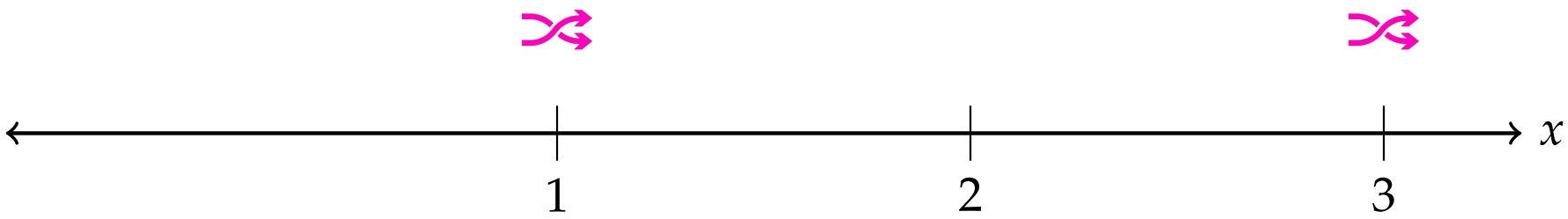


Sign of entire function:

Signs of Factored Functions

SKIP SIGN CHANGES

$$f(x) = \quad (x-1) \quad (x-2)^2 \quad (x-3)$$
$$+$$

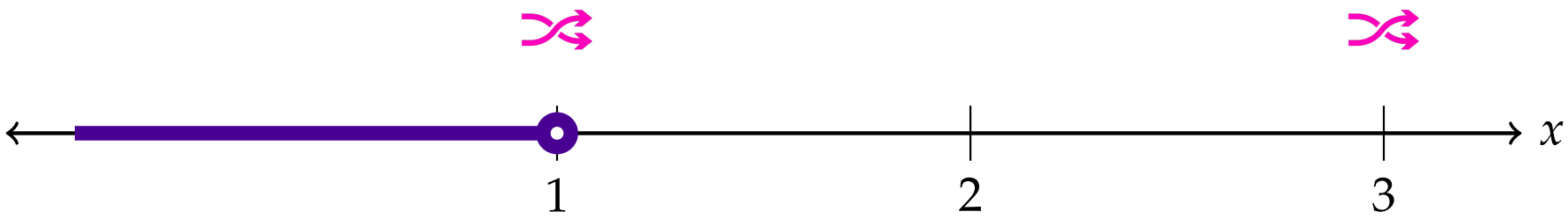


Sign of entire function:

Signs of Factored Functions

SKIP SIGN CHANGES

$$f(x) = \underset{-}{(x-1)} \quad \underset{+}{(x-2)^2} \quad \underset{-}{(x-3)}$$



Sign of entire function:

Signs of Factored Functions

SKIP SIGN CHANGES

$$f(x) = \underset{+}{(x-1)} \quad \underset{+}{(x-2)^2} \quad \underset{-}{(x-3)}$$

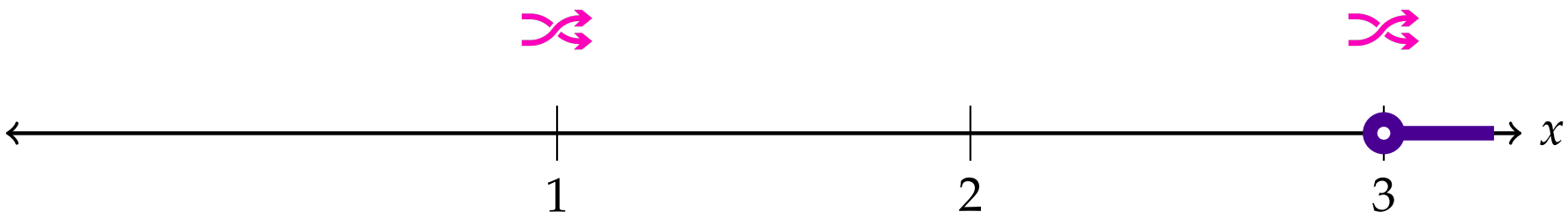


Sign of entire function:

Signs of Factored Functions

SKIP SIGN CHANGES

$$f(x) = \underset{+}{(x-1)} \quad \underset{+}{(x-2)^2} \quad \underset{+}{(x-3)}$$



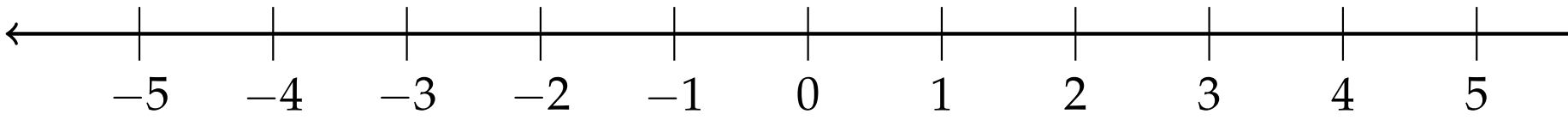
Sign of entire function:

Signs of Factored Functions

SKIP SIGN CHANGES

$$f(x) = (x-3)(x-1)^2x(x+2)^3(x+5)^4$$

Where is $f(x)$ positive? Where is it negative?

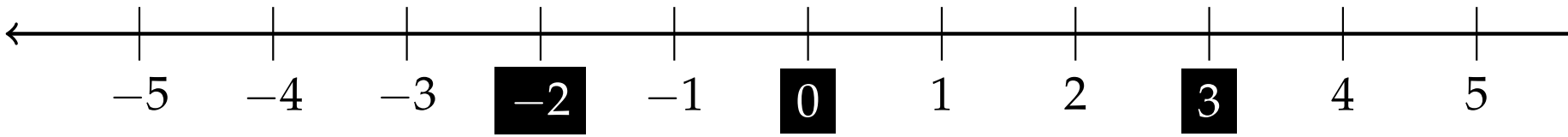


Signs of Factored Functions

SKIP SIGN CHANGES

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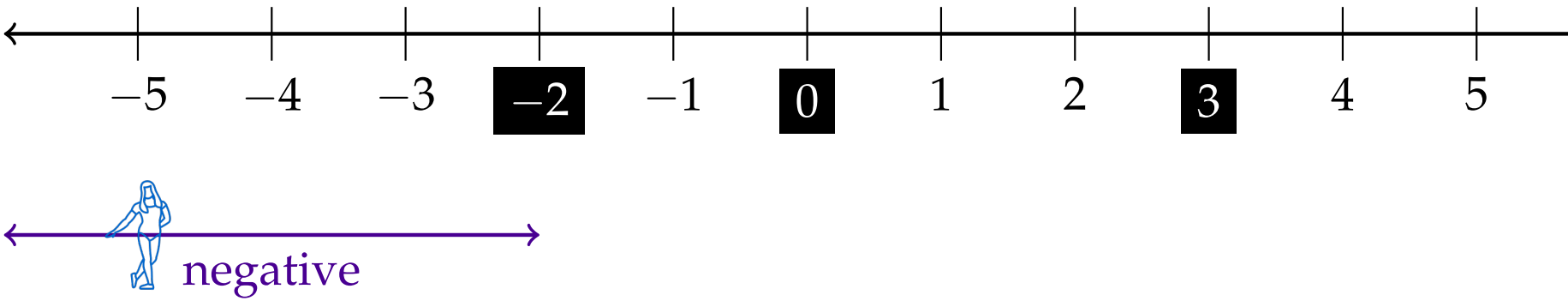


Signs of Factored Functions

SKIP SIGN CHANGES

$$f(x) = (x-3)(x-1)^2x(x+2)^3(x+5)^4$$

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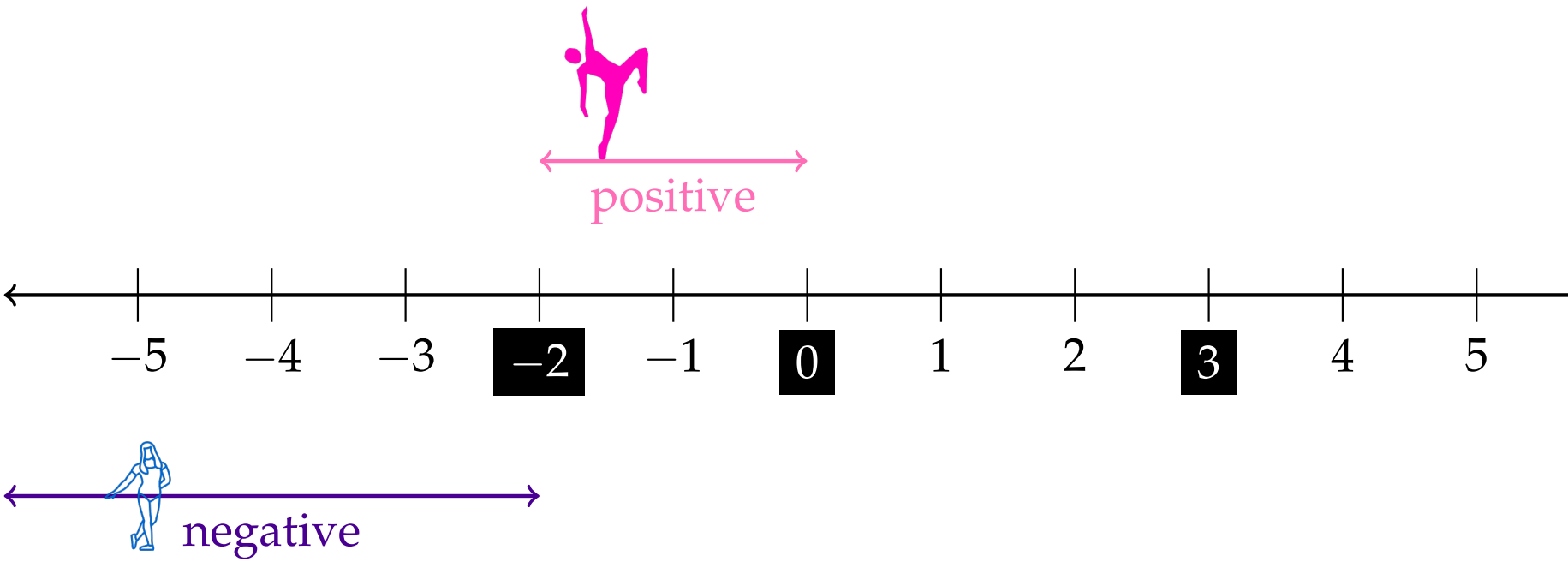


Signs of Factored Functions

SKIP SIGN CHANGES

$$f(x) = (x-3)(x-1)^2x(x+2)^3(x+5)^4$$

Where is $f(x)$ positive? Where is it negative?

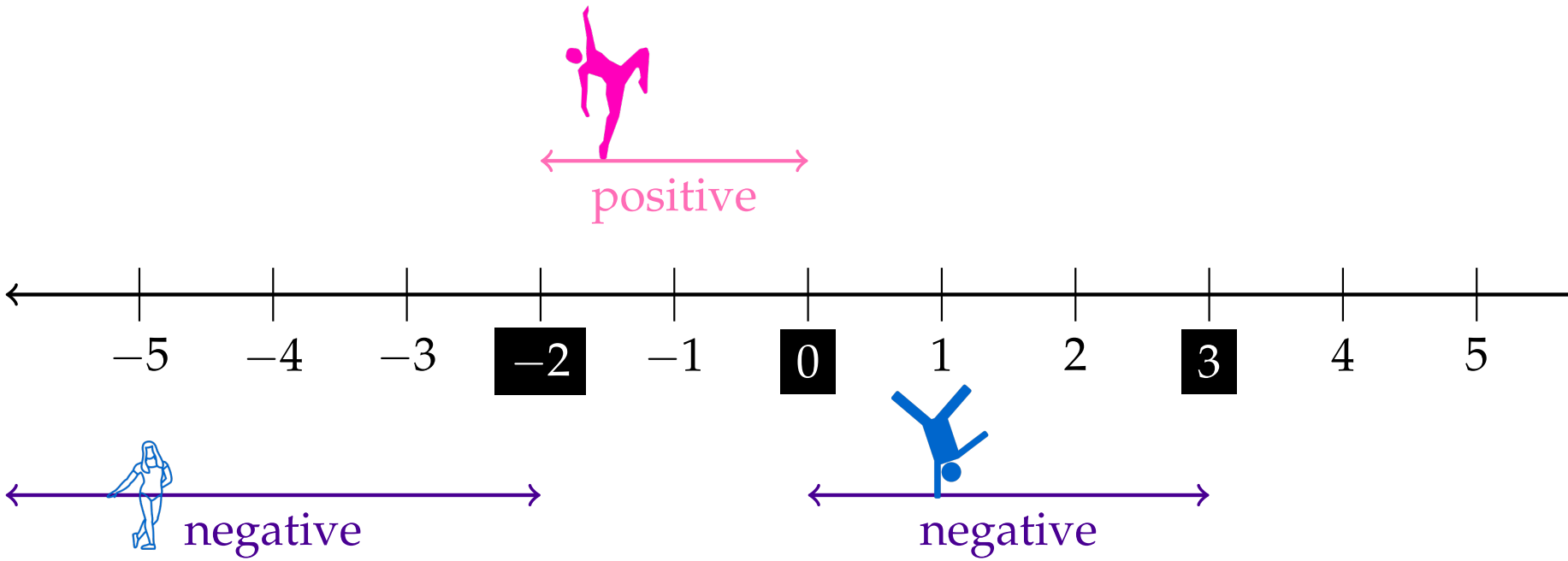


Signs of Factored Functions

SKIP SIGN CHANGES

$$f(x) = (x-3)(x-1)^2x(x+2)^3(x+5)^4$$

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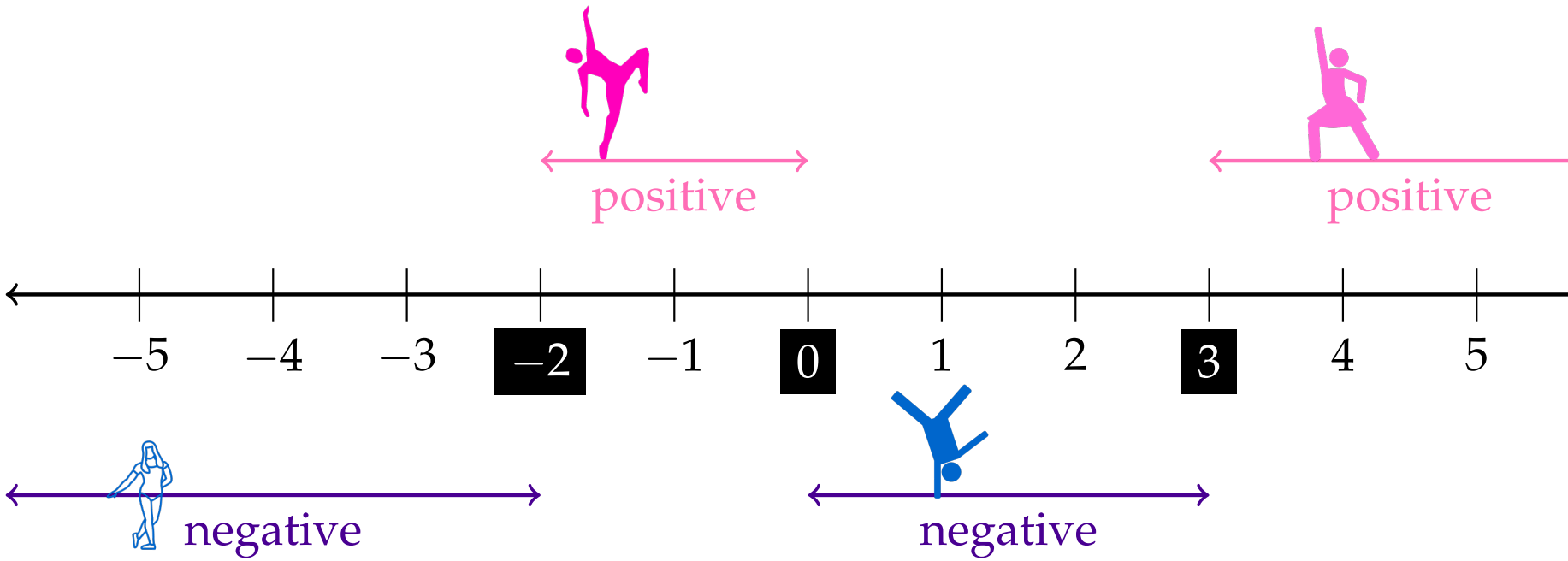


Signs of Factored Functions

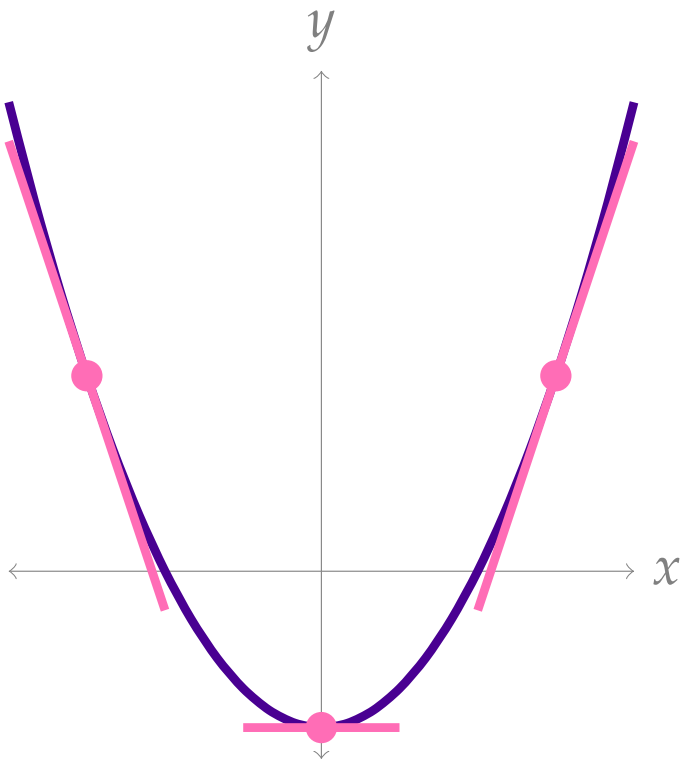
SKIP SIGN CHANGES

$$f(x) = (x-3)(x-1)^2x(x+2)^3(x+5)^4$$

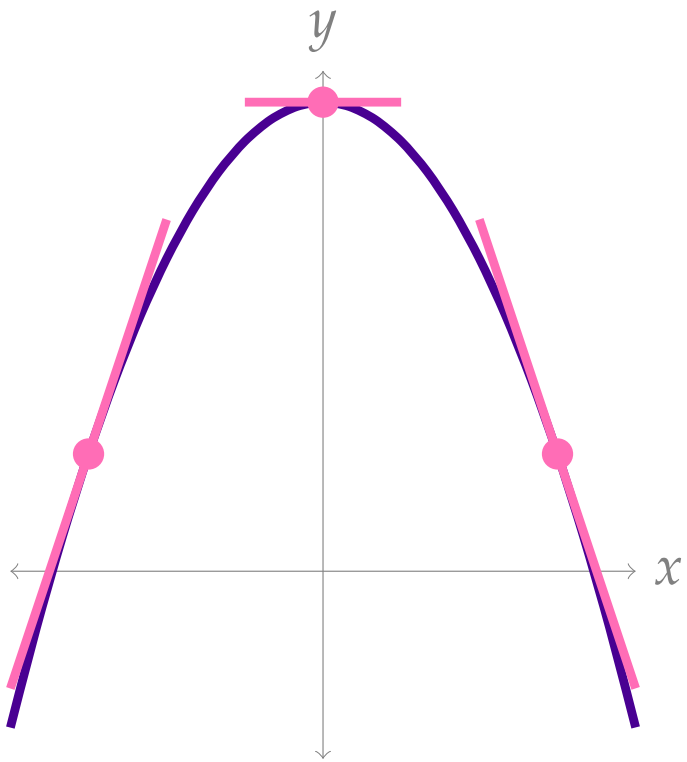
Where is $f(x)$ positive? Where is it negative?



Concavity



- Slopes are increasing
- $f''(x) > 0$
- "concave up"
- tangent line below curve

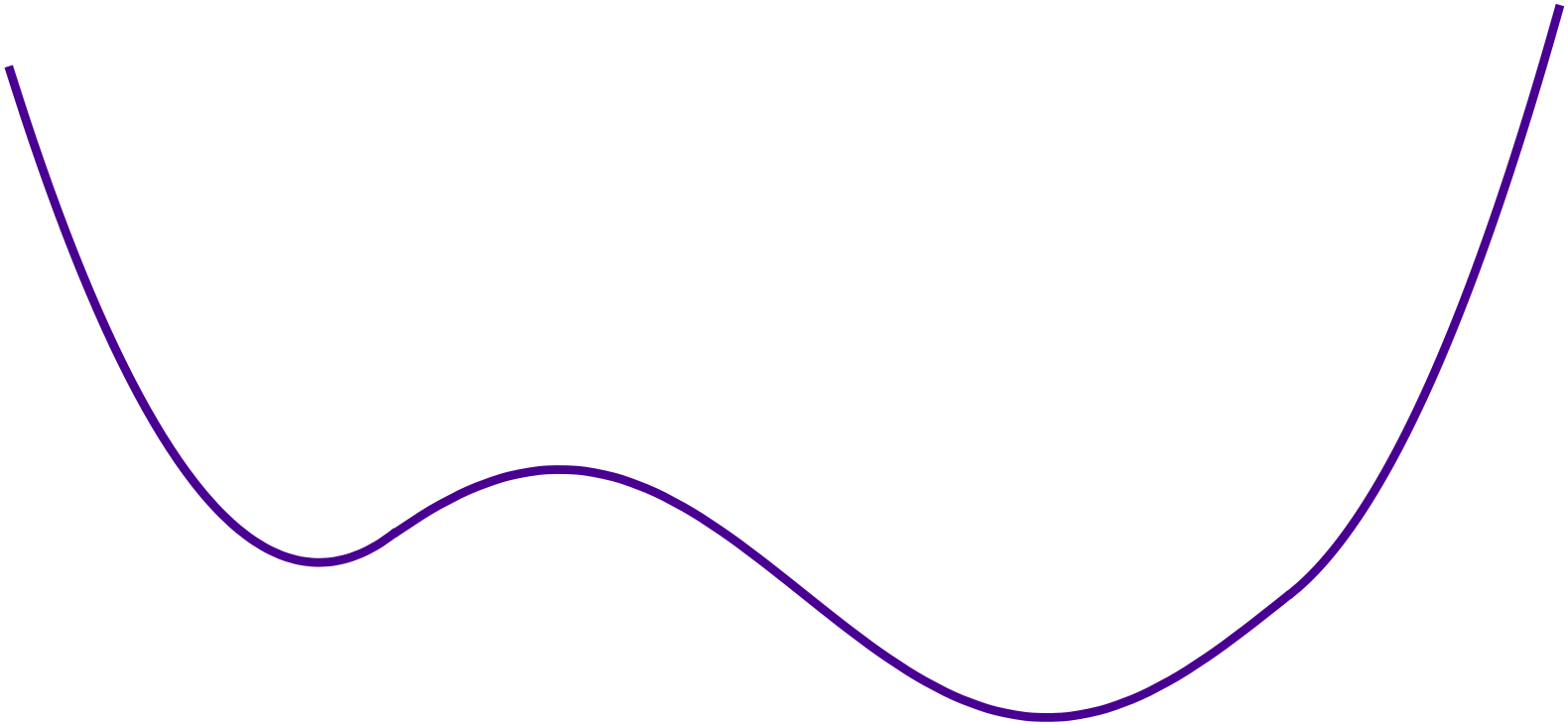


- Slopes are decreasing
- $f''(x) < 0$
- "concave down"
- tangent line above curve

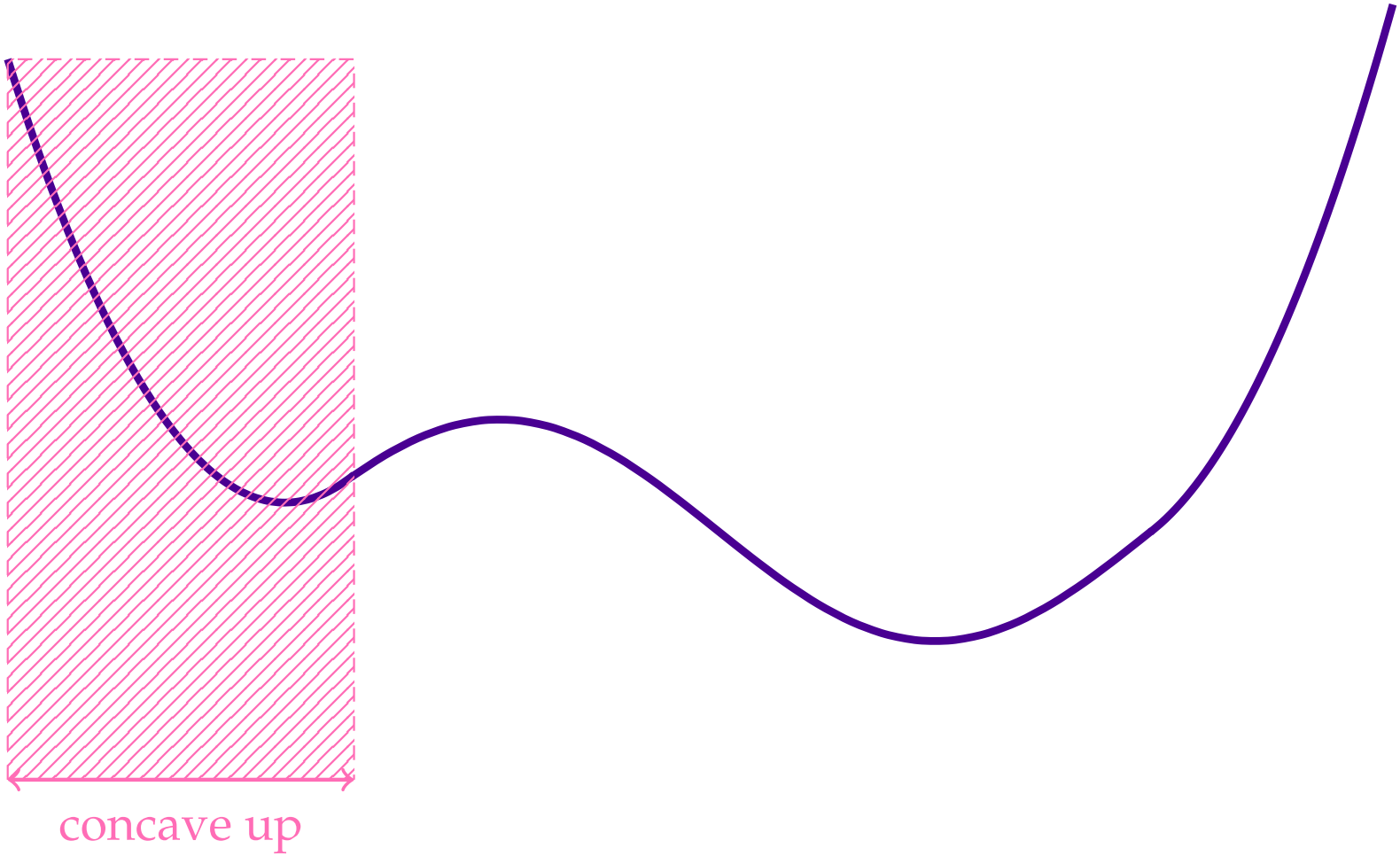
Mnemonic



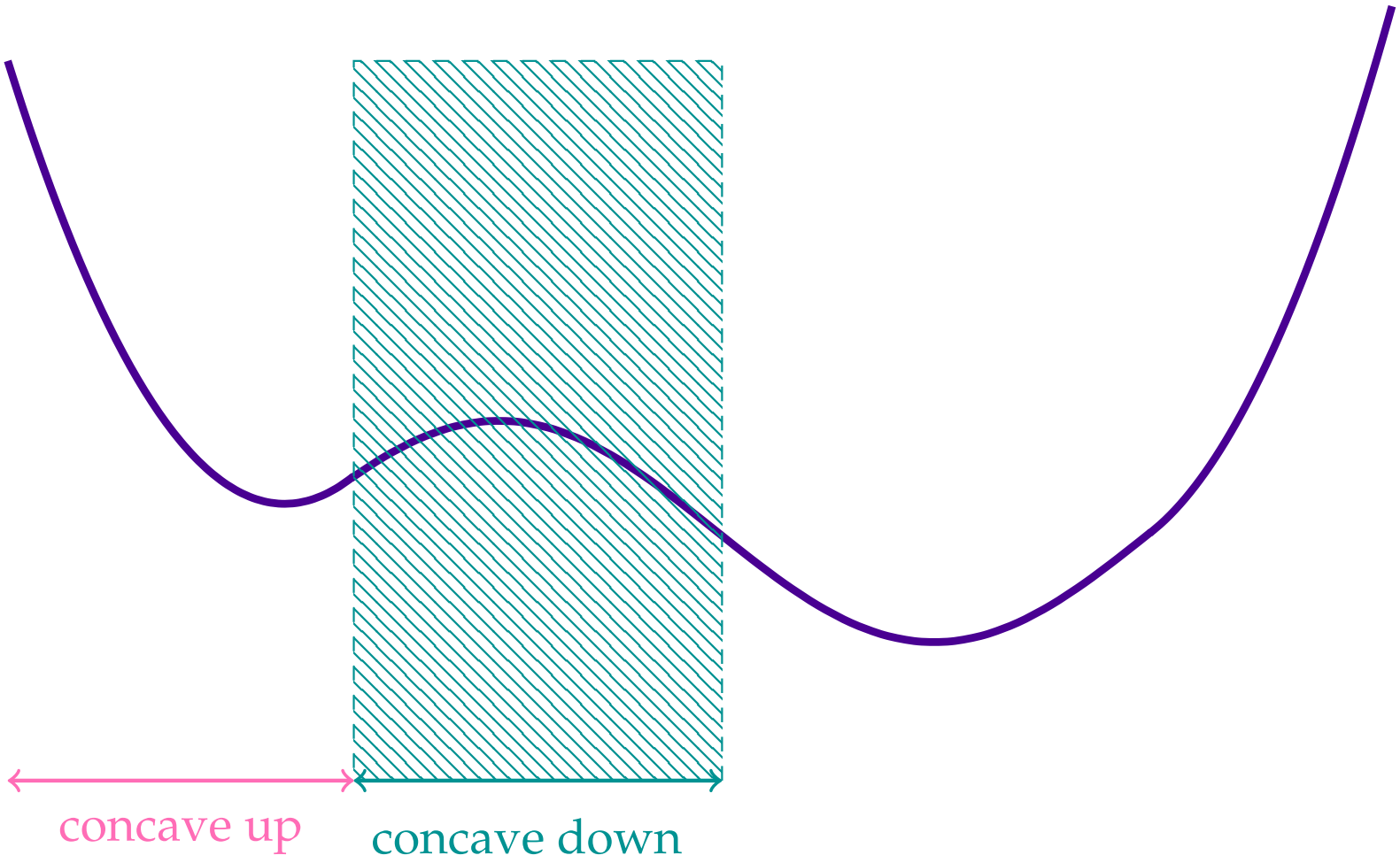
Concavity



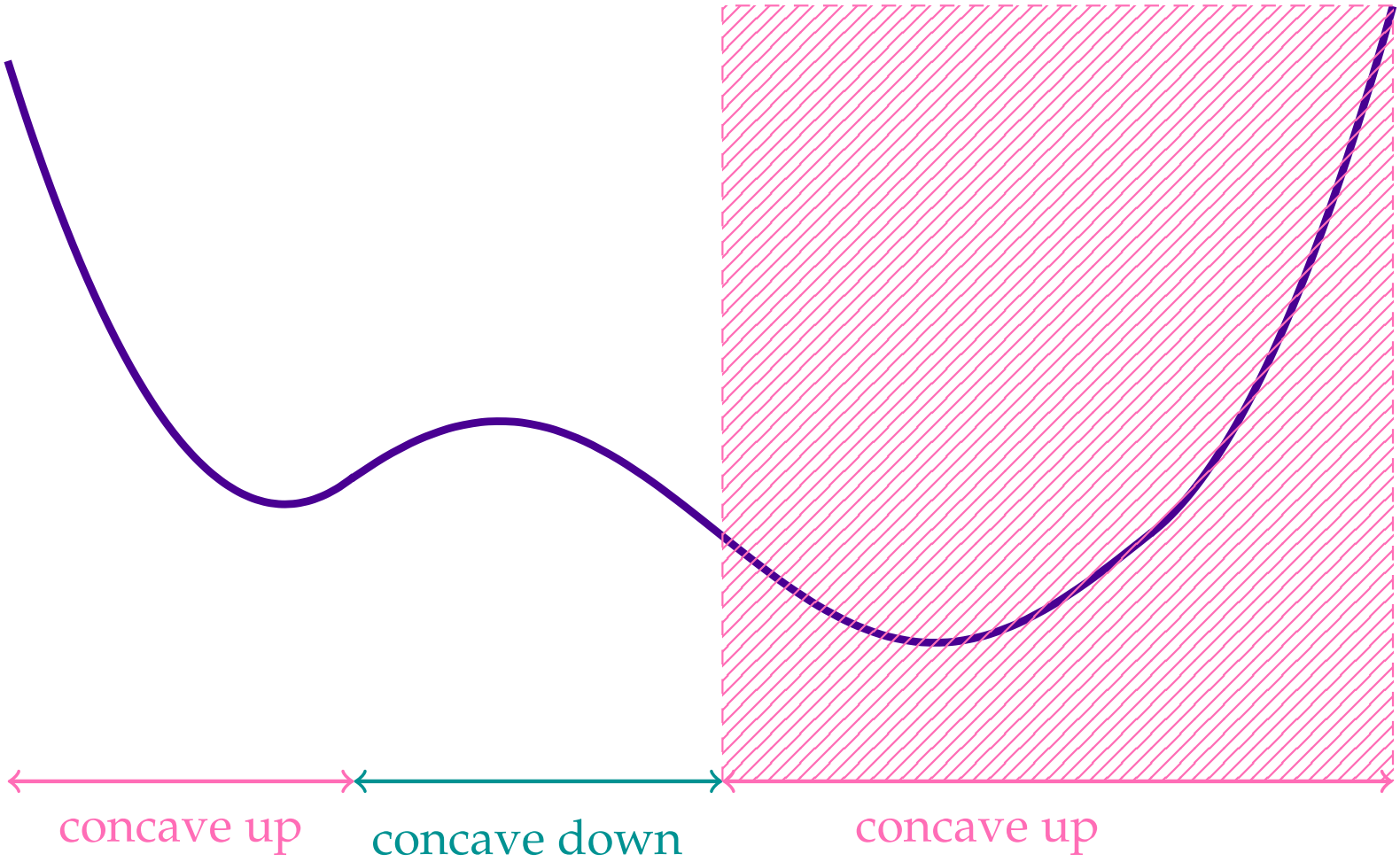
Concavity



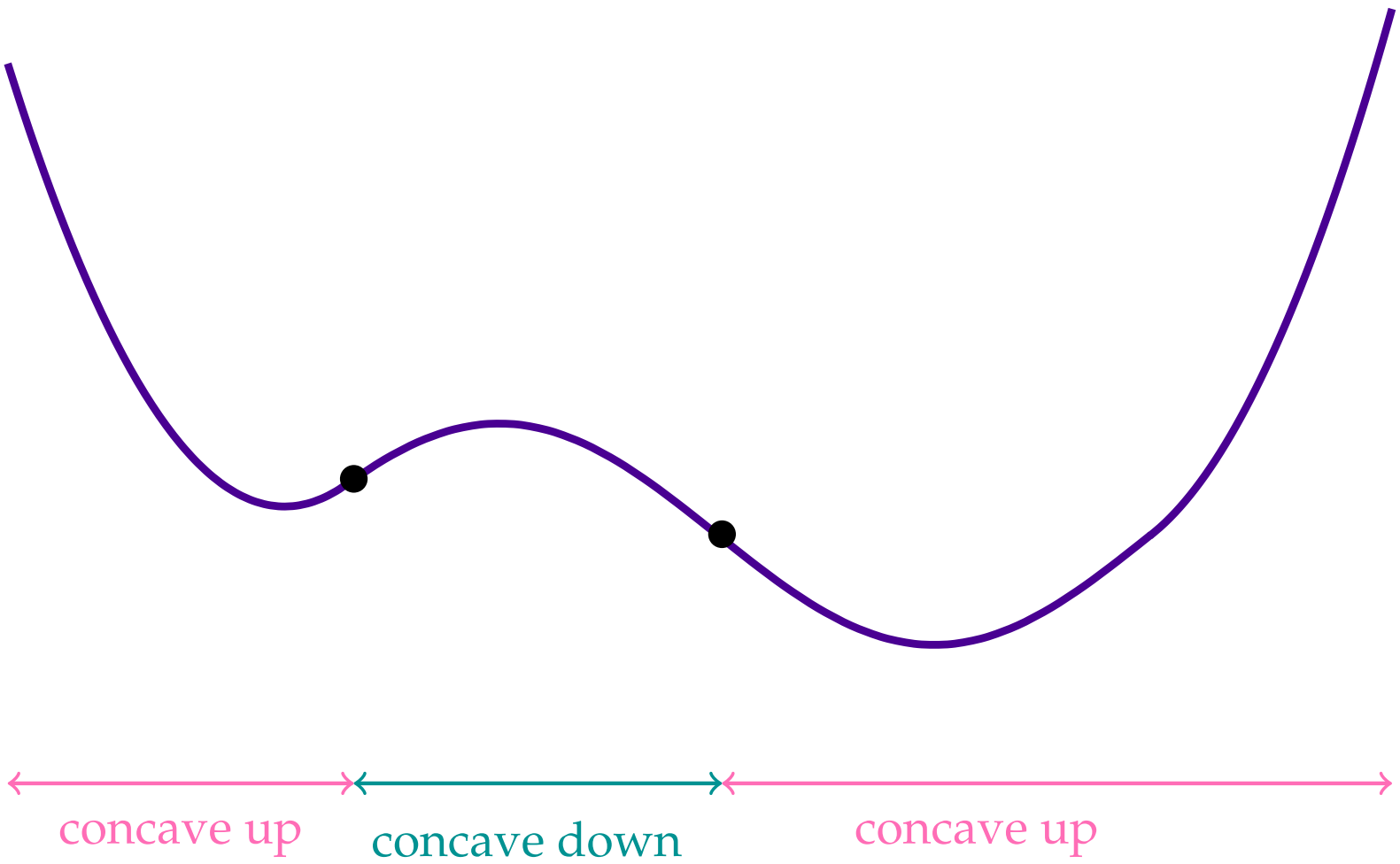
Concavity



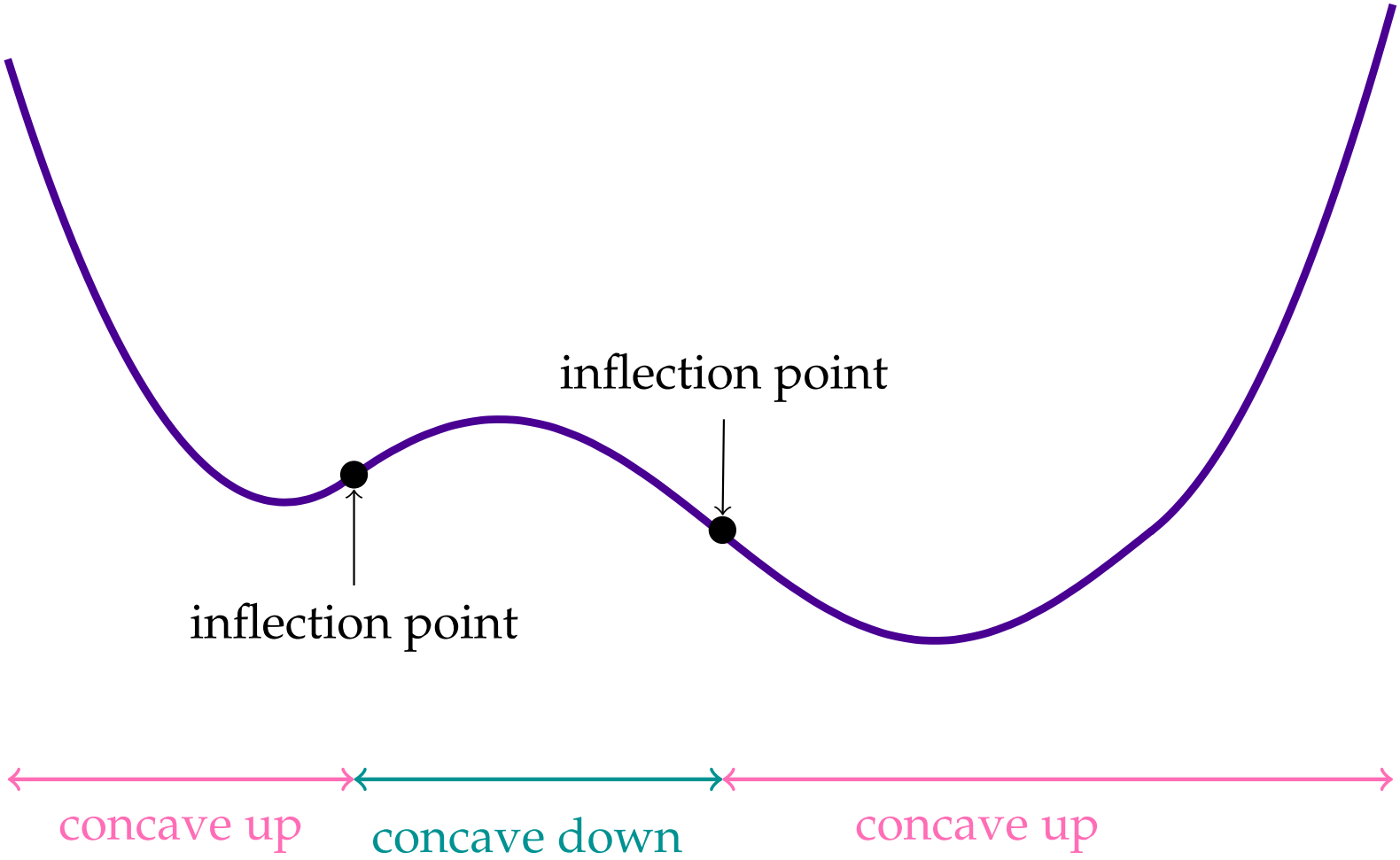
Concavity



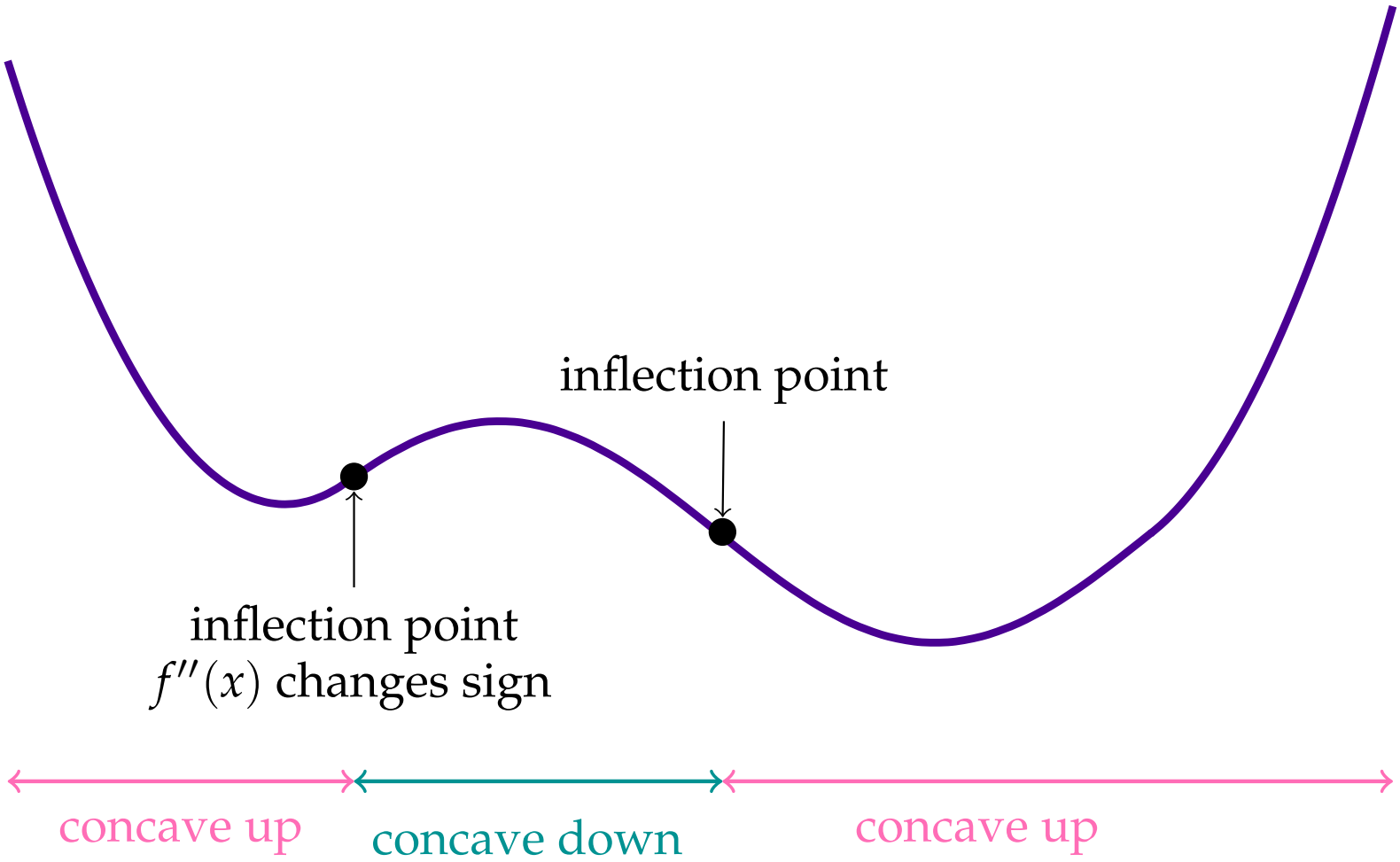
Concavity



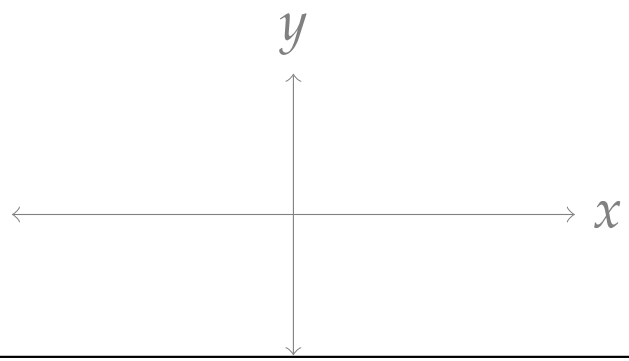
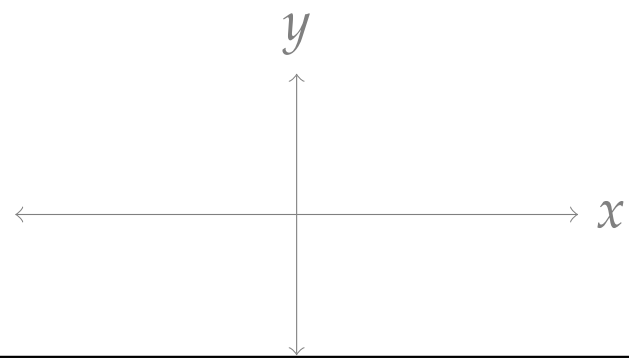
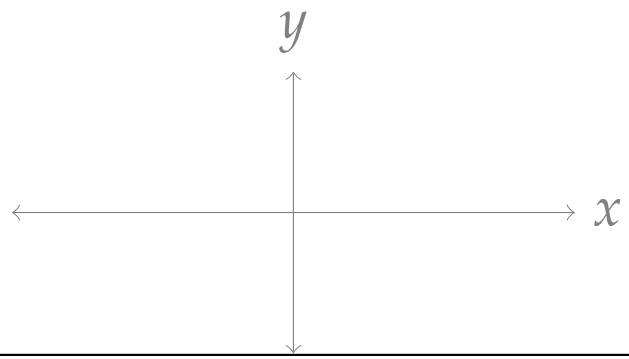
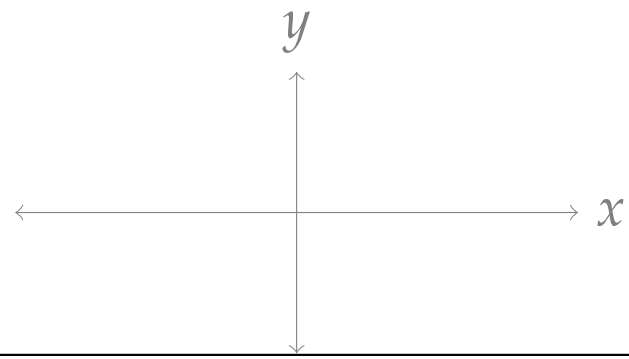
Concavity


inflection point
inflection point
concave up
concave down
concave up

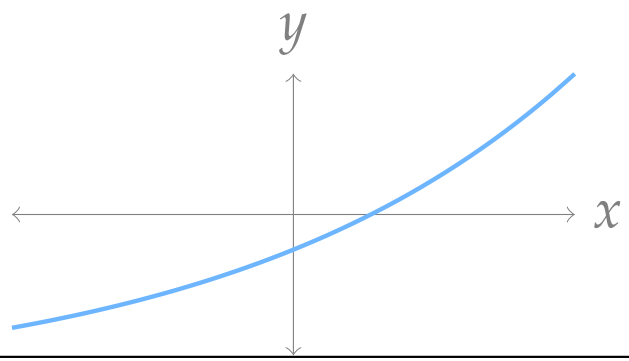
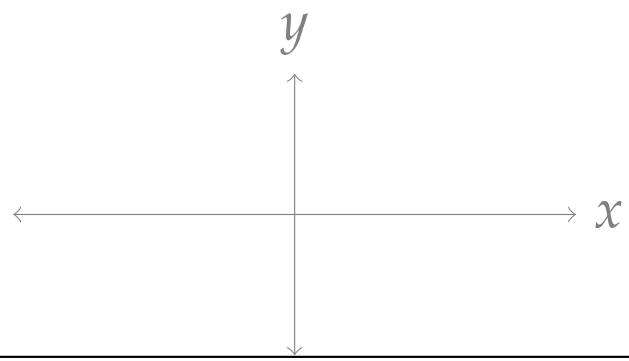
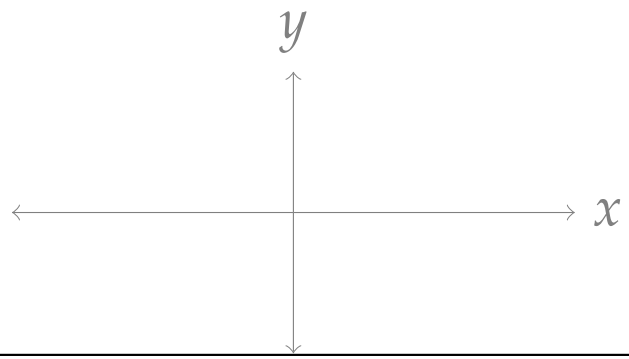
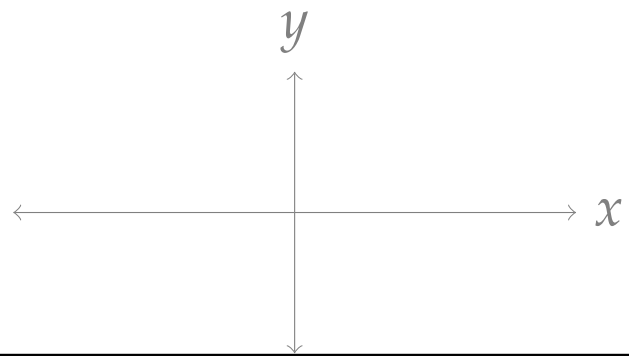
Concavity


inflection point
inflection point
$f''(x)$ changes sign
concave up
concave down
concave up

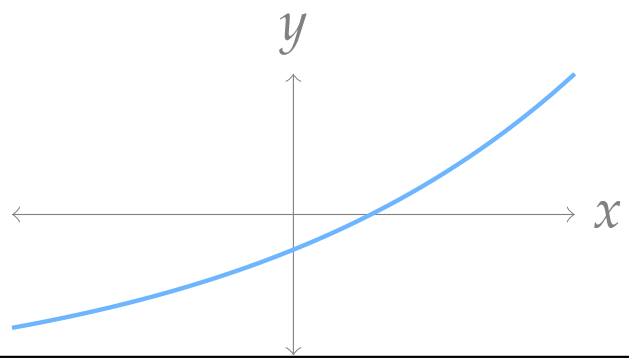
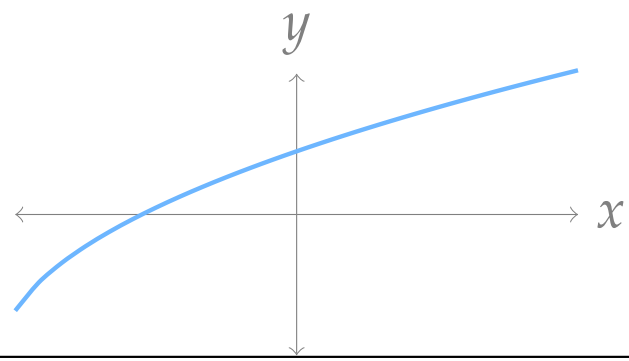
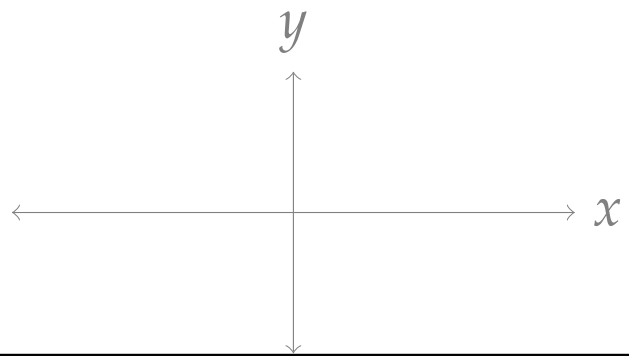
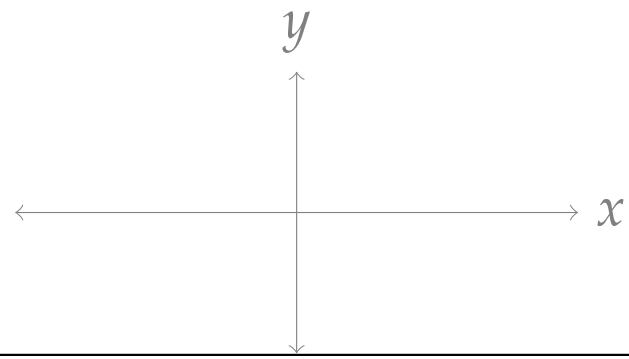
Sketch graphs with the following properties, or explain that none exist.

	concave up	concave down
increasing		
decreasing		

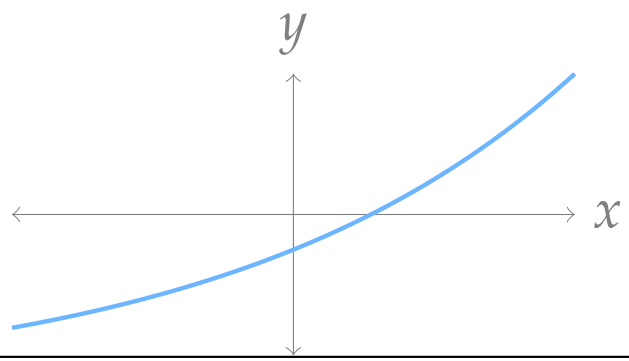
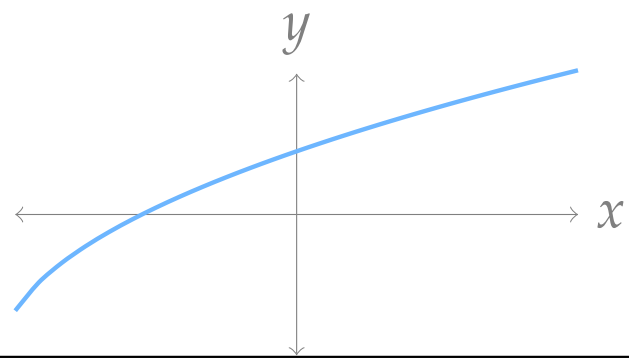
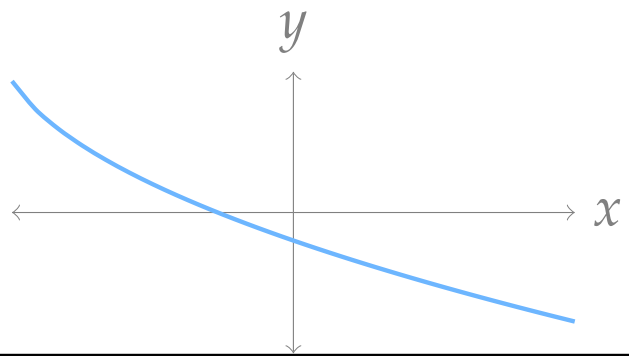
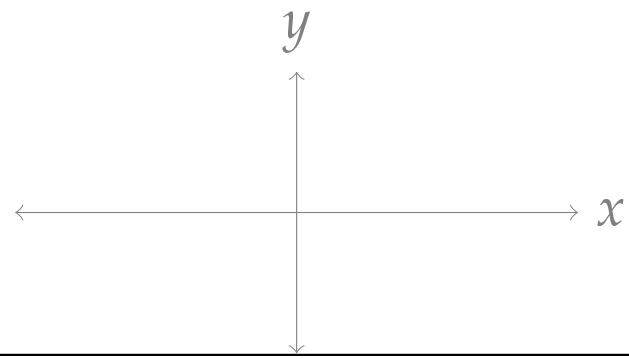
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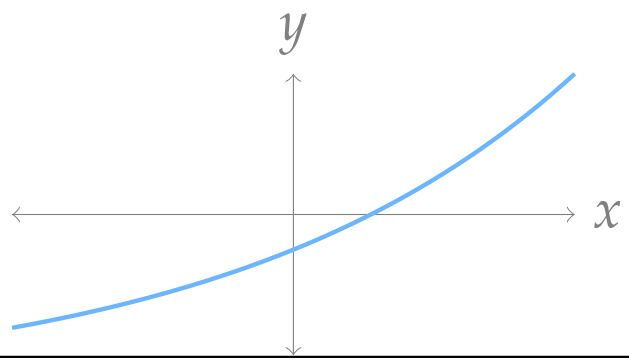
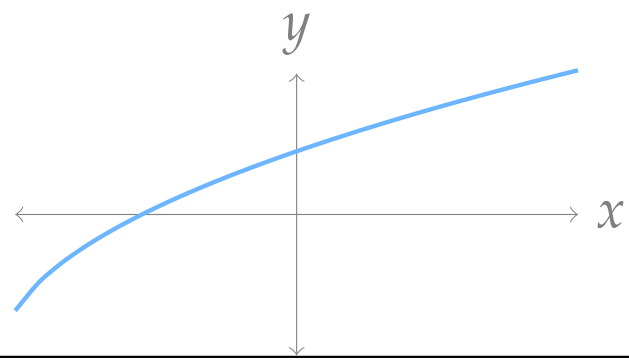
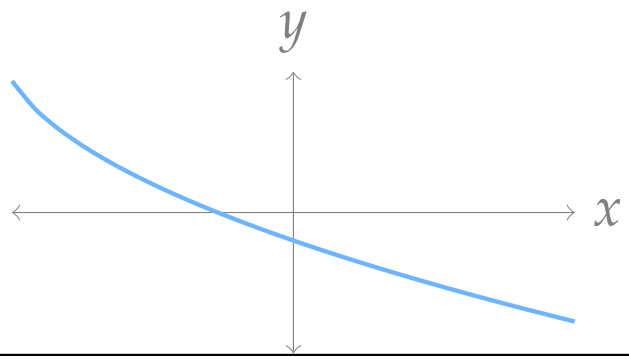
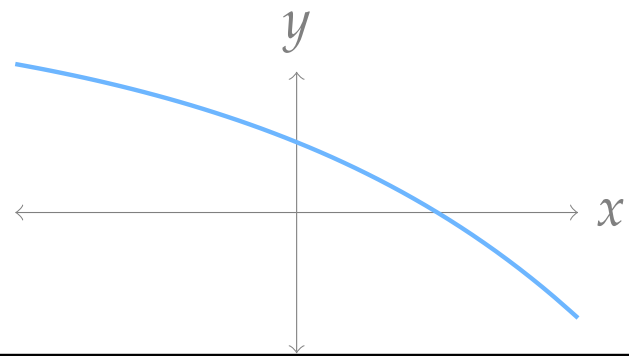
Sketch graphs with the following properties, or explain that none exist.

	concave up	concave down
increasing		
decreasing		

Sketch graphs with the following properties, or explain that none exist.

	concave up	concave down
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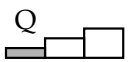
Sketch graphs with the following properties, or explain that none exist.

	concave up	concave down
increasing		
decreasing		

Poll Questions

Describe the concavity of the function $f(x) = e^x$.

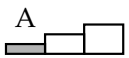
A. concave up
B. concave down
C. concave up for $x < 0$; concave down for $x > 0$
D. concave down for $x < 0$; concave up for $x > 0$
E. I'm not sure



Poll Questions

Describe the concavity of the function $f(x) = e^x$.

A. concave up
B. concave down
C. concave up for $x < 0$; concave down for $x > 0$
D. concave down for $x < 0$; concave up for $x > 0$
E. I'm not sure



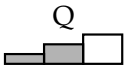
Poll Questions

Describe the concavity of the function $f(x) = e^x$.

A. concave up
B. concave down
C. concave up for $x < 0$; concave down for $x > 0$
D. concave down for $x < 0$; concave up for $x > 0$
E. I'm not sure

Is it possible to be concave up and decreasing?

A. Yes B. No C. I'm not sure



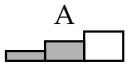
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Is it possible to be concave up and decreasing?

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Poll Questions

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A. concave up
B. concave down
C. concave up for $x < 0$; concave down for $x > 0$
D. concave down for $x < 0$; concave up for $x > 0$
E. I'm not sure

Is it possible to be concave up and decreasing?

A. Yes B. No C. I'm not sure

Suppose a function $f(x)$ is defined for all real numbers, and is concave up on the interval $[0, 1]$. Which of the following must be true?

A. $f'(0) < f'(1)$
B. $f'(0) > f'(1)$
C. $f'(0)$ is positive
D. $f'(0)$ is negative
E. I'm not sure



Poll Questions

Describe the concavity of the function $f(x) = e^x$.

A. concave up
B. concave down
C. concave up for $x < 0$; concave down for $x > 0$
D. concave down for $x < 0$; concave up for $x > 0$
E. I'm not sure

Is it possible to be concave up and decreasing?

A. Yes B. No C. I'm not sure

Suppose a function $f(x)$ is defined for all real numbers, and is concave up on the interval $[0, 1]$. Which of the following must be true?

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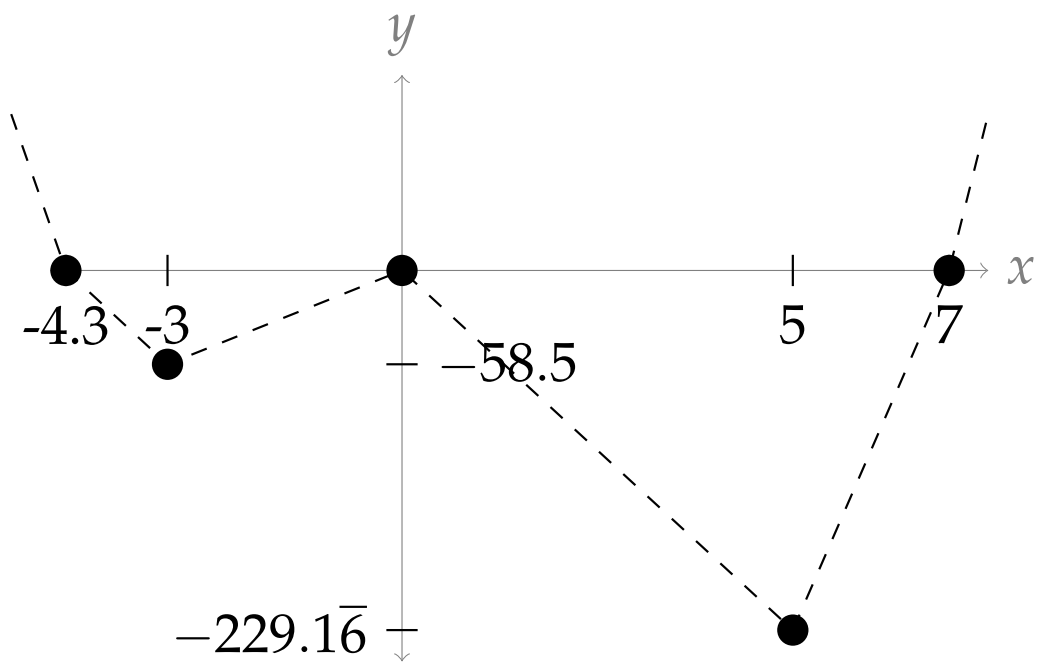
A



Revisiting a previous example

original example

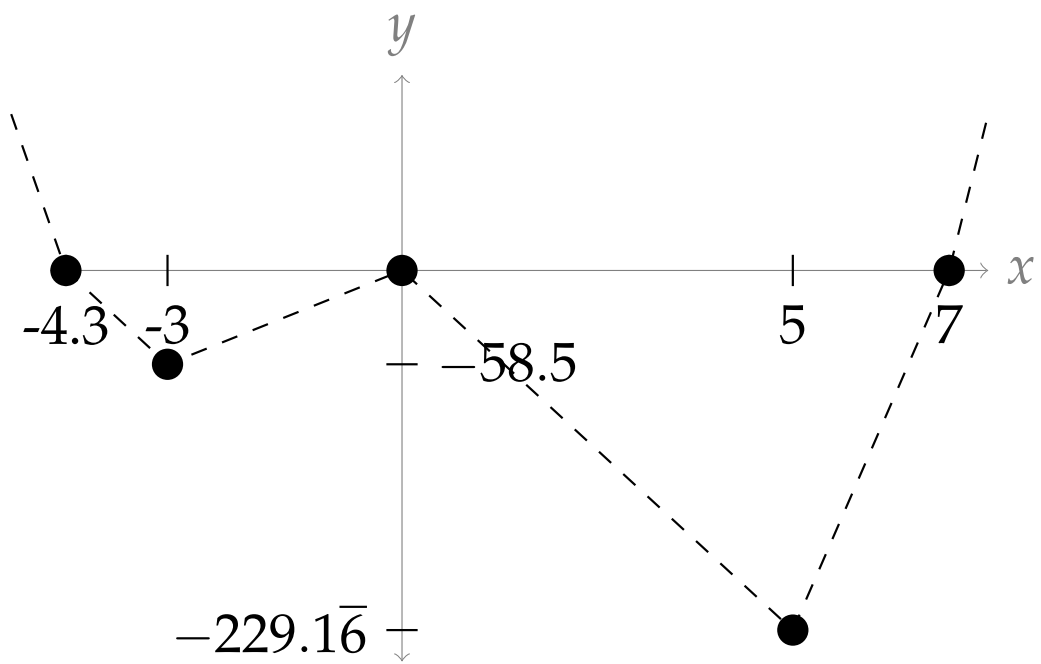
$$f(x) = \frac{1}{2}x^4 - \frac{4}{3}x^3 - 15x^2$$



Revisiting a previous example

original example

$$f(x) = \frac{1}{2}x^4 - \frac{4}{3}x^3 - 15x^2$$

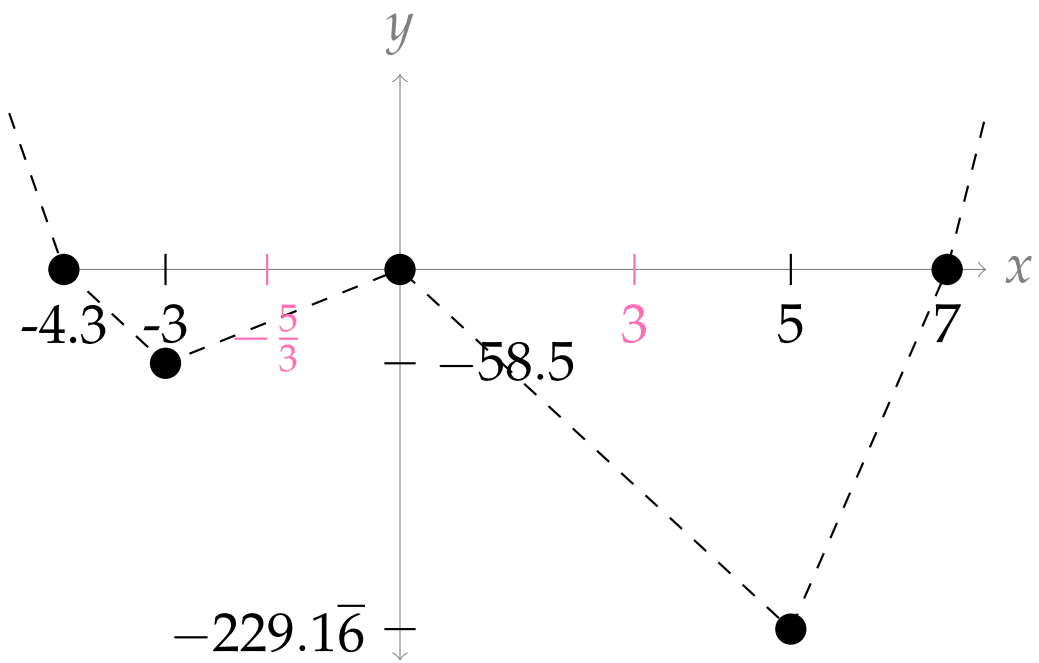


$f''(x) = 6x^2 - 8x - 30 = 2(x-3)(3x+5)$

Revisiting a previous example

original example

$$f(x) = \frac{1}{2}x^4 - \frac{4}{3}x^3 - 15x^2$$

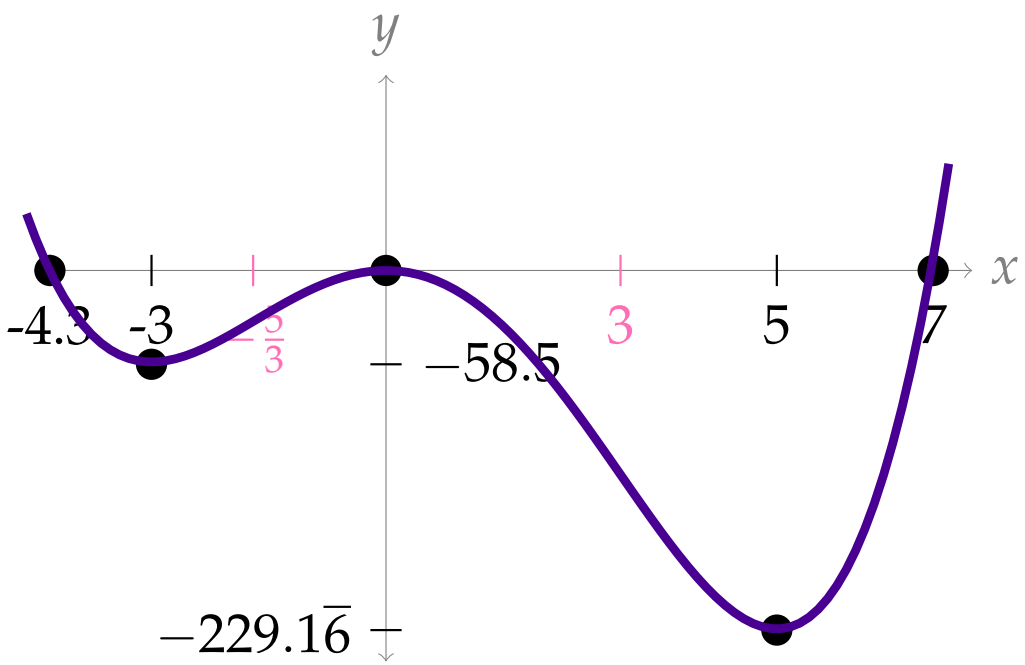


$$f''(x) = 6x^2 - 8x - 30 = 2(x-3)(3x+5)$$

Revisiting a previous example

original example

$$f(x) = \frac{1}{2}x^4 - \frac{4}{3}x^3 - 15x^2$$



$$f''(x) = 6x^2 - 8x - 30 = 2(x-3)(3x+5)$$

Sketch:

$$f(x) = x^5 - 15x^3$$

A

Sketch:

$$f(x) = x^5 - 15x^3$$

Domain	Defined and differentiable for all real numbers.
Intercepts	$f(x) = x^3(x^2 - 15)$: Roots are at $x = 0$ and $x = \pm\sqrt{15} \approx \pm 4$
End behaviour	$\lim_{x \to -\infty} f(x) = -\infty$ and $\lim_{x \to \infty} f(x) = \infty$
Critical points	$f'(x) = 5x^4 - 45x^2 = 5x^2(x^2 - 9)$. So the critical points are $x = 0$, $x = \pm 3$.
Increasing, decreasing	Increasing on $(-\infty, -3)$, decreasing on $(-3, 0)$ and $(0, 3)$, increasing on $(3, \infty)$
Local extrema	From intervals of increase and decrease: local max at $x = -3$ and local min at $x = 3$



Sketch:

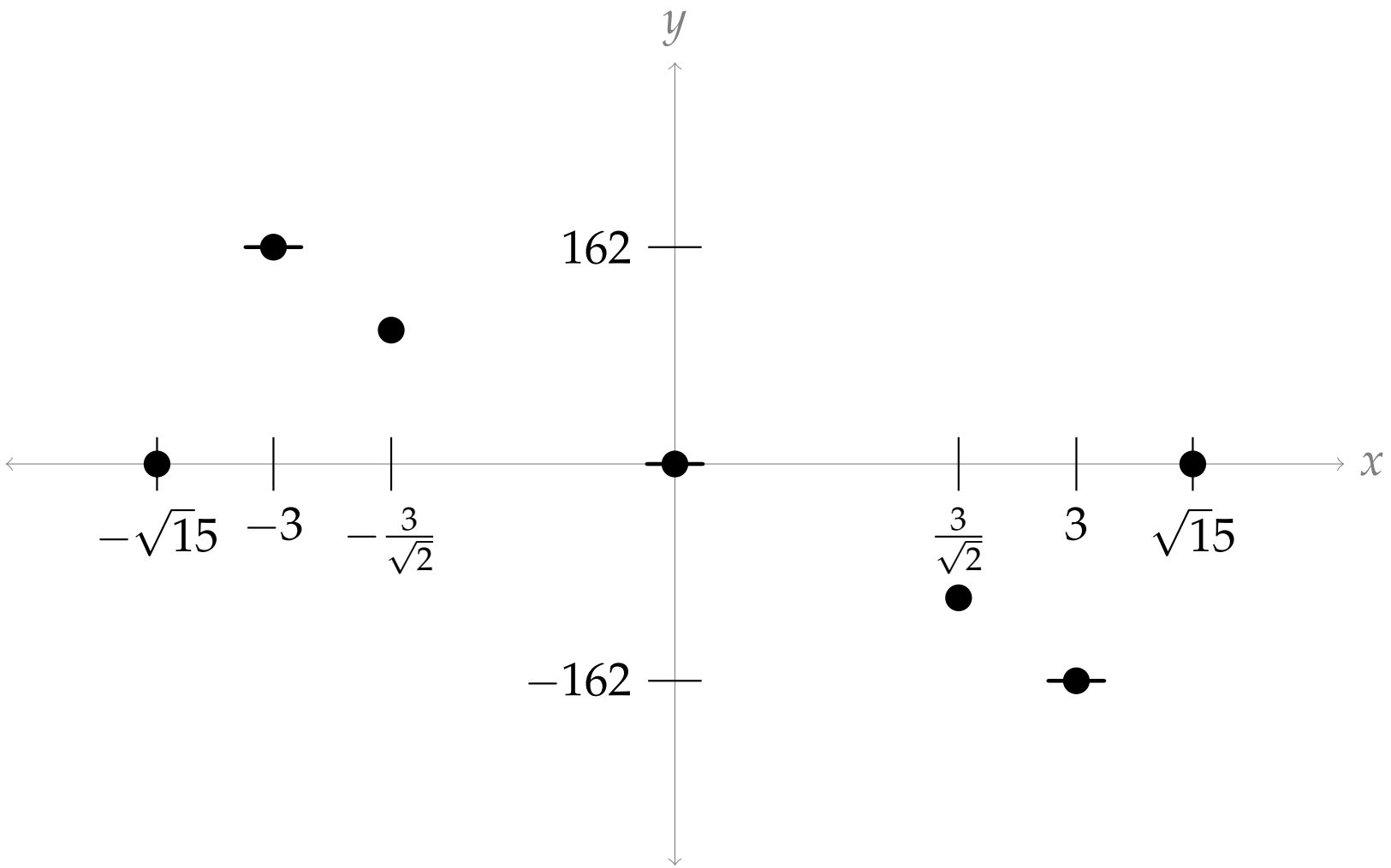
$$f(x) = x^5 - 15x^3$$

Concavity $f''(x) = 20x^3 - 90x = 10x(2x^2 - 9) = 0$ for $x = 0$ and $x = \pm\frac{3}{\sqrt{2}} \approx \pm 2$. All of these are inflection points; concave down $(-\infty, -\frac{3}{\sqrt{2}})$, concave up $(-\frac{3}{\sqrt{2}}, 0)$, concave down $(0, \frac{3}{\sqrt{2}})$, and concave up $(\frac{3}{\sqrt{2}}, \infty)$.

***y*-values of notable points** $f(3) = -162, f(-3) = 162, f(-3/\sqrt{2}) \approx 100, f(3/\sqrt{2}) \approx -100$

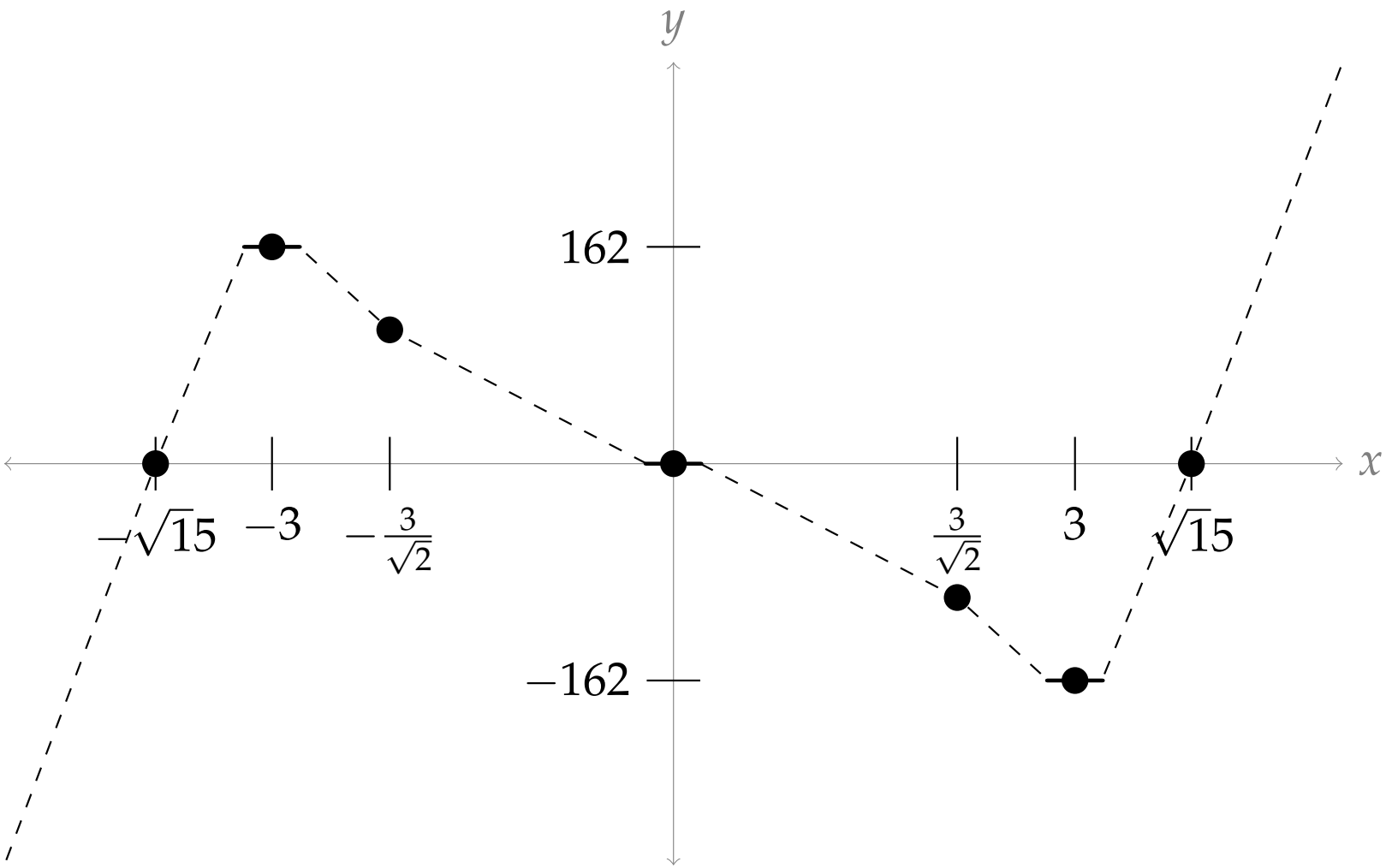
Sketch:

$$f(x) = x^5 - 15x^3$$



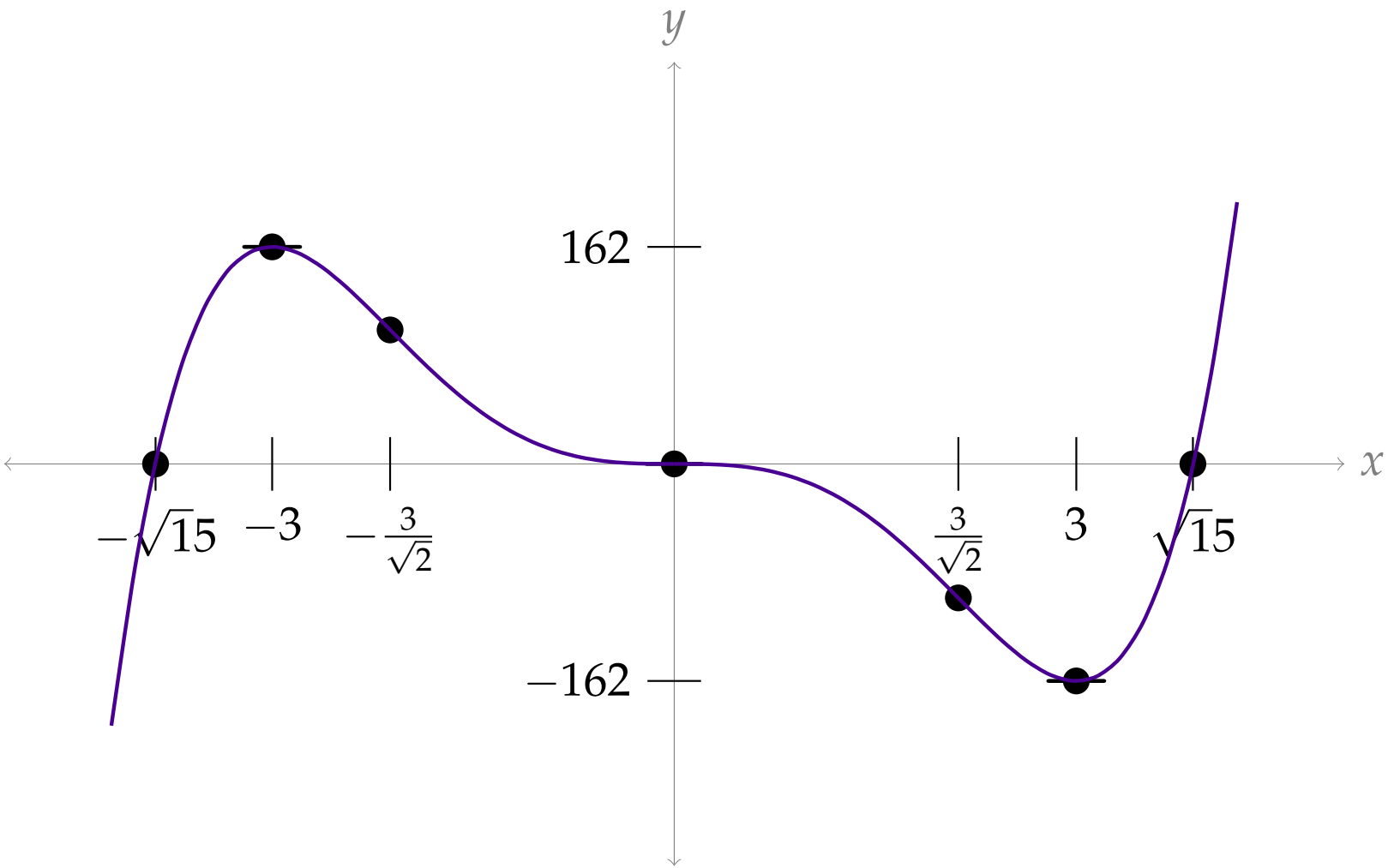
Sketch:

$$f(x) = x^5 - 15x^3$$



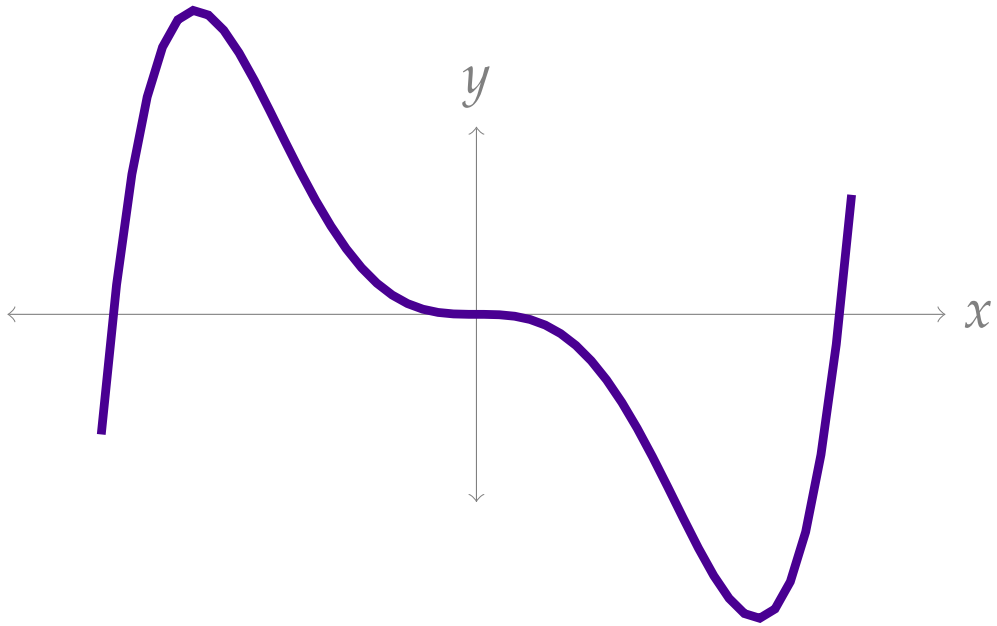
Sketch:

$$f(x) = x^5 - 15x^3$$



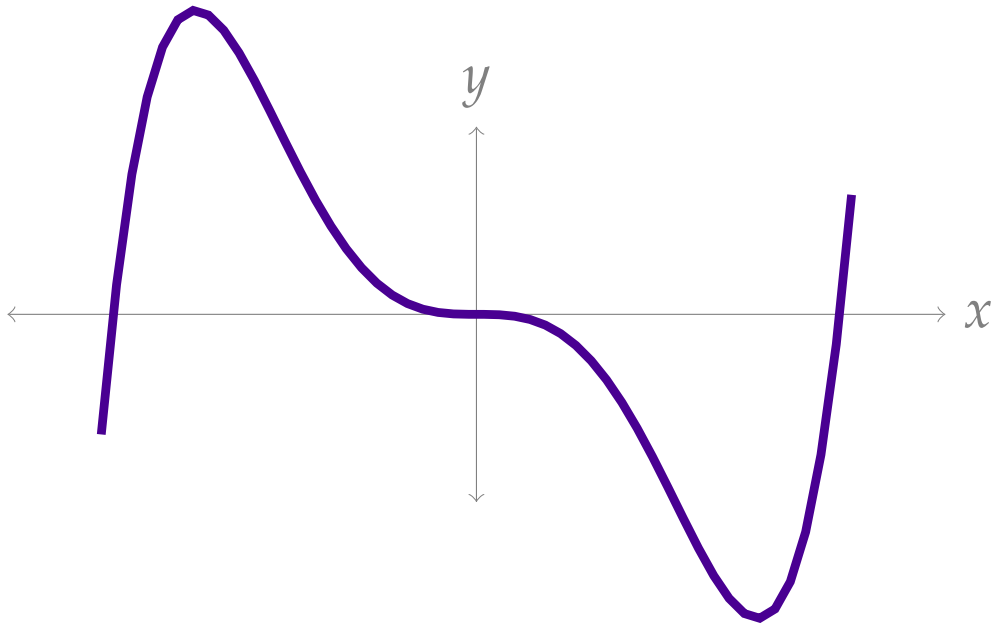
https://www.desmos.com/calculator/uoii6nmgr8

Even and Odd Functions



$$f(x) = x^5 - 15x^3$$

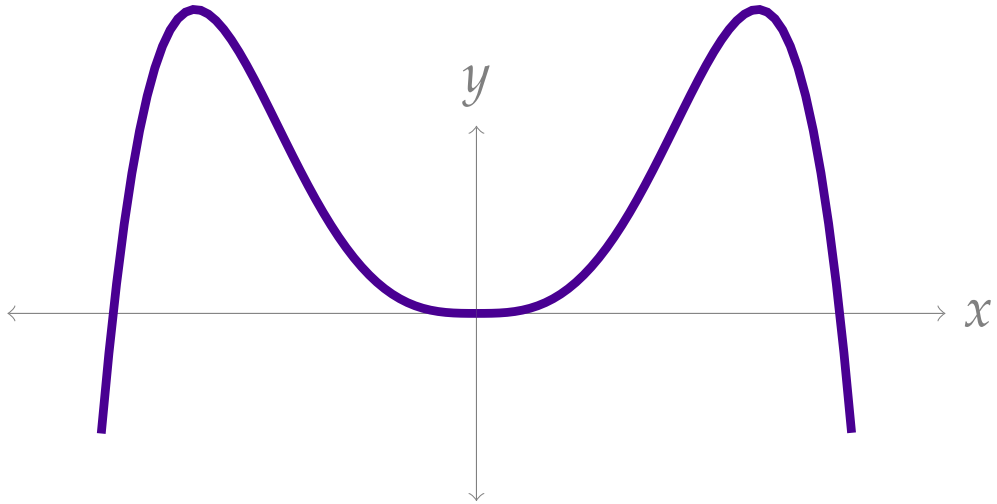
Even and Odd Functions



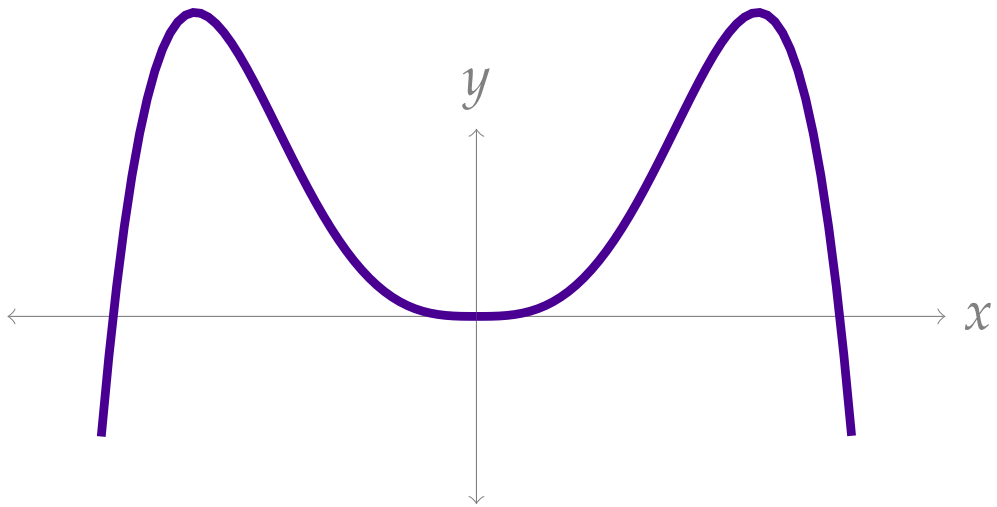
$f(x) = x^5 - 15x^3$

odd function

Even and Odd Functions



Even and Odd Functions

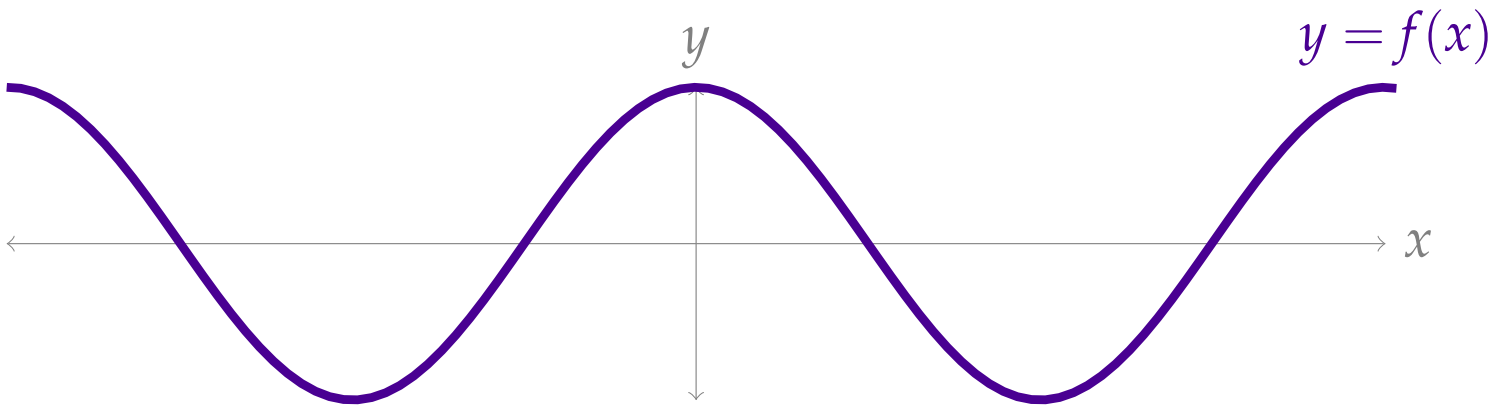


even function

Even Function – Definition 3.6.6

A function $f(x)$ is even if, for all x in its domain,

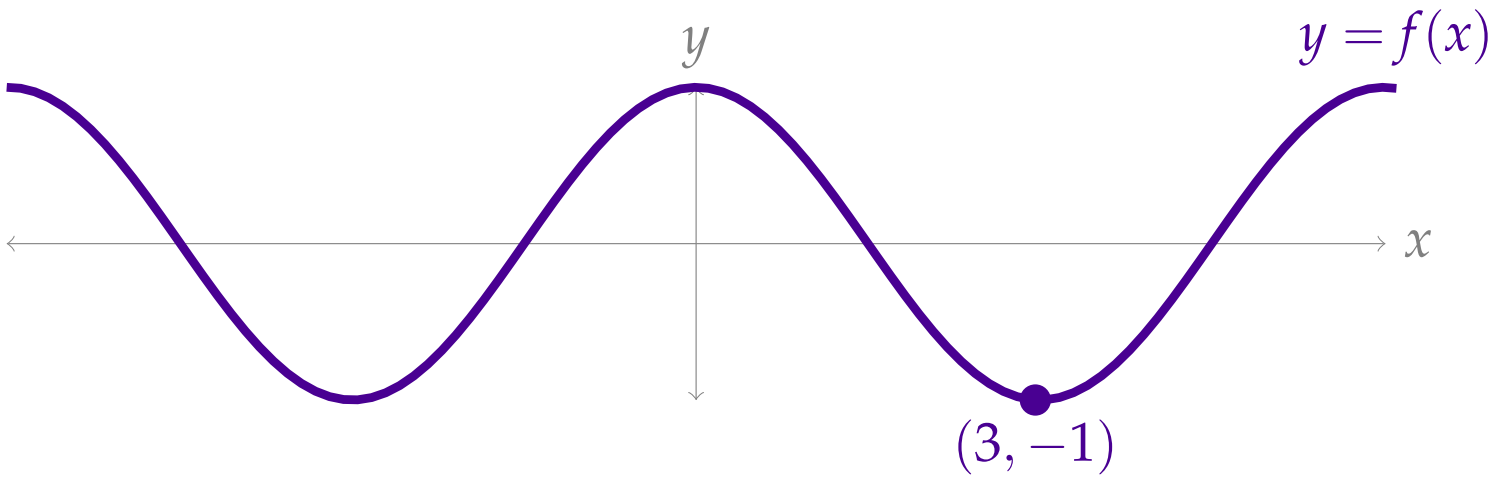
$$f(-x) = f(x)$$



Even Function – Definition 3.6.6

A function $f(x)$ is even if, for all x in its domain,

$$f(-x) = f(x)$$

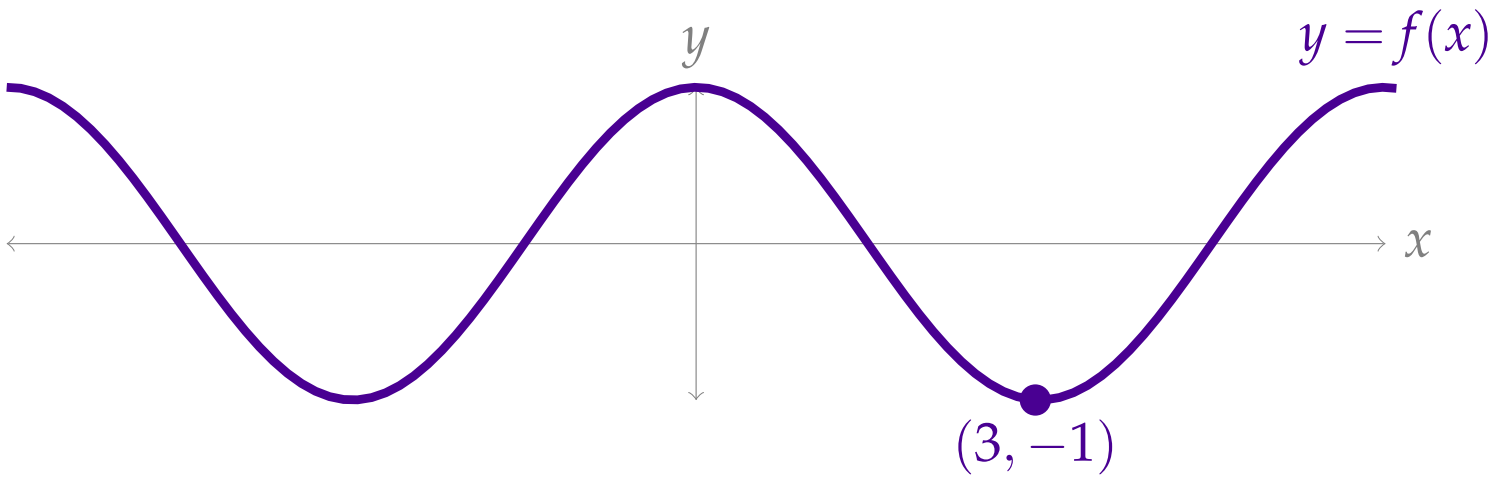


Suppose $f(3) = -1$.

Even Function – Definition 3.6.6

A function $f(x)$ is even if, for all x in its domain,

$$f(-x) = f(x)$$

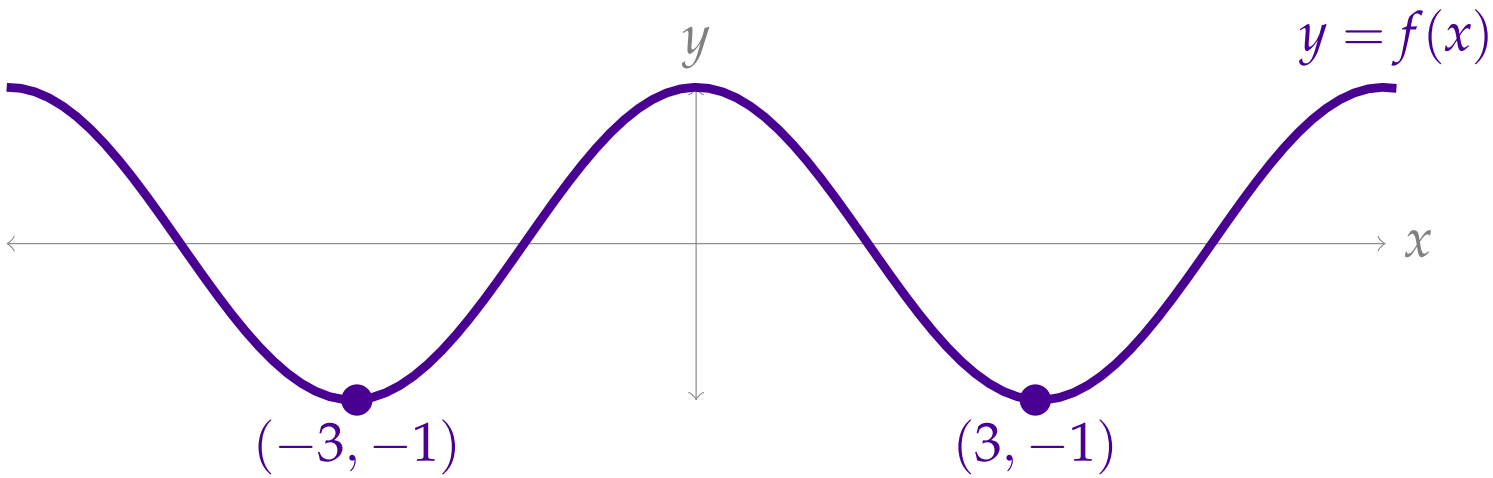


Suppose $f(3) = -1$. Then $f(-3) =$

Even Function – Definition 3.6.6

A function $f(x)$ is even if, for all x in its domain,

$$f(-x) = f(x)$$

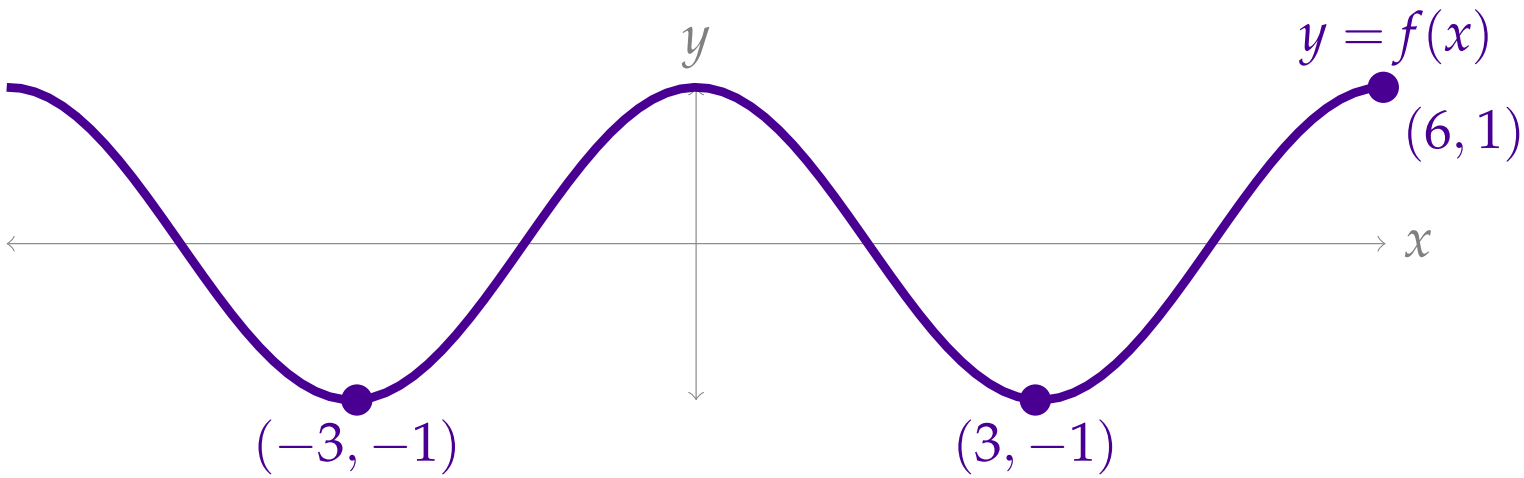


Suppose $f(3) = -1$. Then $f(-3) = -1$ also.

Even Function – Definition 3.6.6

A function $f(x)$ is even if, for all x in its domain,

$$f(-x) = f(x)$$

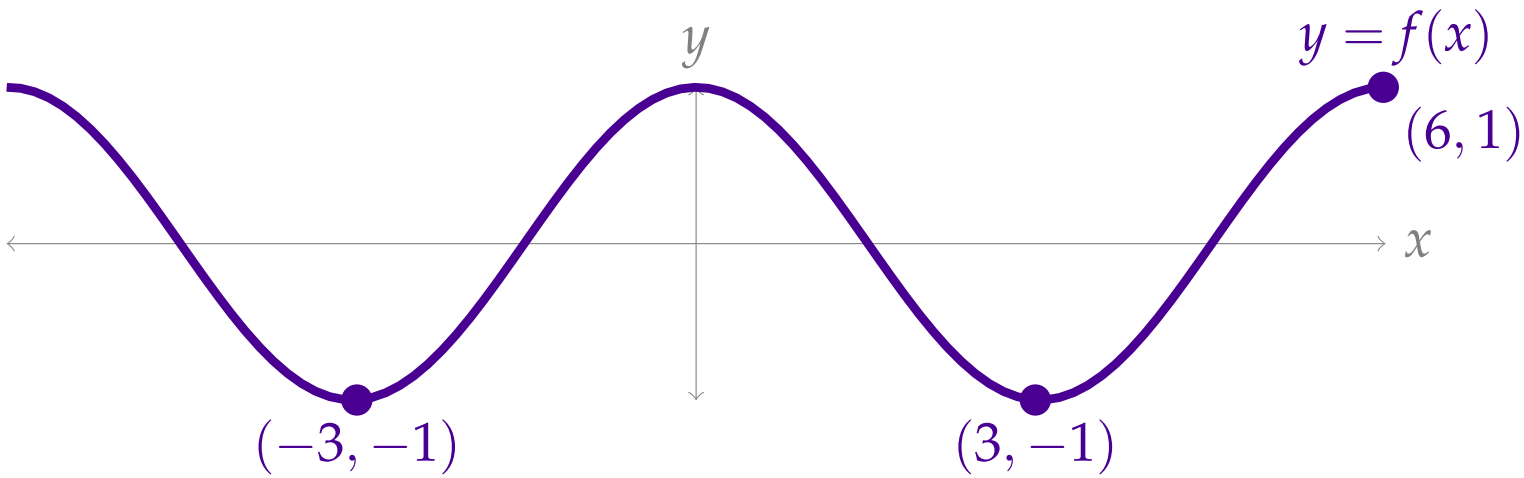


Suppose $f(3) = -1$. Then $f(-3) = -1$ also.
Suppose $f(6) = 1$.

Even Function – Definition 3.6.6

A function $f(x)$ is even if, for all x in its domain,

$$f(-x) = f(x)$$

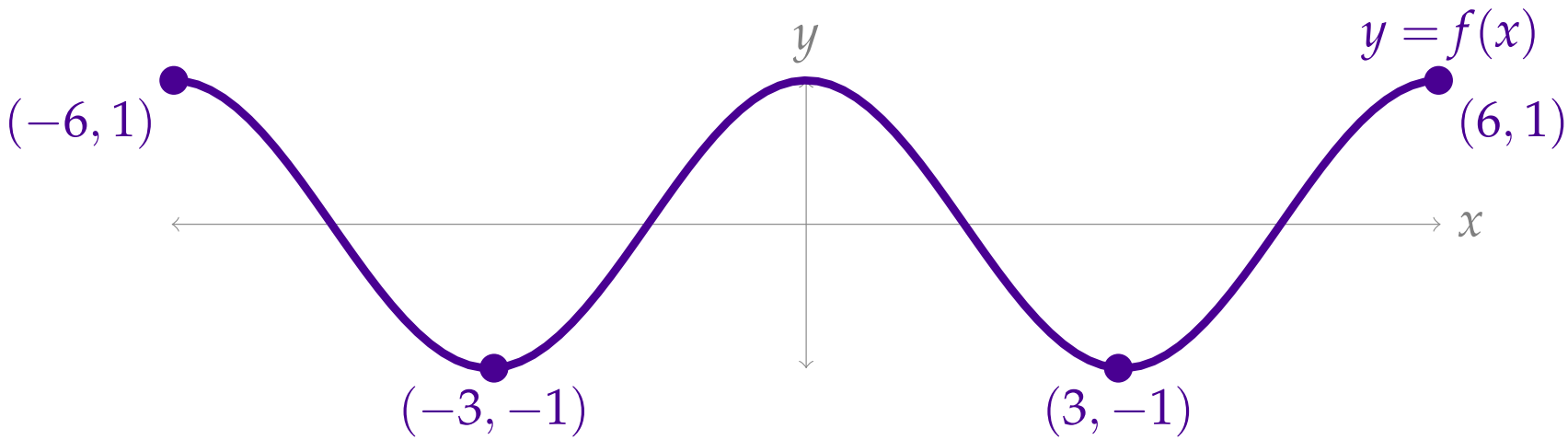


Suppose $f(3) = -1$. Then $f(-3) = -1$ also.
Suppose $f(6) = 1$. Then $f(-6) =$

Even Function – Definition 3.6.6

A function $f(x)$ is even if, for all x in its domain,

$$f(-x) = f(x)$$



Suppose $f(3) = -1$. Then $f(-3) = -1$ also.
Suppose $f(6) = 1$. Then $f(-6) = 1$ also.

Even Functions

Even Function – Definition 3.6.6

A function $f(x)$ is even if, for all x in its domain,

$$f(-x) = f(x)$$

Examples:

Even Functions

Even Function – Definition 3.6.6

A function $f(x)$ is even if, for all x in its domain,

$$f(-x) = f(x)$$

Examples:
$f(x) = x^2$

Even Functions

Even Function – Definition 3.6.6

A function $f(x)$ is even if, for all x in its domain,

$$f(-x) = f(x)$$

Examples:
$f(x) = x^2$
$f(x) = x^4$

Even Functions

Even Function – Definition 3.6.6

A function $f(x)$ is even if, for all x in its domain,

$$f(-x) = f(x)$$

Examples:
$f(x) = x^2$
$f(x) = x^4$
$f(x) = \cos(x)$

Even Functions

Even Function – Definition 3.6.6

A function $f(x)$ is even if, for all x in its domain,

$$f(-x) = f(x)$$

Examples:

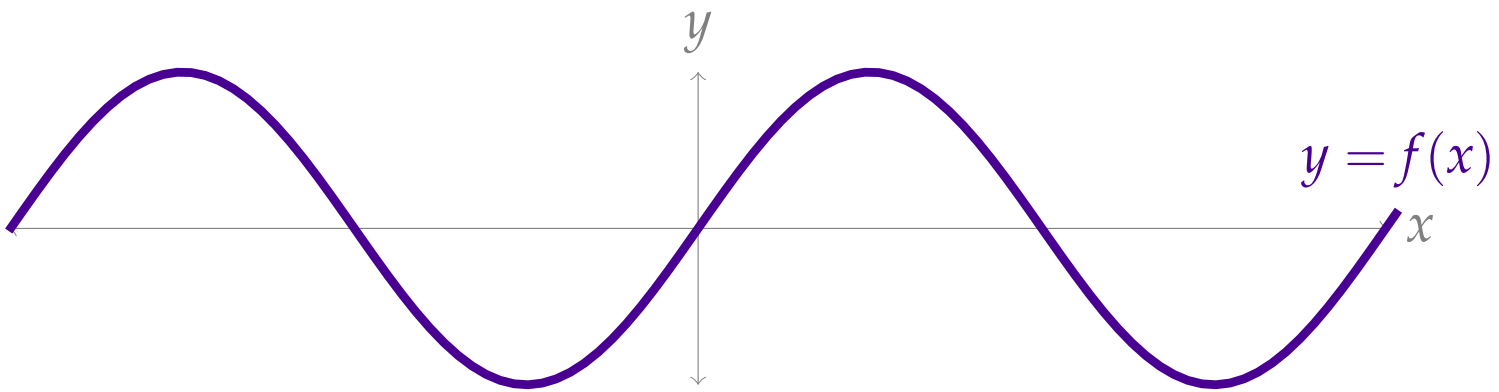
$f(x) = x^2$

$f(x) = x^4$

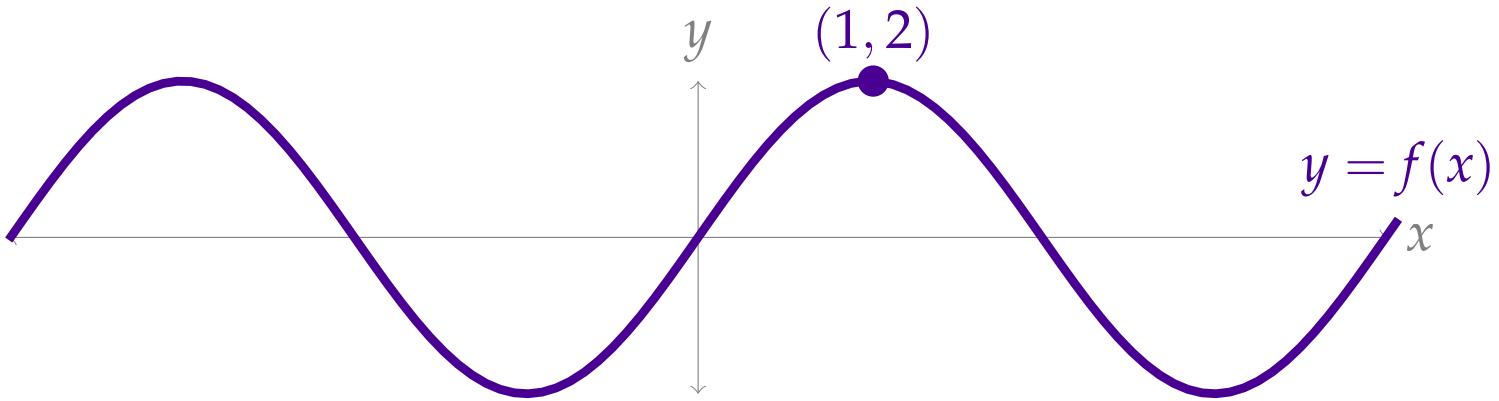
$f(x) = \cos(x)$

$f(x) = \dfrac{x^4 + \cos(x)}{x^{16} + 7}$

Odd Functions

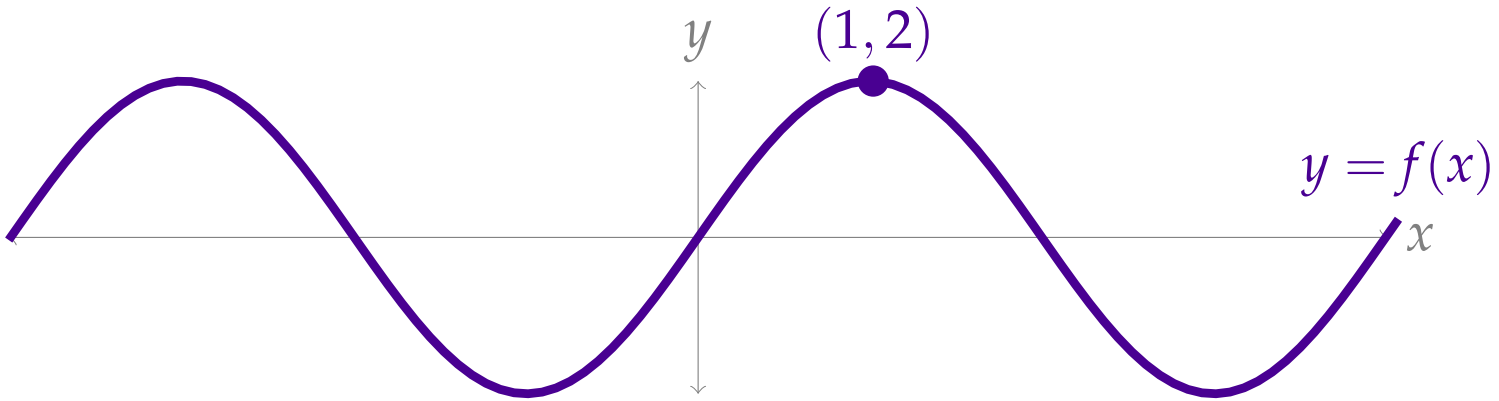


Odd Functions



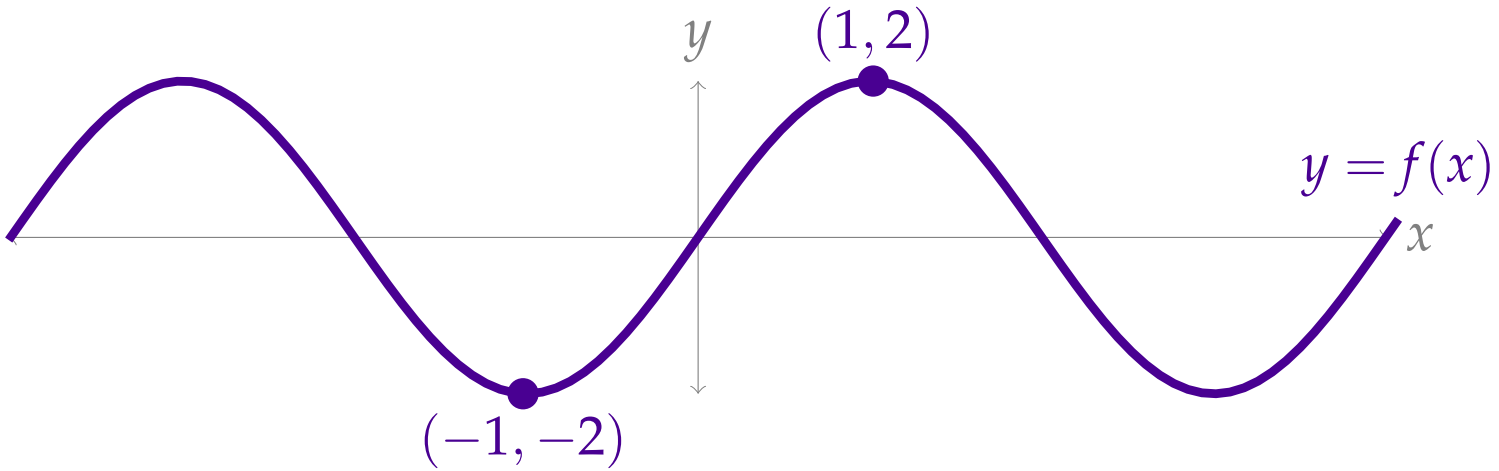
Suppose $f(1) = 2$.

Odd Functions



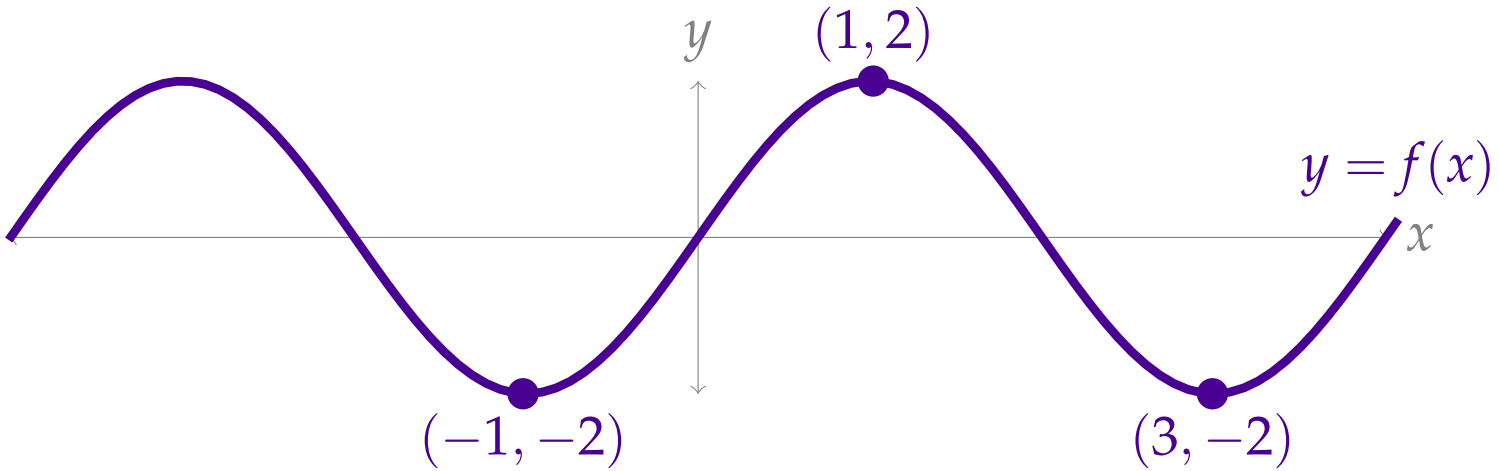
Suppose $f(1) = 2$. Then $f(-1) =$

Odd Functions



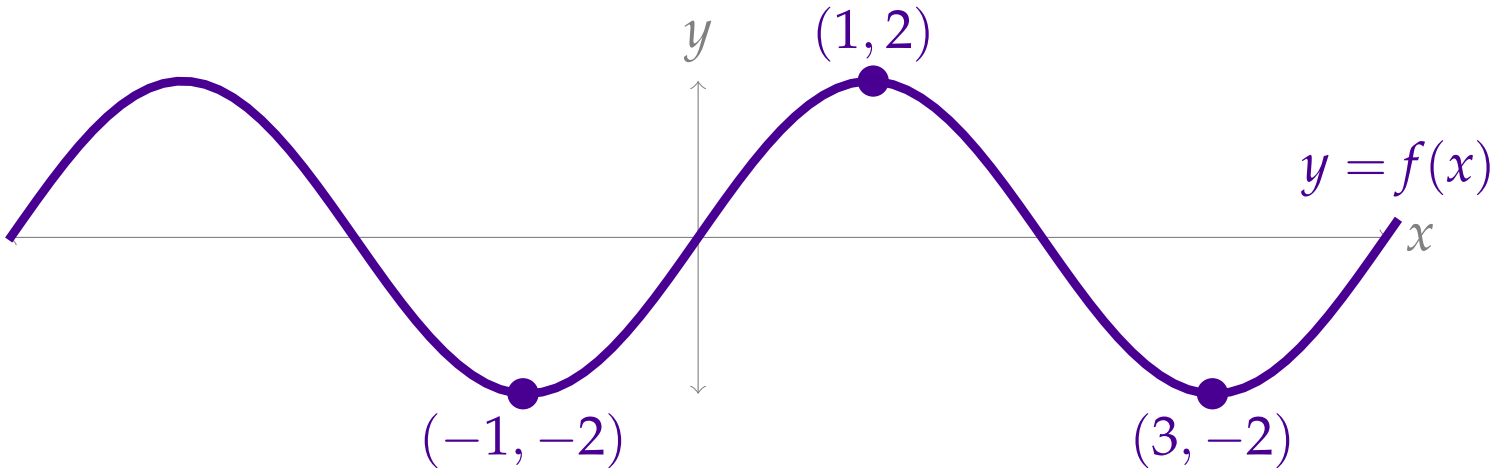
Suppose $f(1) = 2$. Then $f(-1) = -2$.

Odd Functions



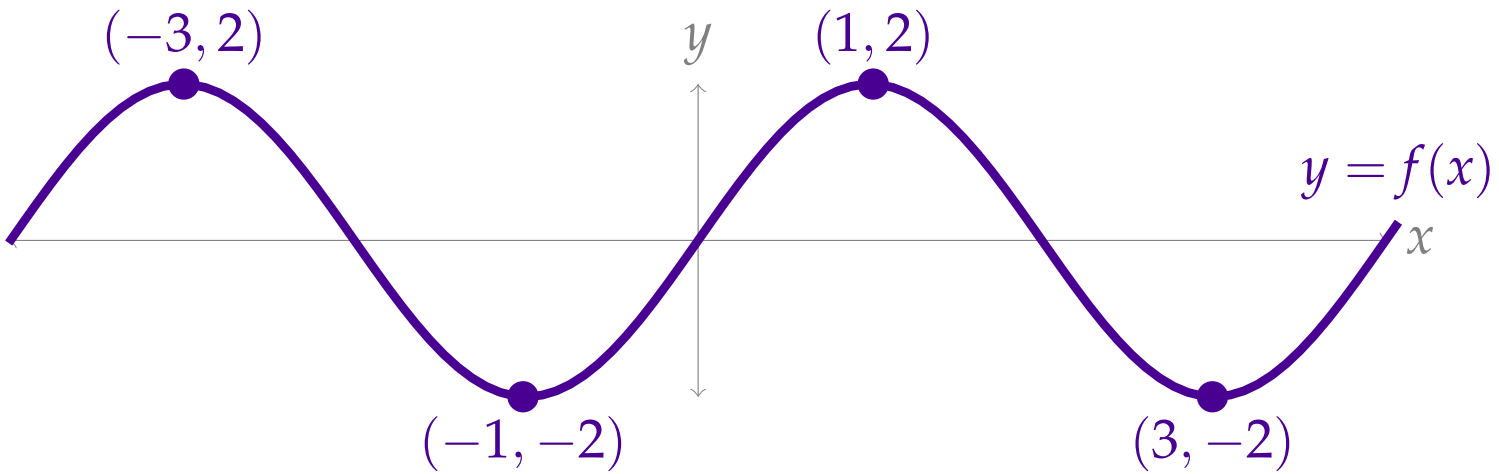
Suppose $f(1) = 2$. Then $f(-1) = -2$.
Suppose $f(3) = -2$.

Odd Functions



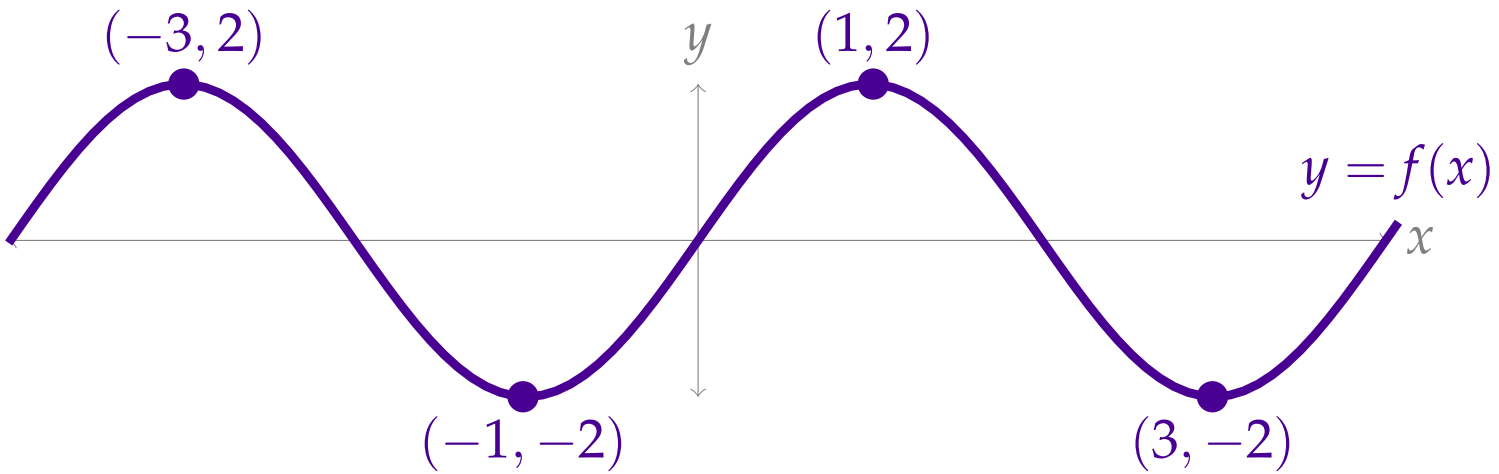
Suppose $f(1) = 2$. Then $f(-1) = -2$.
Suppose $f(3) = -2$. Then $f(-3) =$

Odd Functions



Suppose $f(1) = 2$. Then $f(-1) = -2$.
Suppose $f(3) = -2$. Then $f(-3) = 2$.

Odd Functions

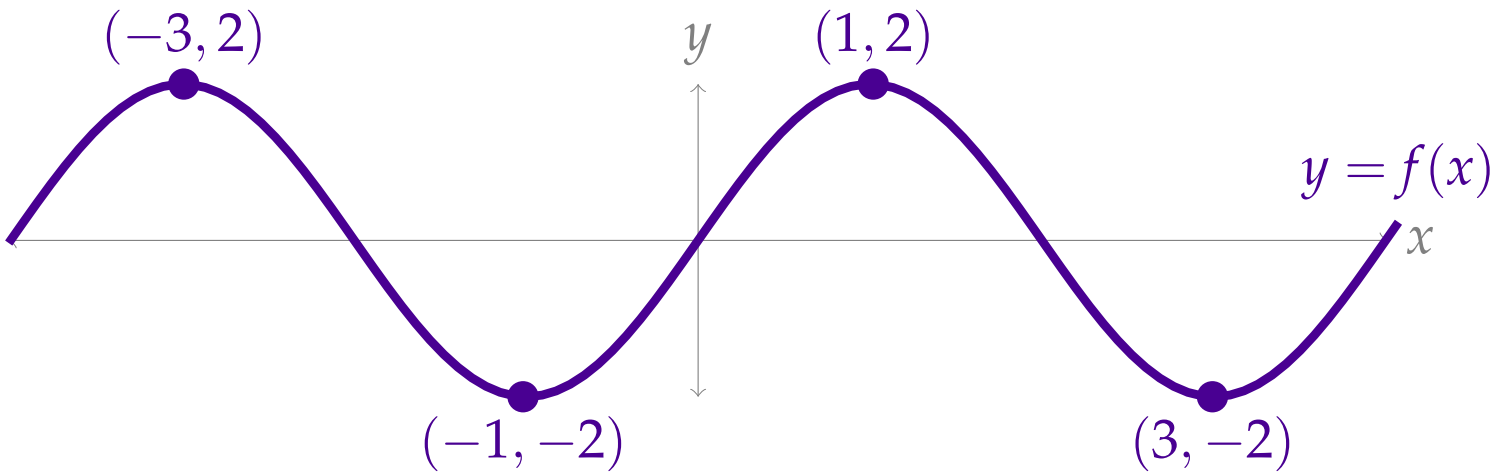


Suppose $f(1) = 2$. Then $f(-1) = -2$.
Suppose $f(3) = -2$. Then $f(-3) = 2$.

Odd Function – Definition 3.6.7

A function $f(x)$ is odd if, for all x in its domain,

Odd Functions



Suppose $f(1) = 2$. Then $f(-1) = -2$.
Suppose $f(3) = -2$. Then $f(-3) = 2$.

Odd Function – Definition 3.6.7

A function $f(x)$ is odd if, for all x in its domain,

$$f(-x) = -f(x)$$

Odd Functions

Odd Function – Definition 3.6.7

A function $f(x)$ is odd if, for all x in its domain,

$$f(-x) = -f(x)$$

Examples:

Odd Functions

Odd Function – Definition 3.6.7

A function $f(x)$ is odd if, for all x in its domain,

$$f(-x) = -f(x)$$

Examples:
$f(x) = x$

Odd Functions

Odd Function – Definition 3.6.7

A function $f(x)$ is odd if, for all x in its domain,

$$f(-x) = -f(x)$$

Examples:
$f(x) = x$
$f(x) = x^3$

Odd Functions

Odd Function – Definition 3.6.7

A function $f(x)$ is odd if, for all x in its domain,

$$f(-x) = -f(x)$$

Examples:

$f(x) = x$

$f(x) = x^3$

$f(x) = \sin(x)$

Odd Functions

Odd Function – Definition 3.6.7

A function $f(x)$ is odd if, for all x in its domain,

$$f(-x) = -f(x)$$

Examples:

$f(x) = x$

$f(x) = x^3$

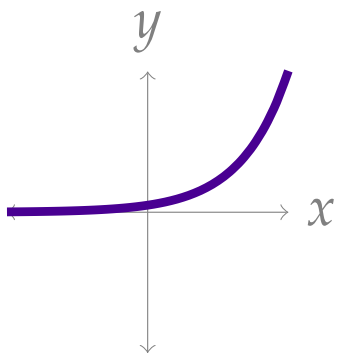
$f(x) = \sin(x)$

$f(x) = \dfrac{x(1+x^2)}{x^2+5}$

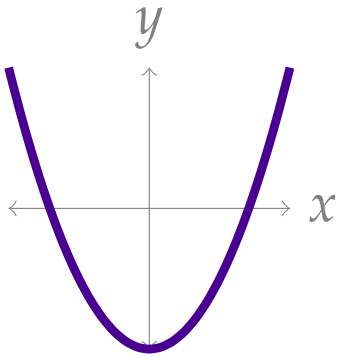
Poll Tiiime

Pick out the odd function.

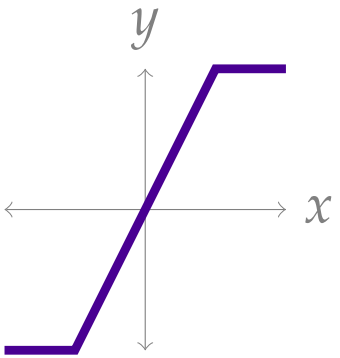
A:



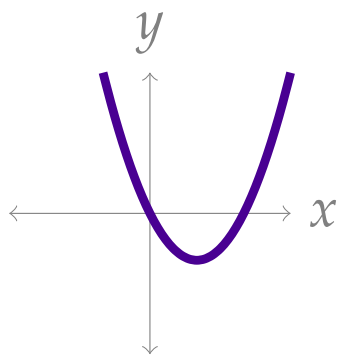
B:



C:



D:

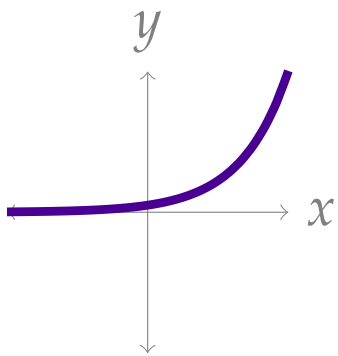


Q

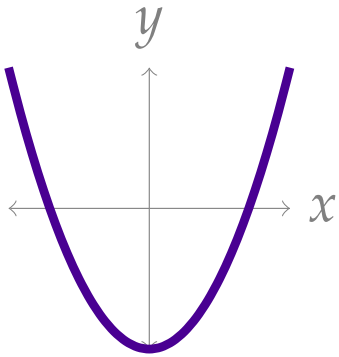
Poll Tiiime

Pick out the odd function.

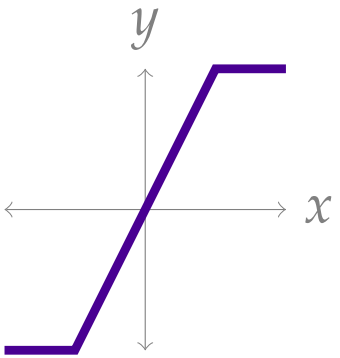
A:



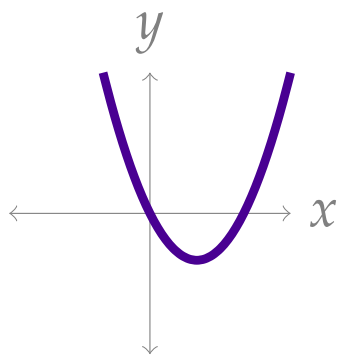
B:



C:



D:

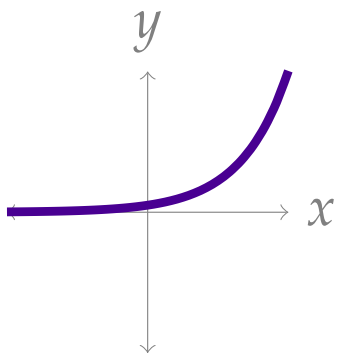


A

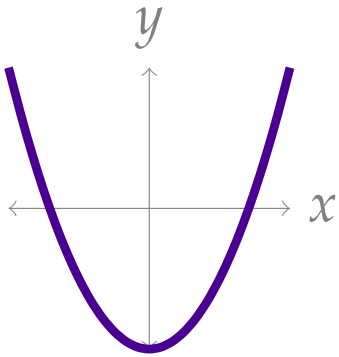
Poll Tiiime

Pick out the even function.

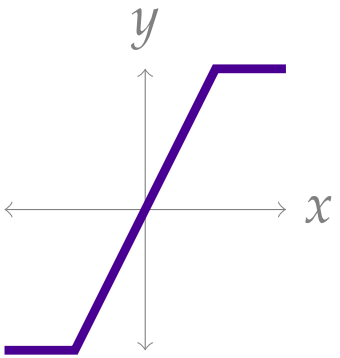
A:



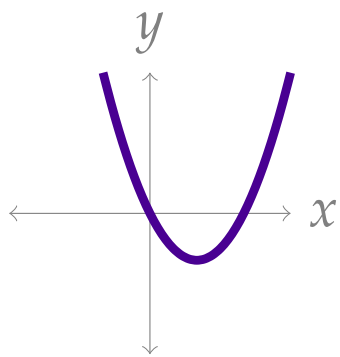
B:



C:



D:

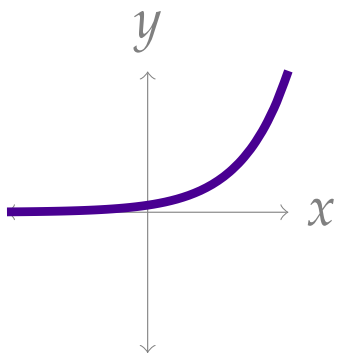


Q

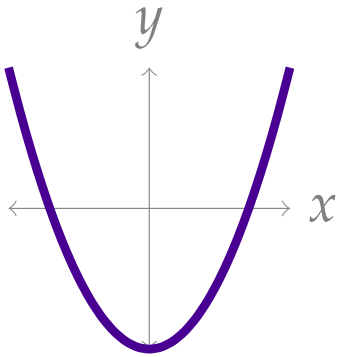
Poll Tiiime

Pick out the even function.

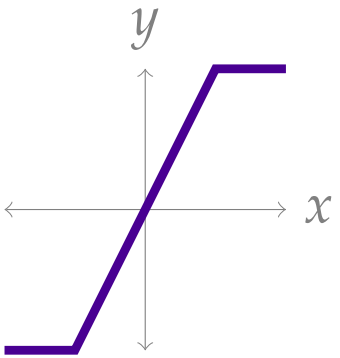
A:



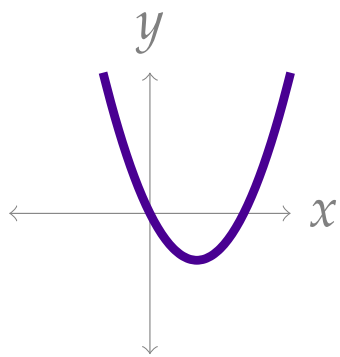
B:

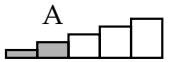


C:



D:





Even more Poll tiiiiime

Suppose $f(x)$ is an odd function, continuous, defined for all real numbers. What is $f(0)$? Pick the best answer.

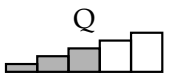
A. $f(0) = f(-0)$

B. $f(0) = -f(0)$

C. $f(0) = 0$

D. all of the above are true

E. none of the above are necessarily true



Even more Poll tiiiiime

Suppose $f(x)$ is an odd function, continuous, defined for all real numbers. What is $f(0)$? Pick the best answer.

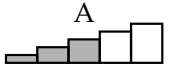
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Even more Poll tiiiiime

Suppose $f(x)$ is an odd function, continuous, defined for all real numbers. What is $f(0)$? Pick the best answer.

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A. $f(0) = f(-0)$ ← true but uninteresting, for all functions

B. $f(0) = -f(0)$ ← only possible for $f(0) = 0$

C. $f(0) = 0$ ← this is equivalent to the choice above

D. all of the above are true

E. none of the above are necessarily true

Even more and more Poll tiiiiime

Suppose $f(x)$ is an even function, continuous, defined for all real numbers. What is $f(0)$? Pick the best answer.

A. $f(0) = f(-0)$

B. $f(0) = -f(0)$

C. $f(0) = 0$

D. all of the above are true

E. none of the above are necessarily true

Q

Even more and more Poll tiiiiime

Suppose $f(x)$ is an even function, continuous, defined for all real numbers. What is $f(0)$? Pick the best answer.

A. $f(0) = f(-0)$

B. $f(0) = -f(0)$

C. $f(0) = 0$

D. all of the above are true

E. none of the above are necessarily true



OK OK... LAST ONE

Suppose $f(x)$ is an even function, differentiable for all real numbers. What can we say about $f'(x)$?

A. $f'(x)$ is also even

B. $f'(x)$ is odd

C. $f'(x)$ is constant

D. all of the above are true

E. none of the above are necessarily true

Q

OK OK... LAST ONE

Suppose $f(x)$ is an even function, differentiable for all real numbers. What can we say about $f'(x)$?

A. $f'(x)$ is also even

B. $f'(x)$ is odd

C. $f'(x)$ is constant

D. all of the above are true

E. none of the above are necessarily true

A



Periodicity

Periodic – Definition 3.6.10

A function is periodic with period $P > 0$ if

$$f(x) = f(x + P)$$

whenever x and $x + P$ are in the domain of f, and P is the smallest such (positive) number

Examples: $\sin(x)$, $\cos(x)$ both have period 2π; $\tan(x)$ has period π.

Ignoring concavity, sketch $f(x) = \sin(\sin x)$.

Challenge: ignoring exact locations of extrema, sketch $g(x) = \sin(2\pi \sin x)$.

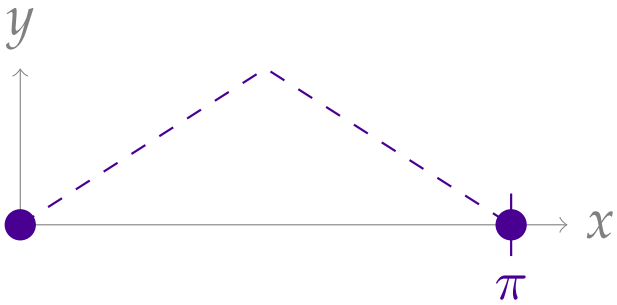
$$f(x) = \sin(\sin x)$$

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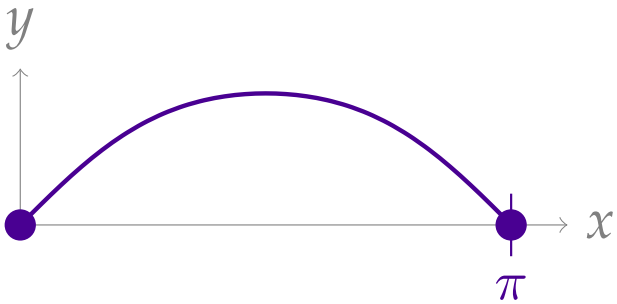
Sin is periodic; since $\sin x = \sin(2\pi + x)$, then $\sin(\sin x) = \sin(\sin(2\pi + x))$, so $f(x)$ is also periodic. It suffices to sketch $f(x)$ for an interval of length 2π, because any such segment will repeat.
Since the function is also odd, if we sketch it on the interval $[0, \pi]$, then we can extrapolate to the interval $[-\pi, 0]$. So we consider the interval $[0, \pi]$.

- Intercepts: $(0, 0)$, $(\pi, 0)$
- First derivative: $f'(x) = \cos(\sin x) \cdot \cos(x)$
  - For $0 \leq x \leq \pi$, we have $0 \leq \sin x \leq 1 < \frac{\pi}{2}$ and hence $0 < \cos(\sin x) \leq 1$.
  - CP: $x = \frac{\pi}{2}$
  - increasing: $(0, \pi/2)$
  - decreasing: $(\pi/2, \pi)$

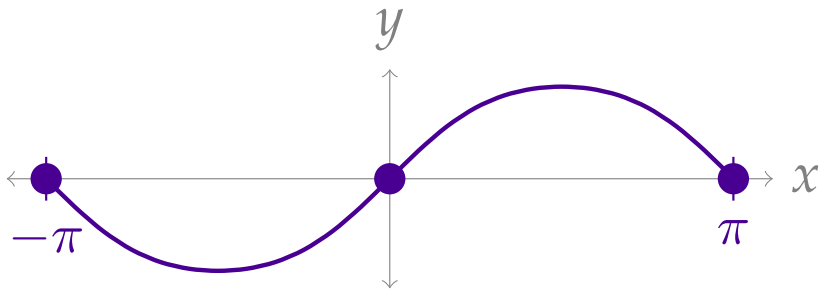
1. Interval $(0, \pi)$ skeleton, based on above work:



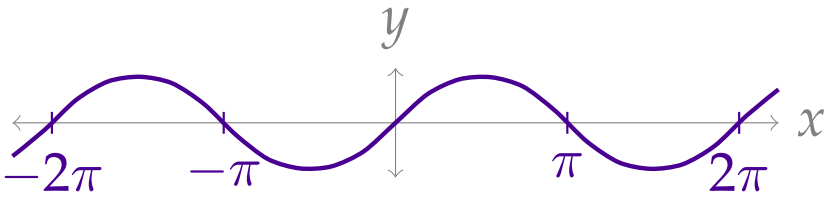
2. Make into a smooth curve:



3. Use odd symmetry to get interval $[-\pi, \pi]$



4. Use periodicity



skip $g(x)$

$$g(x) = \sin(2\pi \sin x)$$

skip $g(x)$

$$g(x) = \sin(2\pi \sin x)$$

Sin is periodic; since $2\pi \sin x = 2\pi \sin(2\pi + x)$, then

$$\sin(2\pi \sin x) = \sin(2\pi \sin(2\pi + x))$$

so $g(x)$ is also periodic. It suffices to sketch $g(x)$ for an interval of length 2π, because any such segment will repeat.

Note

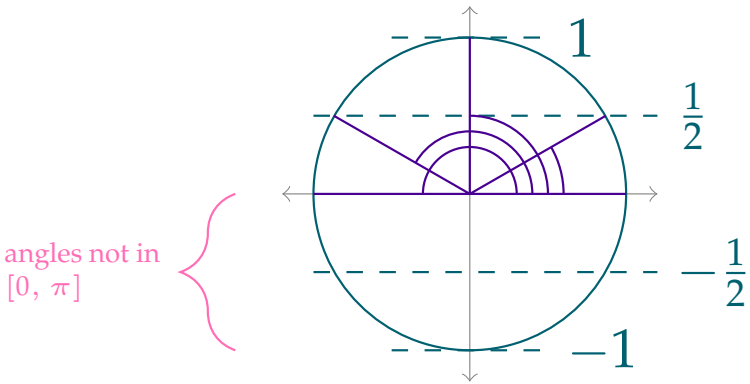
$$\begin{aligned} g(-x) &= \sin\big(2\pi \sin(-x)\big) = \sin\big((2\pi)(-\sin x)\big) \\ &= \sin(-2\pi \sin x) = -\sin(2\pi \sin x) = -g(x) \end{aligned}$$

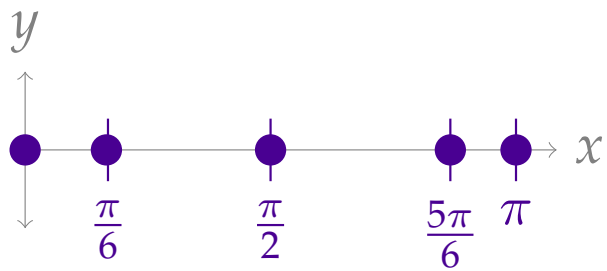
so $g(x)$ is odd. If we sketch it on the interval $[0, \pi]$, then we can extrapolate to the interval $[-\pi, 0]$. So we consider the interval $[0, \pi]$.

skip $g(x)$

Intercepts in $[0, \pi]$:

$$
\begin{aligned}
g(0) &= 0 \\
0 = g(x) &= \sin(2\pi \sin x) \\
\implies 2\pi \sin x &\in \{0, \pm\pi, \pm 2\pi, \pm 3\pi, \ldots\} \\
\implies \sin x &\in \left\{0, \pm\frac{1}{2}, \pm 1, \pm\frac{3}{2}, \ldots\right\} \\
\implies x &\in \left\{0, \frac{\pi}{6}, \frac{\pi}{2}, \frac{5\pi}{6}, \pi\right\}
\end{aligned}
$$





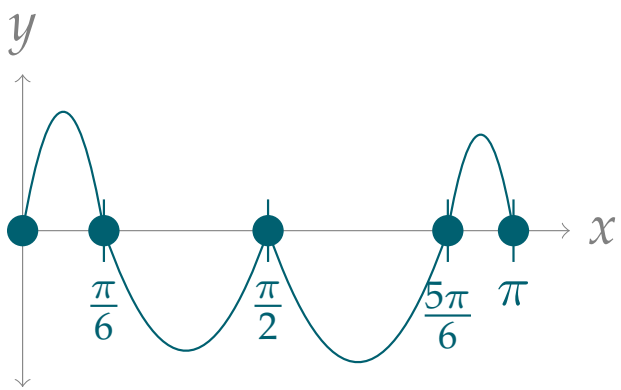
A

skip $g(x)$

Now let's consider the sign of $g(x)$ between the intercepts. Since $g(x)$ isn't given as a factored product, our old shortcut isn't so useful.

interval	$\left(0, \frac{\pi}{6}\right)$	$\left(\frac{\pi}{6}, \frac{\pi}{2}\right)$	$\left(\frac{\pi}{2}, \frac{5\pi}{6}\right)$	$\left(\frac{5\pi}{6}, \pi\right)$
range of $\sin x$	$\left(0, \frac{1}{2}\right)$	$\left(\frac{1}{2}, 1\right)$	$\left(\frac{1}{2}, 1\right)$	$\left(0, \frac{1}{2}\right)$
range of $2\pi \sin x$	$(0, \pi)$	$(\pi, 2\pi)$	$(\pi, 2\pi)$	$(0, \pi)$
sign of $\sin(2\pi \sin x)$	$+$	$-$	$-$	$+$

So, a rough sketch on the interval $[0, \pi]$ is:

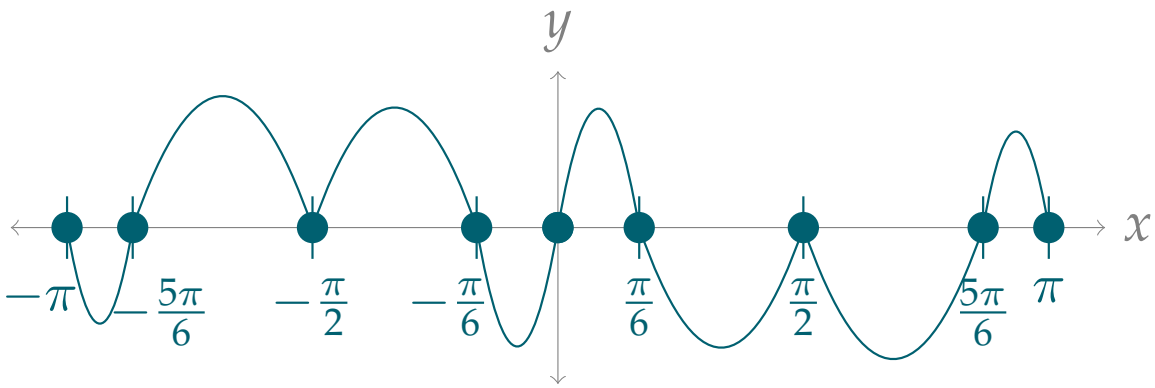


Yes, this is a rough sketch. The curve should be smooth at $\frac{\pi}{2}$.

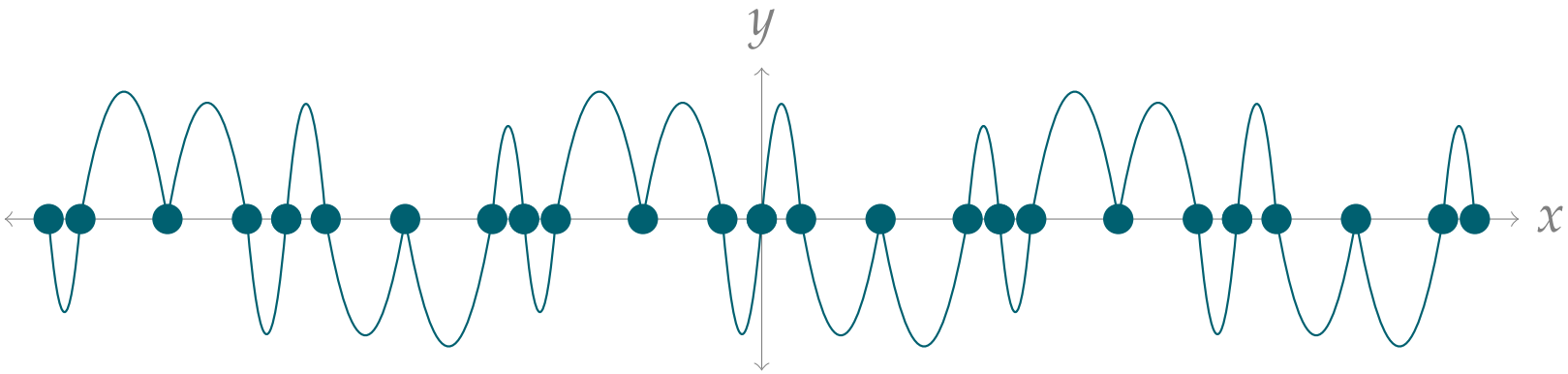
A

skip $g(x)$

Use odd symmetry, we sketch the interval $[-\pi, \pi]$:



Using periodicity:



A

Let's Graph

$$f(x) = (x^2 - 64)^{1/3}$$

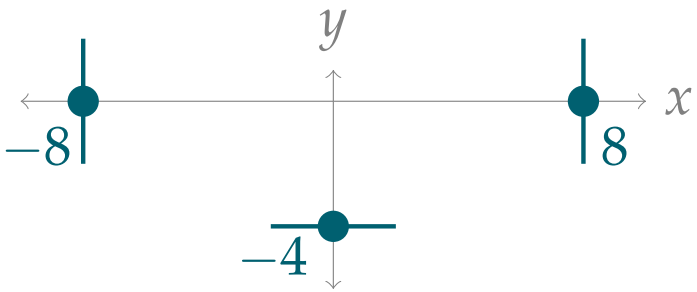
Let's Graph

$$f(x) = (x^2 - 64)^{1/3}$$

$$f'(x) = \frac{2x}{3(x^2 - 64)^{2/3}};$$

$$f''(x) = \frac{-2(\frac{1}{3}x^2 + 64)}{3(x^2 - 64)^{5/3}}$$

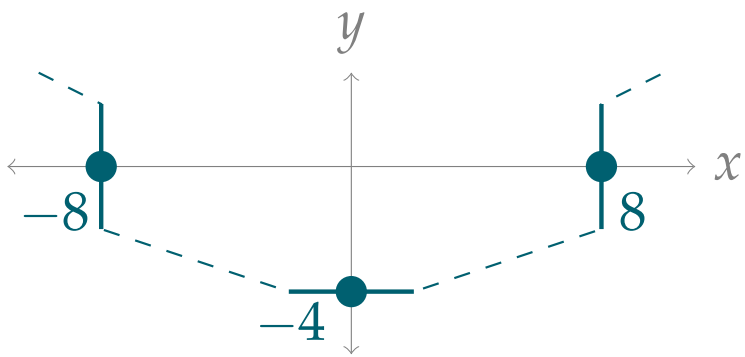
Domain	all real numbers
End behaviour	$\lim_{x\to\infty} f(x) = \infty$ and $\lim_{x\to-\infty} f(x) = \infty$
Intercepts	$(0,-4)$, $(\pm 8, 0)$
Critical point	$(0,-4)$
Singular points	$(-8,0)$, $(8,0)$ Near the singular points, $f'(x)$ gets very large, so $f(x)$ looks vertical.



Increasing, decreasing

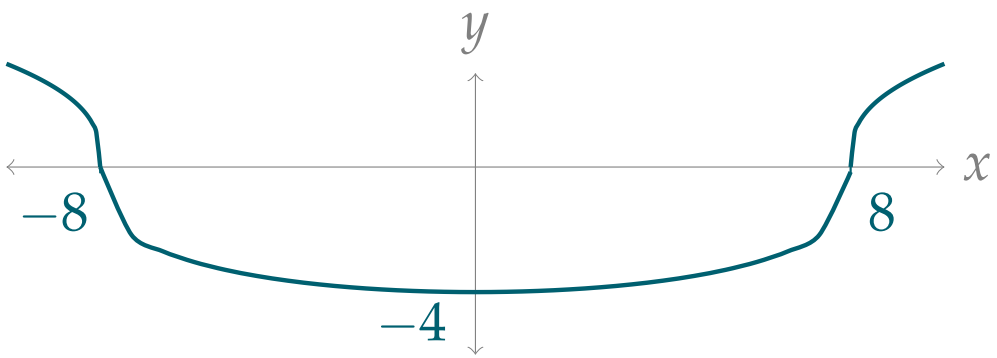
Numerator of $f'(x)$ is positive on $(0, \infty)$ and negative on $(-\infty, 0)$. Denominator is positive where it exists. So $f(x)$ is decreasing on $(-\infty, 0)$ (where it is differentiable) and increasing on $(0, \infty)$ (where it is differentiable).

Skeleton



Concavity

Numerator of second derivative is negative everywhere. Denominator is positive (so $f''(x)$ is negative) on $(-\infty, -8) \cup (8, \infty)$ and denominator is negative (so $f''(x)$ is positive) on $(-8, 8)$.
So our function is concave up on $(-8, 8)$ and concave down on $(-\infty, -8) \cup (8, \infty)$.



Let's Graph

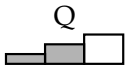
$$f(x) = \frac{x^2 + x}{(x+1)(x^2+1)^2}$$

Q

Let's Graph

$$f(x) = \frac{x^2 + x}{(x+1)(x^2+1)^2}$$

Note: for $x \neq -1, f(x) = \dfrac{x(x+1)}{(x+1)(x^2+1)^2} = \dfrac{x}{(x^2+1)^2}$



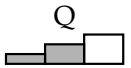


Let's Graph

$$f(x) = \frac{x^2 + x}{(x+1)(x^2+1)^2}$$

Note: for $x \neq -1, f(x) = \dfrac{x(x+1)}{(x+1)(x^2+1)^2} = \dfrac{x}{(x^2+1)^2}$

$$g(x) := \frac{x}{(x^2+1)^2}$$



Let's Graph

$$f(x) = \frac{x^2 + x}{(x+1)(x^2+1)^2}$$

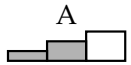
Note: for $x \neq -1, f(x) = \dfrac{x(x+1)}{(x+1)(x^2+1)^2} = \dfrac{x}{(x^2+1)^2}$

$$g(x) := \frac{x}{(x^2+1)^2}$$

$$g'(x) = \frac{1 - 3x^2}{(x^2+1)^3}$$

$$g''(x) = \frac{12x(x^2-1)}{(x^2+1)^4}$$

$$f(x) = \frac{x^2 + x}{(x+1)(x^2+1)^2}$$

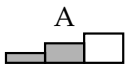


$$f(x) = \frac{x^2 + x}{(x+1)(x^2+1)^2}$$

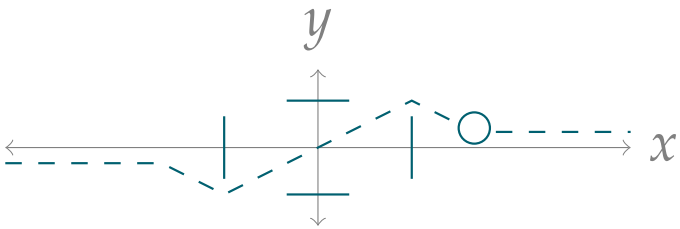
When $x \neq -1, f(x) = g(x)$. So, $f(x)$ looks like $g(x)$ except it has a removable discontinuity (hole) at $x = -1$. Let's graph

$$g(x) = \frac{x}{(x^2+1)^2}$$

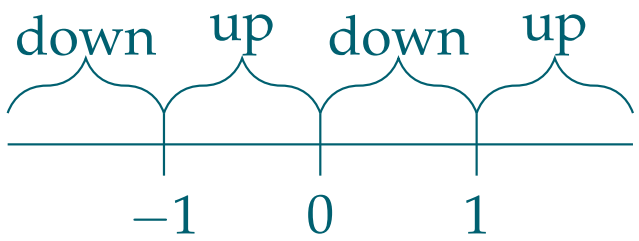
- Domain: all real numbers
- HA: $y = 0$ on both sides
- VA: none
- Intercepts: $(0, 0)$
- Odd symmetry
- CP: $x = \pm\frac{1}{\sqrt{3}}$; associated points: $\left(-\frac{1}{\sqrt{3}}, -\frac{3\sqrt{3}}{16}\right)$ and $\left(\frac{1}{\sqrt{3}}, \frac{3\sqrt{3}}{16}\right)$

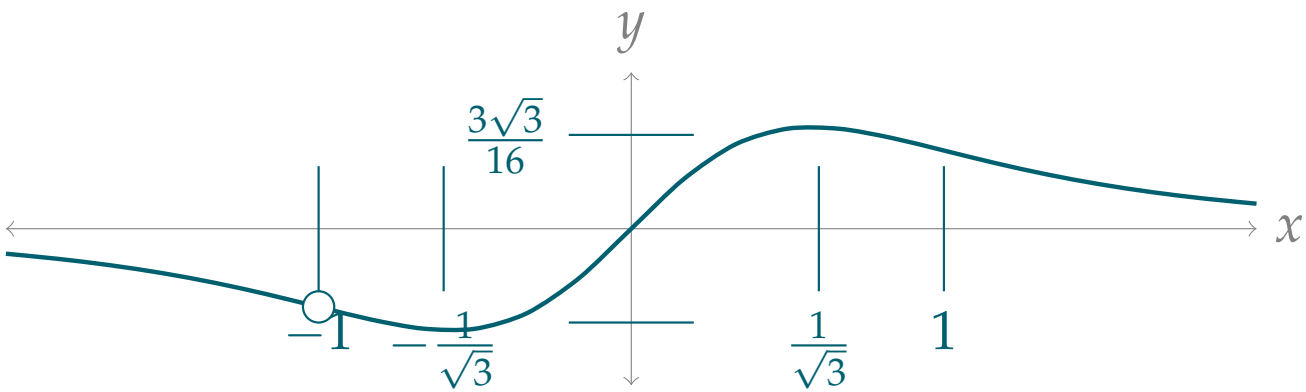


- Increasing on $\left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}\right)$ and decreasing on $\left(-\infty, -\frac{1}{\sqrt{3}}\right) \cup \left(\frac{1}{\sqrt{3}}, \infty\right)$



- g'' is given (nearly) factored, so we can see its sign changes at $x = -1, 0, 1$. Concavity:





Let's Graph

$$f(x) = x(x-1)^{2/3}$$

- $f'(x) = \dfrac{5x-3}{3\sqrt[3]{x-1}}$
- $f''(x) = \dfrac{2(5x-6)}{9(\sqrt[3]{x-1})^4}$



Let's Graph

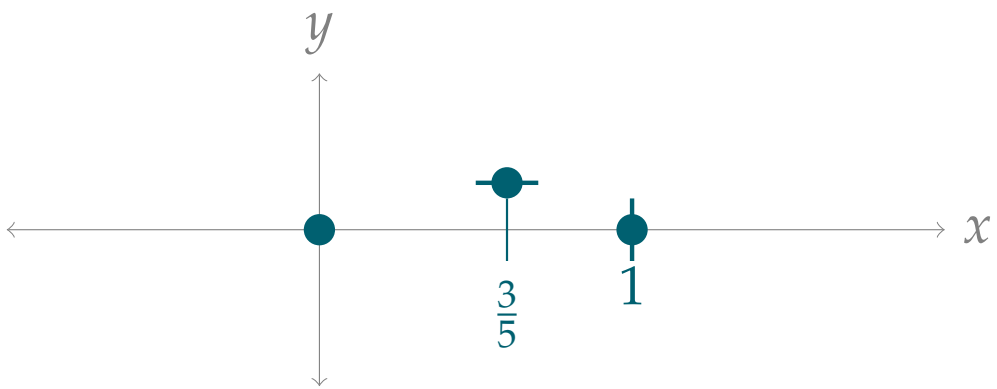
$$f(x) = x(x-1)^{2/3}$$

- $f'(x) = \dfrac{5x-3}{3\sqrt[3]{x-1}}$
- $f''(x) = \dfrac{2(5x-6)}{9(\sqrt[3]{x-1})^4}$

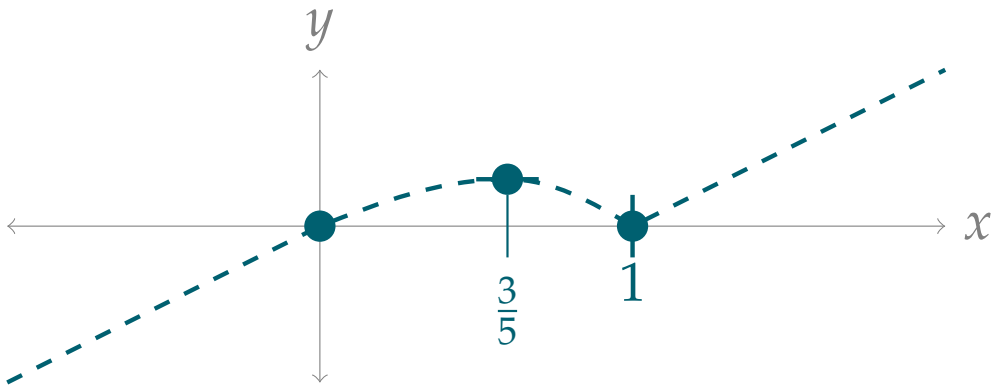
- $f(3/5) \approx 0.3$
- $f(6/5) \approx 0.4$

Q

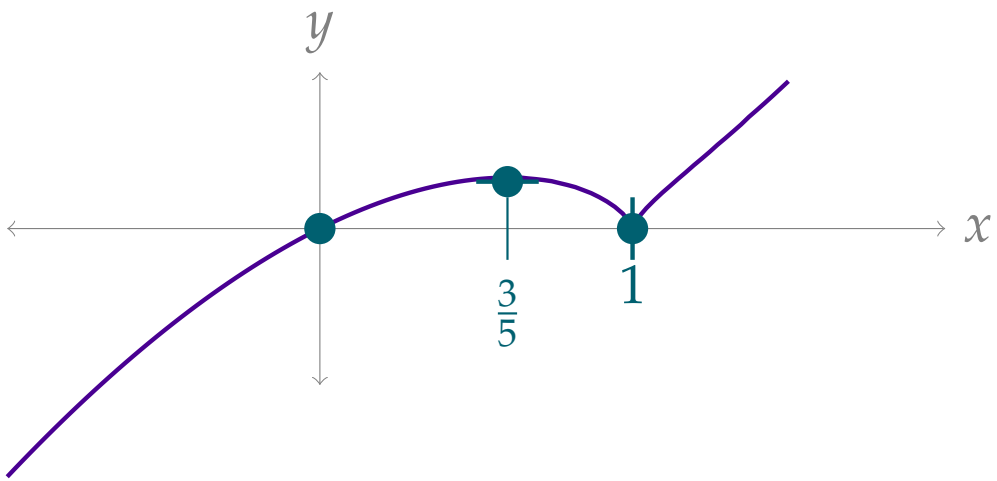
- Domain: all reals
- VA: none
- HA: none
- Intercepts: $(0,0)$, $(1,0)$
- Symmetry: not even, not odd, not periodic
- CP: $x = \frac{3}{5}$ ($y \approx 0.3$); SP $x = 1$ ($y = 0$).
  Near the SP, first deriv is very large, so function looks vertical.



- First derivative changes sign at $x = \frac{3}{5}$ and $x = 1$. Function is increasing on $\left(-\infty, \frac{3}{5}\right) \cup (1, \infty)$ and decreasing on $\left(\frac{3}{5}, 1\right)$.



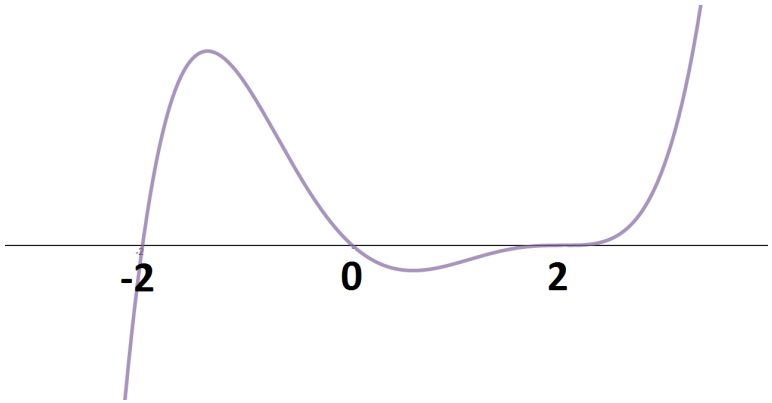
- Second derivative changes sign at $x = \frac{6}{5}$. (Note the denominator has an even power). Function is concave down on $\left(-\infty, \frac{6}{5}\right)$ and concave up on $\left(\frac{6}{5}, \infty\right)$.



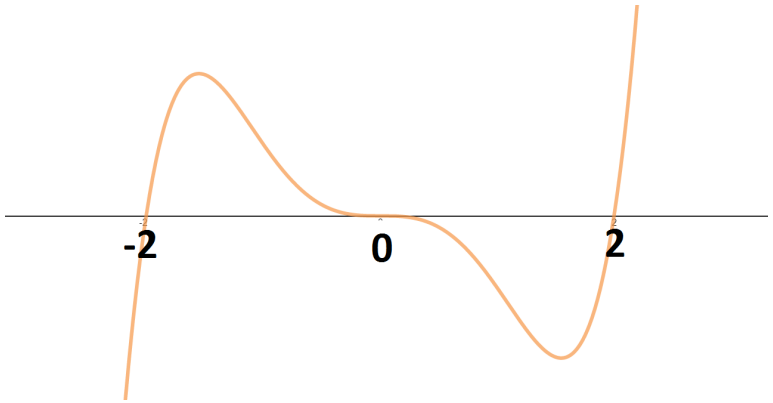
Ch 3.6 Review: matching

Match the Function to its Graph

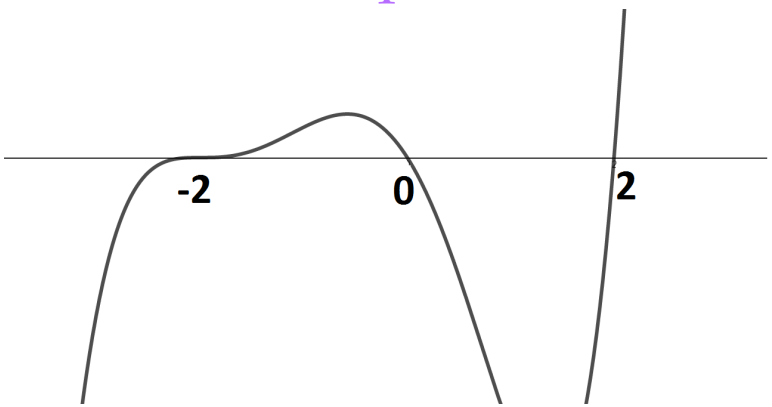
A. $f(x) = x^3(x+2)(x-2) = x^5 - 4x^3$
B. $f(x) = x(x+2)^3(x-2) = x^5 + 4x^4 - 16x^2 - 16x$
C. $f(x) = x(x+2)(x-2)^3 = x^5 - 4x^4 + 16x^2 - 16x$



I



III



II

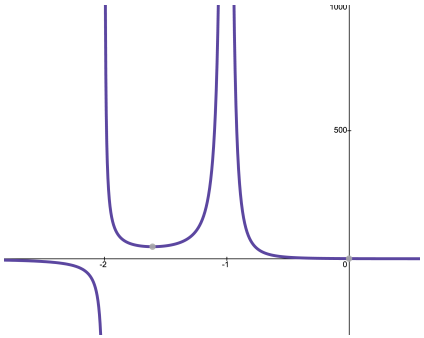
A. $f(x) = \dfrac{x-1}{(x+1)(x+2)}$

B. $f(x) = \dfrac{(x-1)^2}{(x+1)(x+2)}$

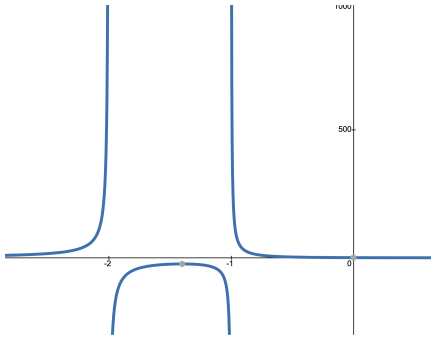
C. $f(x) = \dfrac{x-1}{(x+1)^2(x+2)}$

D. $f(x) = \dfrac{(x-1)^2}{(x+1)^2(x+2)}$

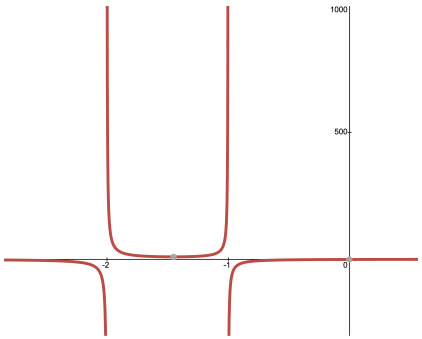
I.



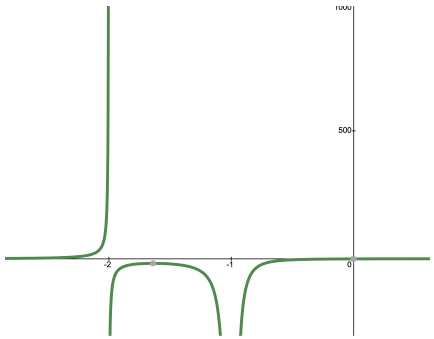
III.



II.



IV.

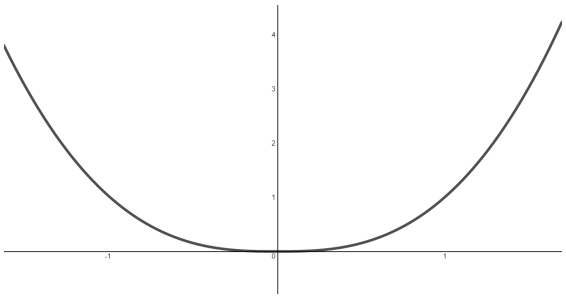


Q

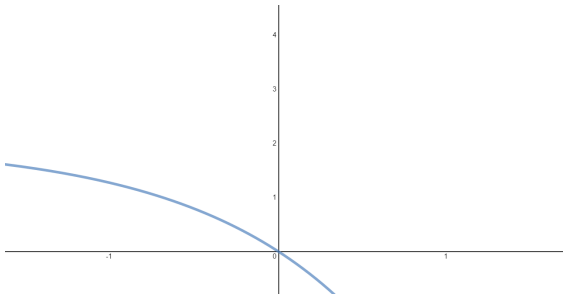
ans

Match the Function to its Graph

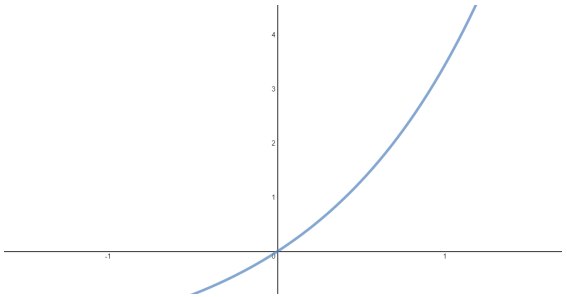
A. $f(x) = |x|^e$ B. $f(x) = e^{|x|}$ C. $f(x) = e^{x^2}$ D. $f(x) = e^{x^4 - x}$



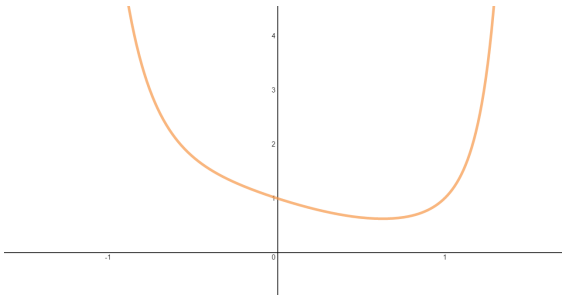
I



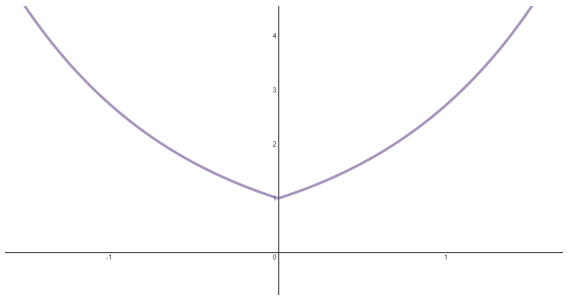
II



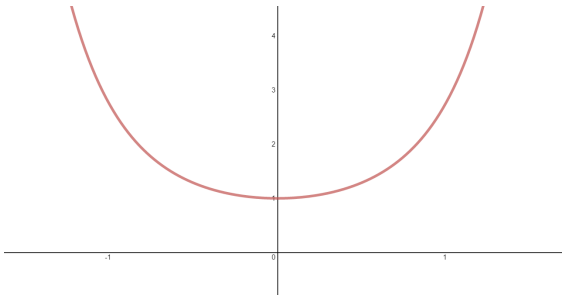
III



IV



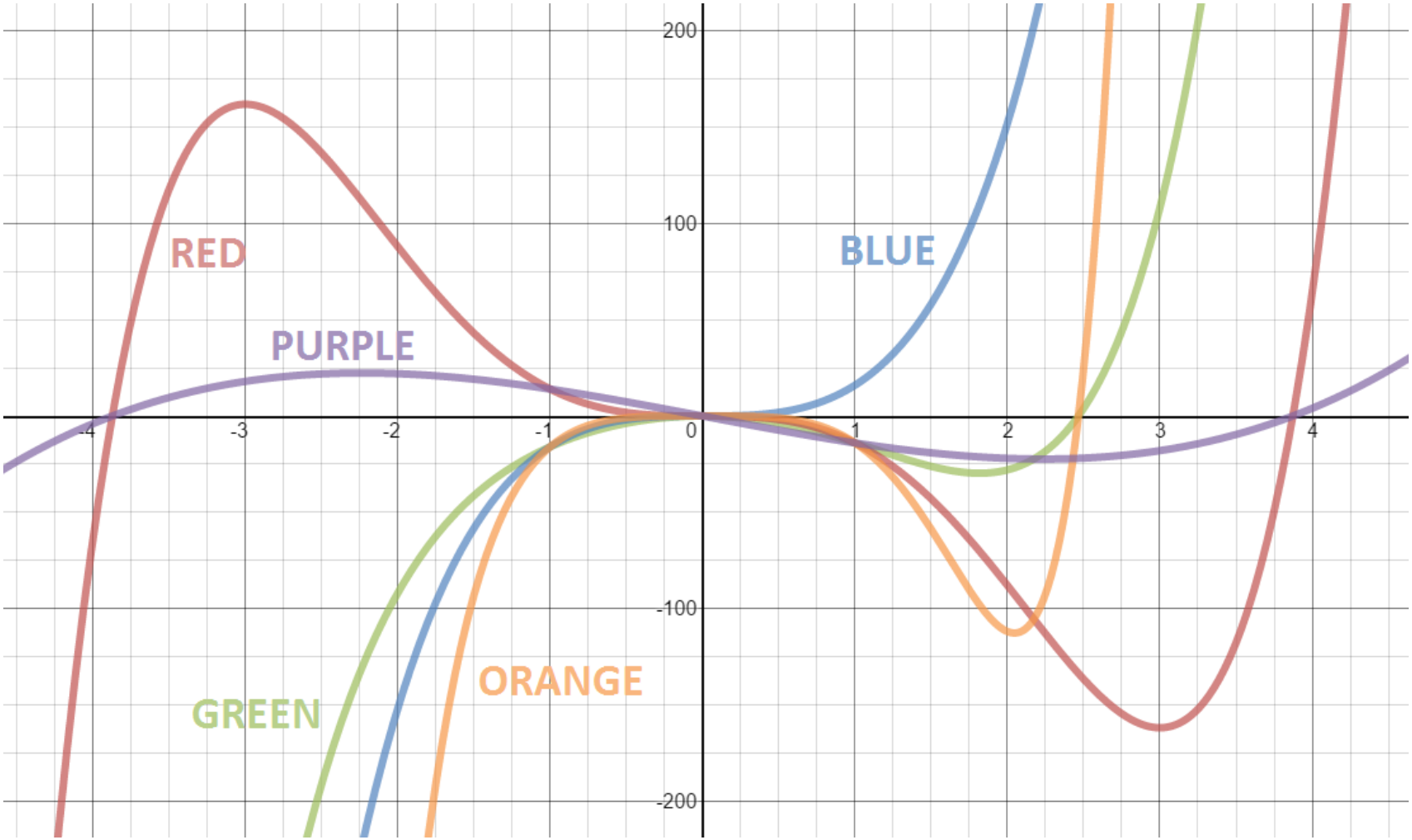
V



VI

A. $f(x) = x^5 + 15x^3$ B. $f(x) = x^5 - 15x^3$ C. $f(x) = x^5 - 15x^2$

D. $f(x) = x^3 - 15x$ E. $f(x) = x^7 - 15x^4$



Q

ans

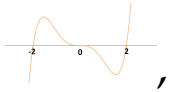
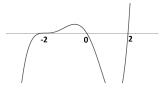
Included Work

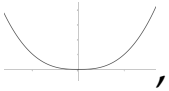


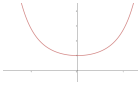
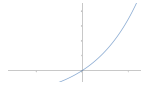
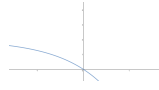
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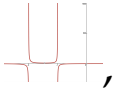
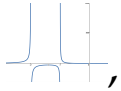
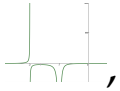
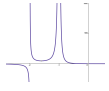


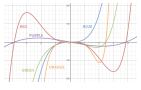
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